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# **GRAVITATIONAL LENSING — A COMPREHENSIVE & RIGOROUS OVERVIEW OF ANALYTICAL RESEARCH TECHNIQUES**

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**A PROJECT REPORT**

submitted by

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in partial fulfillment of the requirements for the award of the PG Diploma in

**ASTRONOMY & PLANETARIUM SCIENCE**

under the guidance of

**DR. BHASWATI MANDAL**



**M. P. Birla Institute of Fundamental Research**

**M. P. Birla Planetarium**

**May 19, 2025**



*To*  
*my Parents*  
*who taught me what excellence meant*  
*and my Teachers*  
*who showed me how to achieve it*



# DECLARATION

All of the work I have submitted is completely original and was finished while enrolled in M P Birla Planetarium's PG Diploma programme.

I have given full credit to those whose work has helped or influenced me. The calculations for mathematical derivations were done by myself from scratch. When results from other authors are used, they are properly attributed in the text or in the corresponding figure.

The plots were created by using Python scripts written by me and the same goes for the schematics which were done in GIMP, Adobe Illustrator and Pinta. The entire dissertation was typeset using  $\text{\LaTeX}$ .

**Partha Pratim Basumallick**

May 19, 2025



# THESIS CERTIFICATE

This is to certify that the thesis titled “Gravitational Lensing—A Comprehensive & Rigorous Overview of Analytical Research Techniques” submitted by Mr. Partha Pratim Basumallick pursuing the PG Diploma course in Astronomy & Planetarium Science at M. P. Birla Planetarium has been carried out under the guidance of Dr. Bhaswati Mandal during the academic session 2023 – 2024. The contents of this project report, in full or parts have not been submitted to any other Institute or University for the award of any degree or diploma.

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# ABSTRACT

Gravitational lensing describes the astrophysical phenomena where the gravitational field of mass distributions in the cosmos affects how the propagation of light is affected. The trajectories that photons would follow in a perfect, homogeneous universe are not followed when they travel cosmic distances because of gravitational potentials linked to mass concentrations. Effectually speaking gravitational lensing or gravitational optics parallels the behaviour of light rays passing through media of varying refractive indices. The current thesis presents a comprehensive account of several analytical techniques that have been developed since the inception of this subject to the most recent advances. From the first forays into the subject by Soldner to the recent novel techniques introduced by Werner-Gibbons & Ono-Ishihara-Asada all these approaches have been discussed in detail sparing none of the mathematical rigor or discussions regarding astronomical observations. In particular the recent approaches incorporating considerations originating from finite distance effects are of prime interest and how in the infinite limit the results reduce to the derivations pre-existing in the literature.



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*Do not Bodies act upon Light at a distance, and by their action bend its Rays; and is not this action strongest at the least distance?*

—Isaac Newton, *Opticks*

*Oh leave the Wise our measures to collate  
One thing at least is certain, LIGHT has WEIGHT,  
One thing is certain, and the rest debate –  
Light-rays, when near the Sun, DO NOT GO STRAIGHT.*

— Arthur Eddington

# 1

## RUDIMENTS OF GRAVITATIONAL LENSING — TRACING OUT A HISTORICAL BLUEPRINT

This first chapter serves as the foundation stone to learning the modern tools of the trade in gravitational lensing. The introduction in [section 1.1](#) presents the scope and possibility of gravitational lensing in the arena of modern astrophysics with emphasis on cosmology [[section 1.1.2](#)]. Furthermore, a concise but detailed discussion on the different observational regimes of gravitational lensing reveals the intricacies and richness of this subject. This serves to establish our motivation for pursuing gravitational lensing as a research topic in general. Next we move onto the historical development of the subject with strong emphasis on its mathematical foundations. Given that the following chapters rely heavily on some intricate mathematical tools this should serve as an apt primer.

In particular since this thesis relies heavily on aspects of spacetime geometry the introductory chapter is structured with due emphasis on these topics. The theoretical framework of lens mapping, magnification and shearing geometry of lensed images, Fermat's principle in general relativity etc have been discussed in detail. On the other hand topics like caustics and cusps are only dealt with cursorily as they do not feature primarily in this dissertation.

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## 1.1 INTRODUCTION

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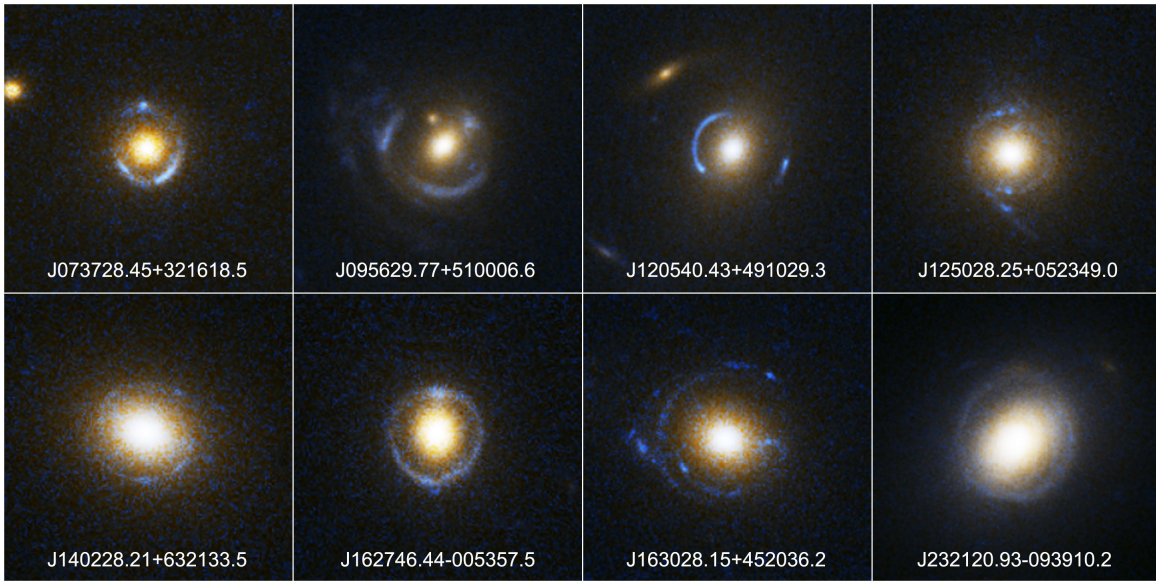
Gravitational lensing, like everything else associated with General Relativity, is corrupted with the seductive grace and elegance of mathematical rigor that lends itself readily to the explanation of breathtaking physical phenomena. This is perhaps one of the strongest reasons why gravitational lensing has emerged as a major field of research in the domain of modern astrophysics. Simply put the deflection of light rays from a far-off source as they approach a massive object, like a galaxy or a black hole, is known as gravitational lensing [Narayan and Bartelmann (1997), Dodelson (2017)]. The massive object’s gravitational field causes spacetime to curve, which results in this bending of light. The deflection angle, a crucial parameter that is essential to both theoretical astrophysics and observational cosmology, describes the degree of this bending. This thesis attempts to provide a detailed explanation of the importance of light bending and how the deflection angle is measured with strong emphasis on the effects of finite distance corrections based on novel mathematical techniques. Far from being an affair of theoretical curiosities lensing is very much an asset for experimental verification and/or designing new analytical frameworks for current and upcoming observations. Ranging from fields such as structural mapping of galaxies and galaxy clusters [Bartelmann (2010), Jalilvand et al. (2020), Torres-Ballesteros and Castañeda (2022), Natarajan et al. (2024), Ferrami and Wyithe (2024), Zamani et al. (2024), Buncher et al. (2025)] to investigating the elusive properties of dark matter and dark energy [Massey et al. (2010), Ellis (2010), Karamazov, Timko, and Heyrovský (2021), Şengül and Dvorkin (2022), Vegetti et al. (2023), Vachher et al. (2024), Gannon et al. (2025)], gravitational lensing encompasses a vast spread of research topics. Although it does not directly interact with light, dark matter, which makes up a sizable amount of the universe’s mass, gravitationally affects passing photons. In order to reconstruct the mass distribution of dark matter halos, scientists examine the deflection angles of light from distant background galaxies. Astronomers can gain a deeper understanding of the enigmatic nature of dark matter by using methods like weak gravitational lensing to infer the large-scale structure of dark matter in cosmic voids and galaxy clusters. In fact gravitational lensing has been successfully employed in studies related to exoplanets as well. The lensing effects of a planet that orbits the lensing star can also be measured. This method is employed to find planets that are far from Earth. [Tsapras (2018), Turyshev and Toth (2020; 2022), Toth and Turyshev (2023), Mróz and Poleski (2024)].

The deflection of null geodesics—trajectories followed by massless waves, such as electromagnetic waves (EMW) and gravitational waves (GWs)—as they propagate through the curved spacetime surrounding a massive object falls very much within the purview of gravitational lensing as well [Li et al. (2018), Diego, J. M. et al. (2019), Pagano, G. et al. (2020), Bulashenko and Ubach (2022), Tambalo et al. (2023), Meena (2024), Shan et al. (2025)]. Needless to say the gravitational potential of the intervening mass acts as the primary agent causing the said deflection, in accordance with the predictions of general relativity. The resulting lensing effect can generate intricate interference patterns and distinct image configurations or caustics, which

depend on the mass distribution of the lens and the geometric alignment between the source, lens, and observer. Hence, we can claim that via gravitational lensing, we gain a deeper understanding of how both electromagnetic radiation and gravitational waves are detected and interpreted from distant astrophysical sources. Moreover gravitational lensing also plays a fundamental role as a cosmological probe. But before we delve into that discussion it is essential to gain some fundamental understanding of the different regimes of gravitational lensing.

### 1.1.1 Observational regimes of gravitational lensing

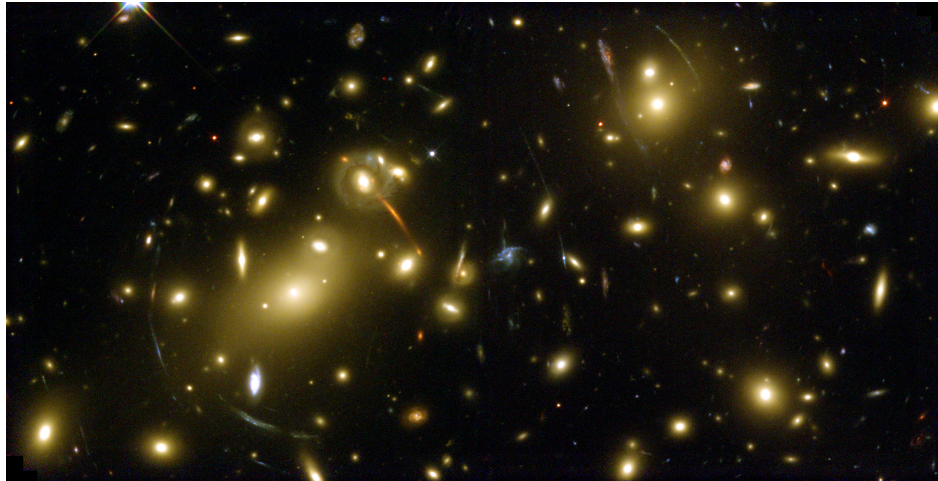
Gravitational lensing can be classified into three distinct categories — strong lensing, weak lensing, and micro lensing — depending on how much the lensing object influences the light rays coming from the source towards the observer. A cursory description of these phenomena is given below with relevant images illustrating the observations made and in some cases the corresponding geometric alignment.



**Figure 1.1:** An assortment of Einstein rings photographed by Hubble Space Telescope in 2005 as part of the Sloan Lens Arc Survey. [Bolton et al. (2006)]

- (a) Strong lensing** – In scenarios where the mass of the lensing object is sufficiently large and can generate a strong gravitational field strong lensing effects may be induced. Of course the geometry involving the trinity of source, lens & observer must conspire accordingly in order for this effect to be observed. In cases of strong lensing there are usually a multitude of images formed of the source and the deflection angle of the light rays are comparatively larger. Detailed discussions on the subject may be found in [Etherington et al. (2023), Kumar et al. (2023), Lemon et al. (2023), Saha et al. (2024), McCarty and

Connor (2024)]. Since it only affects a small percentage of distant sources, multiple-imaging caused by strong lensing is a relatively uncommon phenomenon [Press and Gunn (1973)]. Subarcsecond resolutions and, depending on the application, knowledge of the source and deflector’s redshifts are typical requirements for the measurements. Photometric redshifts are frequently used as estimates when redshift information is required and cannot be obtained through spectroscopy. However, because light from the background source and the foreground object mix, these are challenging to compute in crowded fields [Treu (2010), Jouvel, S. et al. (2014), Molino et al. (2017)]. Furthermore, they may occasionally turn out to be insufficiently accurate for the purposes in question [González et al. (2018)]. Strong lensing may also be utilised for determining cosmological parameters like the Hubble-Lemaitre parameter  $H_0$  [Narlikar (1996), Bernal et al. (2016)]. Distortions of background galaxies or other light sources are highly noticeable in strong gravitational lensing, producing arcs, multiple images, or so called Einstein rings fig. [1.1] first proposed in Chwolson (1924) and later by Einstein (1936).



**Figure 1.2:** Lensing of distant galaxy by the cluster Abell 2218 as an instance of weak lensing. In the image provided above there are quite a few instances of multiple images or arcs bearing testimony for strong lensing. However, the background galaxies tell a different story with their distorted images. Although these distortions/magnifications are quite difficult to measure [section 1.6, section 1.7].

- (b) Weak lensing** – In spite of the lensing object having a significant mass (either concentrated locally or distributed on large scales) the geometrical alignment of the aforementioned trinity may be less than ideal. As a consequence of this the image of the source suffers distortion to some extent (via magnification and shearing far from *critical lines*) and in some cases mild arcing (or arclet formation) as well [Fort et al. (1988), Tyson et al. (1990)]. This effect is known as *shear* in the literature of gravitational lensing. The existence of the shearing effect implies that the alignment of the objects in the background appear to be non-random. As a result of this the shearing effect can be measured statistically even if

the distortions of individual objects are too insignificant for direct identification. In general formation of small arclets detectable directly is quite rare. The measurement of the slight distortions that lensing causes in the form of background galaxies when photons pass through large-scale structures is the basis of the cosmic shear technique. [Bartelmann and Schneider (2001), Refregier (2003), Hoekstra (2013), Giblin et al. (2021), Li et al. (2022), Zhang et al. (2024), Adam et al. (2025)]. It is worth mentioning in this context that CMB lensing is classified as weak lensing and can accommodate the formation of multiple images for very small deflections provided the source is also very small [Lewis and Challinor (2006)]. A further sub-division of weak lensing is **statistical weak lensing** for cases which rely solely on statistical analysis and is devoid other observable characteristics. Foreground galaxy clusters and cosmic web filaments, as well as dark matter along the photon path, are examples of large-scale matter inhomogeneities that weakly lens high-redshift background galaxies. In these cases the lensing induced shear is observed statistically via averaging over a large number of galaxies [Kaiser (1992), Kaiser and Squires (1993), Köhlinger et al. (2017), Congdon and Keeton (2018), ParticleDataGroup et al. (2022)].

Because there are more dark-matter structures acting as lenses between us and the light sources, the distortions caused by weak gravitational lensing become more noticeable the farther we look. A three-dimensional layout of the distribution of dark matter in our universe will be provided by Euclid, which will measure the warped forms of billions of galaxies over ten billion years of cosmic history [Troja et al. (2022), Collaboration et al. (2024)]. This will clarify the nature of the exotic dark-matter element. As such weak lensing has emerged as a potent tool in modern cosmology[[section 1.1.2](#)].

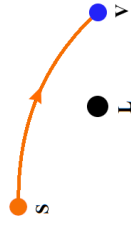
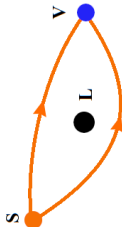
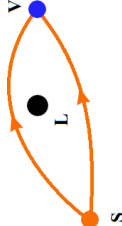
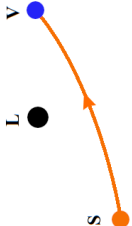
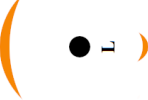
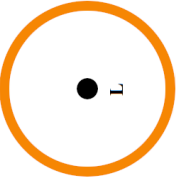
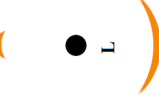
- (c) **Microlensing** – Microlensing is typically associated with lensing objects belonging to the category of stellar mass objects(e.g. normal stars, brown dwarfs or stellar remnants [white dwarfs, neutron stars and black holes]) . In this scenario the geometrical alignment is extremely favourable but the image splitting is too small (of the order of milliarcseconds in the local group) to be resolved by ground based telescopes. The effects that we can register are changes in magnification as a function of time. For the duration of the lensing event the image of the source appears to brighten. Note, that a specific reference to *time duration* is mentioned in the context of microlensing. This is due to the fact that the trinity of source, lens & observer all have relative motions and the alignment for which microlensing takes place is temporary. Microlensing has been employed as a tool for searching MACHO'S in the halo of the galaxy. For further discussions the reader is referred to the following works [Mao (2012), Gaudi (2012), Suyu et al. (2024)]. In fact signatures of microlensing in gravitational waves is also an active frontier of research [Diego, J. M. et al. (2019)].

Regarding the formation of Einstein rings and the importance of geometrical alignment in gravitational lensing a detailed description is provided with appropriate schematics in [table. 1.1](#).

A comparative description of the different types of lensing, the source and lensing object responsible, the resulting observation or detection and finally the research domain pertinent to them are provided in [table. 1.2](#).

**Table 1.1:** The table provided below presents a series of schematics depicting the formation of Einstein rings and arcs of varying dimensions. The source (S), lensing object (L) and Viewer (V) are denoted by the colours orange, black and blue respectively. The corresponding alignment scenario with their respective astronomical observations allow for a readymade conceptualisation of the mechanics of image formation by gravitational lensing. A comparative study with the panel given in [fig. \[1.1\]](#) should help us understand the several arcs present alongside the actual rings and how the geometrical alignment of the source, lens and observer must have been for those images.

Scenario III presents the most ideal perfectly symmetrical alignment of the source, lens and viewer giving rise to the Einstein ring. The scenarios I & V depict a situation where the source is so far off the central axis connecting the lens and viewer that only light rays from one side can reach the viewer after suffering gravitational lensing. As such there is only one image that can be viewed from that location. Depending on the direction in which the source shifts from the perfect alignment the corresponding image also shifts accordingly. Scenario II & IV presents only a slight shifting of the source from the perfect alignment. In this case there are two images formed. Depending on the direction of the shift of the source the locations of the elongated and diminished images are fixed accordingly. Note that whenever the source is offset from the ideal position the elongated image is formed slightly outside the Einstein radius while the smaller image is formed slightly within. [\[fig. \[1.6\]\]](#)

| Scenario         | I                                                                                     | II                                                                                    | III                                                                                  | IV                                                                                  | V                                                                                   |
|------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Lensing geometry |    |    |    |   |   |
| Observed image   |  |  |  |  |  |

**Table 1.2:** Classification of gravitational lensing phenomena based on observation/detection related characteristics and their corresponding research area. (Inspired from Table 7.1 of [Peter and Uzan \(2013\)](#)). Typical source and lens pair responsible for each lensing regime are also listed accordingly.

| Lensing                         | Source                  | Lens                      | Observations                                    | Pertinence                                                |
|---------------------------------|-------------------------|---------------------------|-------------------------------------------------|-----------------------------------------------------------|
| <b>Strong lensing</b>           | Quasar                  | Galaxy                    | Multiple images, Einstein rings, time delays    | Mass of galaxies, substructures, $H_0$ , $\Omega_\Lambda$ |
|                                 | Galaxy                  | Cluster                   | Giant arcs, multiple images                     | Total mass of the cluster, $\Omega_\Lambda$ , redshifts   |
| <b>Weak lensing</b>             | Galaxy                  | Cluster                   | Arclets, magnification bias                     | Cluster mass profile, cluster morphology                  |
| <b>Statistical weak lensing</b> | Galaxy                  | Galaxy                    | Ellipticity bias, shear-galaxy correlation      | Galaxy parameters, halo characteristics                   |
|                                 | Galaxy                  | Large-scale structure     | Cosmic shear, shear-shear correlation           | Cosmological parameters, power spectra                    |
|                                 | Quasar                  | Large-scale structure     | Cosmic magnification, quasar-galaxy correlation | Cosmological parameters, power spectra, bias              |
|                                 | Last scattering surface | Large-scale structure     | Smoothing of $C_\ell$ , $B$ -modes              | Cosmological parameters, power spectra                    |
| <b>Microlensing</b>             | Star (Galactic)         | Star (Galactic)           | Light curves                                    | Extra-solar planets, Milky Way inner structure            |
|                                 | Star (Extra-galactic)   | Compact object (Galactic) | Light curves                                    | Dark matter in Milky Way halo                             |

### 1.1.2 Gravitational lensing – A cosmologist’s toolkit

A unique feature of gravitational lensing is that it is independent of the nature of matter or its state. As such its effects are not subject to peculiarities of baryonic matter or dark matter or even if they are in equilibrium or not. Consequently gravitational lensing can be employed to measure the mass of lensing object (galaxy, cluster, black holes and even dark matter distributions). As a result of this gravitational lensing offers its services readily to investigations in the field of cosmology [[Blandford and Narayan \(1992\)](#), [Campbell et al. \(2013\)](#)]. Furthermore, there also exists the potential of cosmology feeding back studies in gravitational

lensing as well. An instance of this might be, assuming that  $H_0$  is well known the time delay data can then be used to constrain the mass distribution of the lensing object [Ng (2024)].

Quite a few techniques have been developed over the past few years for investigating cosmological queries with the help of gravitational lensing. Out of these the most significant research methodologies employing gravitational lensing in the field of cosmology are:–

- (a) **Strong lensing cosmography** – Employs strong lensing by clusters of multiple sources. [Lizardo et al. (2021), Caminha et al. (2022), Jana et al. (2024)]
- (b) **Time delay cosmography** – The time delay [section 1.9] between multiple images of strongly lensed, time-variable sources is carefully determined. This allows us to infer a dimensional distance ratio along with absolute distances which in turn facilitates the determination of cosmological parameters like  $H_0$ . [Treu and Marshall (2016), Ng and Lewis (2019), Ng (2023), Birrer et al. (2024)]
- (c) **Weak lensing by large-scale structure of the CMB.** [Amblard et al. (2004), Hirata et al. (2008)]
- (d) **Cosmic shear analysis** – Weak lensing by large-scale structures in the foreground of background galaxies is used to probe the development of structures and the expansion of the universe. [Schneider et al. (2002), Asgari et al. (2021)]
- (e) **Microlensing as a probe of dark matter in the Milky Way halo.** [Evans and Jijina (1994), Holopainen et al. (2006), Hawkins, M. R. S. (2015), Fedorova et al. (2016), Lin et al. (2025)]

Regarding the use of gravitational lensing as a probe in cosmology the actual observable quantities and observation techniques of the lensing geometry merit a separate discussion. For strong lensing, spectroscopy provides information about the kinematics of the lens while high resolution imaging contributes to modelling the lens environment (external convergence or line-of-sight effects). With the advent of advanced instruments in space-based and ground-based telescopes both spectroscopy and high resolution imaging have improved significantly. As such the prospect of these research techniques have improved significantly as well [Treu et al. (2022)]. It is worth mentioning in this context that regarding strong lensing the relative image positions constitute the most important data for the lens system. In certain cases additional constraints may be found from the deformation of an extended source in the form of arcs and rings. The occurrence of an Einstein ring on top of the other discrete images is capable of providing even more robust caveats regarding the lensing system.

As far as the case of weak lensing goes formation of only a single image occurs. Consequently the mild deformation of the source objects is the primary observable in these cases.

A rough classification of observable quantities and their corresponding observation techniques in gravitational lensing phenomena pertaining to cosmology has been provided in table. 1.3.

**Table 1.3:** Broad classification of observable quantities and their corresponding observation techniques in gravitational lensing phenomena pertaining to cosmology

| Observation technique | Observable/Computable quantities                                                                                                                                         |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Spectroscopic         | light spectra or redshifts of the source and lensing objects                                                                                                             |
| Photometric           | apparent brightness and relative flux of individual images                                                                                                               |
|                       | apparent brightness and relative flux as a function of time, <i>i.e.</i> , light curves between images facilitating the determination of time delays of variable sources |
| Imaging               | <i>astrometric observations</i> – relative positions of the lens and images, no. of images                                                                               |
|                       | dimensions of images for extended sources                                                                                                                                |

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## 1.2 EARLY BEGINNINGS

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Like several fundamental physical concepts that are the cornerstones of modern physics, the idea of light rays being influenced by a gravitational field can be traced back to Isaac Newton. Indeed the quote at the very beginning of this chapter was from his 1704 treatise *Opticks*. However, it seems there was a debt owed to time and it was only in 1784 that Henry Cavendish computed the deflection angle of light due to a point mass. The two major assumptions behind his derivation were Newton’s theory of gravitation and Newton’s corpuscular theory of light. John Michell a friend and colleague of Cavendish had a pivotal role to play in this chapter of history as well. Michell in a 1783 correspondence to Cavendish laid down a proposal to determine the mass of stars by measuring the reduction in speed of the light corpuscles traveling from the star to Earth [Michell (1784)]. This letter turned out to be a key motivator for Cavendish’s subsequent work on the deflection of light in a gravitational field. Strangely enough Cavendish was reluctant to publish his calculations and his work gained recognition only after he had passed away. The 1783 letter from Michell reserves its place in the history of science for another reason as well. It was here that he ventured the idea that the universe might host a certain variety of “dark stars” commanding such an enormous gravitational field that not even light can escape from their surface. This is now widely accepted as the very first instance of anyone proposing the concept of a black hole. Pierre-Simon Laplace made independent contributions to the subject over a decade later and ascertained the conditions that would render the escape velocity from the surface of an object greater than the velocity of light [Laplace (2009)].

Laplace’s work, it seems, had a strong influence on Johann Georg von Soldner, which motivated him to compute detailed calculation of the deflection of light under the gravitational influence of a massive object [Jaki (1978), Treder and Jackisch (1981)]. von Soldner’s results showed that for an object of mass  $M$  a ray of

light passing close to the object is deflected by

$$\alpha_{\text{Newton}} = \frac{2GM}{c^2 b} \quad (1.1)$$

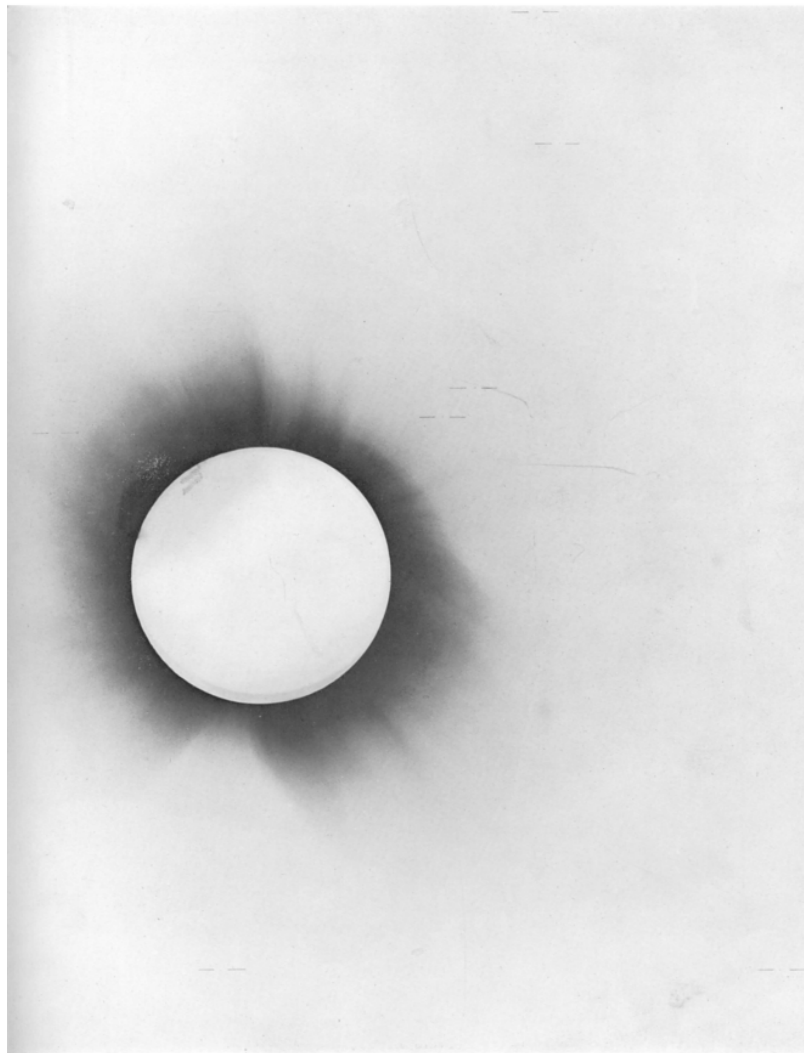
where symbols having their usual meaning. Einstein in 1911 came up with the same results as Soldner employing the equivalence principle of special theory of relativity [Einstein (1911)]. Following this Einstein persuaded his friend, the German astronomer Erwin Freundlich to lead an expedition to the Crimean peninsula with the objective of measuring the deflection of light during a solar eclipse. Unfortunately or (perhaps fortunately?), the onset of WWI prevented Freundlich's team from fulfilling their undertaking. This setback in the meantime allowed Einstein to perfect his *magnum opus*—The General Theory of relativity. Armed with the tools of general relativity he revisited the problem of deflection of light in a gravitational field and found that the bending angle was

$$\alpha_{\text{Einstein}} = \frac{4GM}{c^2 b} \quad (1.2)$$

*i.e.*, twice the value in Newtonian regime [Einstein (1915)]. While the discrepancy in Mercury's orbit was successfully addressed by Einstein's theory, the experimental verification of bending angle of light near a massive object was to be the acid test that would establish general relativity as an accurate physical theory. The stage was set.

A total solar eclipse was due to occur in May, 1919. Frank Dyson was put in charge of organizing a British expedition. The decision was made to send two groups of two observers each. One team traveled to Sobral, Brazil, and included employees from the Royal Observatory in Greenwich. 220 kilometers off the coast of West Africa, the Island of Principe is a small volcanic island that was visited by the second group from the Cambridge Observatory. Andrew Crommelin and Charles Davidson, his less experienced colleague, made up the Greenwich group. Edwin Cottingham, who maintained the Observatory's clocks, and Arthur Eddington, the director of the Observatory, made up the Cambridge group. The joint standing eclipse committee of the Royal Society and Royal Astronomical Society provided support and some funding for the expeditions. It is interesting to note that the previous recorded solar eclipse by Dyson was in 1905 and the few stars that were observed during that event were insufficient for any conclusive evidence either in favour or against general relativity. As far as other attempts at experimental verification are concerned the attempt by a group of astronomers at Lick observatory during the total eclipse of June 1918 had also been unsuccessful.

Eddington's experiment [fig. \[1.3\]](#) not only cemented Einstein's theory as the most beautiful and precise theory of gravitation but it also set a precedent of international collaboration amongst scientists rising above political animosity [Dyson et al. (1920), Will (2014)]. After all, the times were that of WWII and it was a British astronomer toiling after the predictions of a German theoretician. Needless to say, the successful union of elegant mathematical structure with clinical experimental validation at the solar eclipse of May 1919 ensured the passing over of the baton from Newton to Einstein. Ironically, it took an eclipse for physics to come out of the shadow of an intellectual giant by the name of Newton.



(a)

\*\*\*  
**LIGHTS ALL ASKEW  
 IN THE HEAVENS**

**Men of Science More or Less  
 Agog Over Results of Eclipse  
 Observations.**

**EINSTEIN THEORY TRIUMPHS**

**Stars Not Where They Seemed  
 or Were Calculated to be,  
 but Nobody Need Worry.**

**A BOOK FOR 12 WISE MEN**

**No More in All the World Could  
 Comprehend It, Said Einstein When  
 His Daring Publishers Accepted It.**

Special Cable to THE NEW YORK TIMES.  
**LONDON, Nov. 9.**—Efforts made to  
 put in words intelligible to the non-  
 scientific public the Einstein theory of  
 light proved by the eclipse expedition  
 so far have not been very successful. The  
 new theory was discussed at a recent  
 meeting of the Royal Society and Royal  
 Astronomical Society, Sir Joseph Thom-  
 son, President of the Royal Society, de-  
 clares it is not possible to put Einstein's  
 theory into really intelligible words, yet  
 at the same time Thomson adds:

(b)

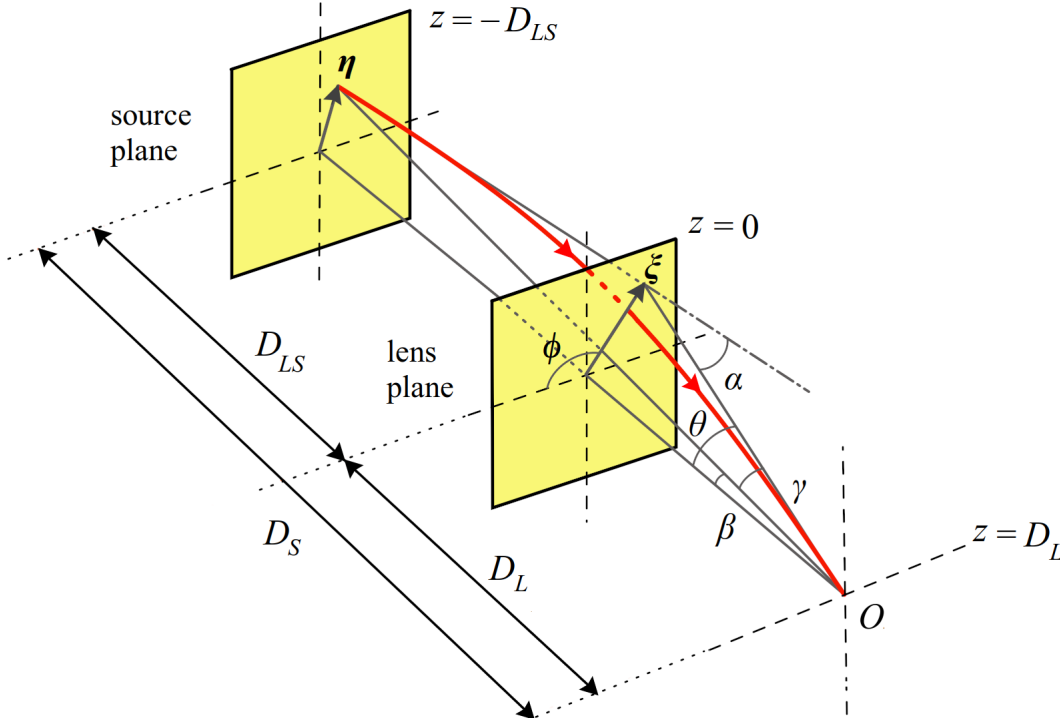
**Figure 1.3: (a)** Photographic plate taken from Dyson et al. (1920). Original caption: From the report of Sir Arthur Eddington<sup>1</sup> on the expedition to verify Albert Einstein's prediction of the bending of light around the sun. In Plate 1 is given a half-tone reproduction of one of the negatives taken with the 4-inch lens at Sobral. This shows the position of the stars, and, as far as possible in a reproduction of this kind, the character of the images, as there has been no retouching. A number of photographic prints have been made and applications for these from astronomers, who wish to assure themselves of the quality of the photographs, will be considered as far as possible acceded to.

**(b)** Excerpt from the issue of The New York Times published on Monday, November 10, 1919 declaring the monumental triumph of Einstein's theory.

<sup>1</sup>Despite being most closely linked to this eclipse and its accomplishment of verifying Einstein's prediction, Eddington's findings would have been mostly useless in the absence of Crommelin's party's observations from Sobral.

### 1.3 BENDING ANGLE OF LIGHT IN NEWTONIAN MECHANICS

While we now know that as far as the bending of light is concerned Newton's theory is off by a factor of 2 it is still instructive to follow the actual derivation in Newtonian mechanics. Furthermore, the derivation with a few tweaks is also capable of yielding the results compatible with general relativity [Congdon and Keeton (2018), Meneghetti (2022)].



**Figure 1.4:** Illustration depicting the bending of light under the influence of gravitational field. The actual path of the light ray is traced in red. The lensing object is an object of mass  $M$  at  $z = 0$ . The deflection angle is given by the angle  $\alpha = (\vec{\xi}, \phi)$  where  $\vec{\xi}$  denotes the distance of closest approach for the light ray. The other symbols depicted here will come in useful while deriving the lens equation and other essential quantities. But for the time being we will concentrate on the variables necessary for deriving the expression for deflection angle. **Note:** The mass  $M$  may be a point mass as well as an extended distribution of mass in which case  $\vec{\xi}$  denotes the radius of the extended mass.

As depicted in fig. [1.4] the lens is located at  $z = 0$  and with respect to it the observer and the source are located at  $z = D_L$  and  $z = -D_{LS}$  respectively (the negative sign indicates distance being measured in the opposite sense from that of the observer). The distance between the source and the observer is  $D_S$ . In this particular scenario we have chosen the cylindrical coordinates  $(\xi, \phi, z)$  where the  $z$ -axis is the central axis passing through the different planes shown in fig. [1.4]. Let us now assume that the velocity of light

ray, traveling initially parallel to the  $z$ -axis is given as  $c\hat{e}_z \equiv \vec{v}_{st}$  where  $\hat{e}_z$  is the unit vector along the  $z$ -axis and  $c$  of course is the speed of light. It is important to note in this context that  $\vec{\xi}$  is the coordinate along the axis perpendicular to  $z$ -axis. Given that we are working with the underlying framework of corpuscular theory and Newtonian gravitation, we can stipulate that the lens induces an acceleration of  $\vec{a} = -\vec{\nabla}\Phi(\vec{\xi})$ .  $\Phi$  represents the gravitational potential at location  $\vec{\xi}$ . Therefore, the corresponding change in velocity of the light corpuscle along its trajectory from the source to the observer is,

$$\Delta\vec{v} = \frac{1}{c} \int_{-D_{LS}}^{D_L} \vec{a} dz = \frac{1}{c} \int_{-D_{LS}}^{D_L} \vec{\nabla}\Phi dz \quad (1.3)$$

Note, that we have simply swapped out the integration w.r.t time coordinate  $t$  by using the relation  $z = ct$ . Furthermore, a crucial point to notice is that the limits of integration in eq. (1.3) actually mean that the integration is not performed along the light trajectory but along the straight line connecting the source to the observer. However, this issue regarding the approximate nature of the integral can be circumvented by considering the first order Born approximation [Strutt (2009)]. The validity and accuracy of the Born approximation for a single lens plane is dealt with rigorously in Yarimoto and Oguri (2025). This essentially means that the light trajectory in the absence of gravitational field can be taken as zeroth-order path from which we can determine the first order correction which can be fed into a second-order correction in turn.

The gradient of the gravitational potential can be decomposed into the components along  $z$  and  $\xi$  as,

$$\vec{\nabla}\Phi = \frac{d\Phi}{dz}\hat{e}_z + \frac{d\Phi}{d\xi}\hat{e}_\xi = \vec{\nabla}_\parallel\Phi\hat{e}_z + \vec{\nabla}_\perp\Phi\hat{e}_\xi \quad (1.4)$$

Similarly, the change in velocity of the light corpuscle will have two components along  $\hat{e}_z$  and  $\hat{e}_\xi$  as well.

$$\Delta\vec{v} = \Delta v_\parallel\hat{e}_z + \Delta v_\perp\hat{e}_\xi \quad (1.5)$$

For the perpendicular component of change in velocity eq. (1.3) yields,

$$\Delta v_\perp = \frac{1}{c} \int_{-D_{LS}}^{D_L} \frac{d\Phi(\xi)}{d\xi} dz \quad (1.6)$$

As for the parallel component (*i.e.*, along the line of sight) the amount of change in velocity is given by

$$\begin{aligned} \Delta v_\parallel &= \frac{1}{c} \int_{-D_{LS}}^{D_L} \frac{d\Phi(\xi)}{dz} dz \\ &= \frac{1}{c} [\Phi(\xi, D_L) - \Phi(\xi, -D_{LS})] \end{aligned} \quad (1.7)$$

Now, we may consider that the distance between the lens and source as well as the distance between lens and observer are both extremely large. As such  $|-D_{LS}|$  &  $|D_L|$  may be considered to be  $\infty$ . Consequently as  $\lim_{|z| \rightarrow \infty} \Phi = 0$ , which yields  $\Delta v_\parallel = 0$ . This means that no additional velocity is acquired by the light corpuscle along the direction of motion *i.e.*, the velocity along the  $z$ -axis remains as  $c$ .

Therefore, the velocity of light after getting deviated due to the gravitational field of the lens will be given by,

$$\vec{v}_{\text{dev}} = \vec{v}_{\text{st}} + \Delta \vec{v} = \Delta v_{\perp} \hat{e}_{\xi} + c \hat{e}_z \quad (1.8)$$

where  $\vec{v}_{\text{st}}$  represents the initial velocity of light. The potential  $\Phi$  has the well known form  $-\frac{GM}{\sqrt{\xi^2 + z^2}}$ . Substituting, this in [eq. \(1.6\)](#) with appropriate limits we get

$$\Delta v_{\perp} = -\frac{GM\xi}{c} \int_{-\infty}^{\infty} (\xi^2 + z^2)^{-\frac{3}{2}} dz \quad (1.9)$$

We can now easily solve this integral by means of a variable change  $\frac{z}{\xi} = \tan \theta$ .

$$\Delta v_{\perp} = -\frac{GM}{c\xi} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos \theta d\theta = -\frac{2GM}{c\xi} \quad (1.10)$$

This when substituted in [eq. \(1.8\)](#) yields

$$\vec{v}_{\text{dev}} = c \hat{e}_z - \frac{2GM}{c\xi} \quad (1.11)$$

Therefore, the angle of deflection is given by,

$$\alpha_{\text{N}} \left( \vec{\xi} \right) = \hat{e}_{\text{st}} - \hat{e}_{\text{dev}} = \frac{2GM}{c^2\xi} \hat{e}_{\xi} \quad (1.12)$$

Here, we have used the magnitude of  $|v_{\text{dev}}|$  as

$$|v_{\text{dev}}| = \sqrt{c^2 + \frac{4G^2M^2}{c^2\xi^2}} \approx c \quad (1.13)$$

This is precisely half the value of the result obtained from general relativistic calculations [[Misner, Thorne, and Wheeler \(1973\)](#), [Carroll \(2003\)](#)].

$$\alpha_{\text{E}} \left( \vec{\xi} \right) = \hat{e}_{\text{st}} - \hat{e}_{\text{dev}} = \frac{4GM}{c^2\xi} \hat{e}_{\xi} \quad (1.14)$$

If we substitute the value of mass as solar mass and the radius as solar radius in [eq. \(1.12\)](#), then we have a scenario where the light ray gets deflected while grazing the surface of the Sun.

**For Newtonian case—**

$$\alpha_{\text{N}\odot} = \frac{2G \times 1.989 \times 10^{30}}{c^2 \times 6.96 \times 10^8} \approx 0.875'' \quad (1.15)$$

**For Einsteinian case—**

$$\alpha_{\text{E}\odot} = \frac{4G \times 1.989 \times 10^{30}}{c^2 \times 6.96 \times 10^8} \approx 1.75'' \quad (1.16)$$

Another interesting case to consider is the case for an extended mass distribution. In particular the introduction of terms such as surface mass density etc will prove extremely essential in defining some of the key concepts related to gravitational lensing.

For any arbitrary 2-dimensional surface mass distribution, the mass term in the expression for deflection angle of light can be expressed as,  $dM = \Sigma(\vec{\xi}) d^2\xi$ . The term  $\Sigma(\vec{\xi})$  denotes the surface mass density enclosed in area  $d\xi$ . Introducing the correction due to general relativity the deflection angle for this case is given by

$$\alpha(\vec{\xi}) = \frac{4G}{c^2} \int_{\mathcal{R}^2} d^2\xi' \Sigma(\vec{\xi}') \frac{\vec{\xi} - \vec{\xi}'}{|\vec{\xi} - \vec{\xi}'|^2} \quad (1.17)$$

where the limits of integration are so chosen as to encompass the entire mass distribution in the lens plane. The [eq. \(1.15\)](#) is valid so long as the following two conditions are satisfied,

- (a) The gravitational fields are weak, *i.e.*, the deflection angle is small.
- (b) The stationary matter distribution of the lensing object should be significantly smaller than  $c$ .

In general for almost every conceivable astrophysical scenario these conditions are applicable.

The following sections will essentially provide derivations of some of the key concepts/reasons regarding — lens geometry, multiple imaging, magnification ratio and time delay.

---

## 1.4 THE LENS EQUATION

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In order to derive the lens equation let us first define some of the essential elements of lensing geometry.

$D_L$  = distance from the observer to the lens

$D_S$  = distance from the observer to the light source

$D_{LS}$  = distance from the lens to the source

$\vec{\beta}$  = actual angle between the lens and the source

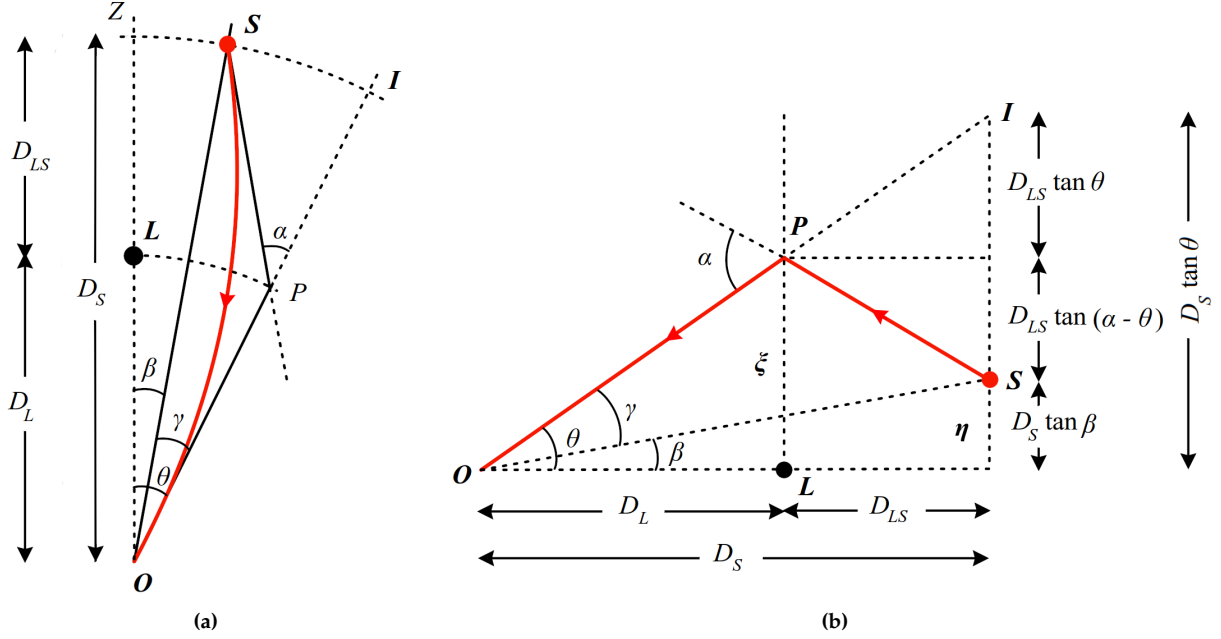
$\vec{\theta}$  = observed angle between the lens and the source *i.e.*, angle between the observer and the image

$\vec{\xi}$  = distance from the lens to photon trajectory at the lens plane

$\vec{\alpha}$  = the angle of deflection

Here,  $\vec{\beta}$ ,  $\vec{\theta}$ ,  $\vec{\alpha}$ , and  $\vec{\xi}$  are all vector quantities and our derivations should include both negative as well as positive angles. In case of simple point source lens objects (needless to say an approximation) the vectorial aspects of the angles don't play any significant role. However, for complex lensing systems like galaxies or clusters vectorial treatment becomes necessary.

We can invoke the small angle approximation as  $\beta$ ,  $\theta$  and  $\alpha$  are all very small. Furthermore, since  $D_S$ ,  $D_{LS}$ , and  $D_L$  are all considerably larger than  $\xi$  this approximation holds good [[fig. \[1.5\]](#)].



**Figure 1.5: (a)** Illustration depicting the projection of the lensing geometry onto a 2-D spatial slice. Here we consider the spacetime to be Friedmann-Lemaitre-Robertson-Walker (FLRW) and the coordinates to be comoving. As such we are at liberty to project the spacetime diagram onto a spatial slice. The source  $S$  emits a photon which follows the path (spatial projection) indicated by the red solid line and reaches the observer  $O$ . The lens is marked by the black circle at  $L$ . The source is at an angular position  $\beta$  w.r.t the observer whereas the image  $I$  is at an angular position  $\theta$ . As we consider the flat sky approximation the spheres at the source and lens can be replaced by their tangent planes, i.e., the arcs  $LP$  and  $SI$  can be replaced by their tangents at  $L$  and  $S$  respectively.

**(b)** For thin lens approximation the actual spatial path is replaced by the piecewise spatial path given by  $SPO$ . The magnitude of the deflection angle is given by the angle between  $SP$  and  $PI$ .

With reference to [fig. \[1.5\]](#) we can write the geometric relation

$$D_S \tan \beta = D_S \tan \theta - D_{LS} \tan \alpha \quad (1.18)$$

As a consequence of the small angle approximation we have ( $\tan \theta \approx \theta$ ). Therefore, we can write [eq. \(1.18\)](#) as

$$\begin{aligned} D_S \beta &= D_S \theta - D_{LS} \alpha \\ \Rightarrow \beta &= \theta - \frac{D_{LS}}{D_S} \alpha \end{aligned} \quad (1.19)$$

In vector notation this reads as

$$\vec{\beta} = \vec{\theta} - \frac{D_{LS}}{D_S} \alpha \left( \vec{\xi} \right) \quad (1.20)$$

We can define a scaled deflection angle  $\alpha \left( \vec{\theta} \right) = \frac{D_{LS}}{D_S} \alpha \left( \vec{\xi} \right)$  and express the lens equation given by

eq. (1.20) as

$$\vec{\beta} = \vec{\theta} - \alpha(\vec{\theta}) \quad (1.21)$$

Similarly, if we want to express the lens equation in terms of displacement vectors  $\vec{\xi}$  and  $\vec{\eta}$  then we have

$$\vec{\eta} = \frac{D_S}{D_L} \vec{\xi} - D_{LS} \alpha(\vec{\xi}) \quad (1.22)$$

In eq. (1.21) the dependent variable has been changed from  $\xi$  to  $\theta$ . The physical interpretation of the lens equation here is that an image of a source at true position  $\vec{\beta}$  will be observed at angular positions  $\vec{\theta}$  so long as eq. (1.21) is satisfied. Multiple images are produced for cases where the lens equation has multiple solutions.

A common practice in gravitational lensing is to write the lens equation in its dimensional form by defining a length scale  $\xi_0$  on the lens plane and a corresponding length scale  $\eta_0 = \frac{\xi_0 D_S}{D_L}$  on the source plane and using them to define the dimensional (scaled) vectors

$$\vec{x} \equiv \frac{\vec{\xi}}{\xi_0}, \quad \vec{y} \equiv \frac{\vec{\eta}}{\eta_0} \quad (1.23)$$

Hence, the dimensionless lens equation is,

$$\vec{y} = \vec{x} - \alpha(\vec{x}) \quad (1.24)$$

where

$$\alpha(\vec{x}) = \frac{1}{\pi} \int_{\mathcal{R}^2} d^2 x' \kappa(\vec{x}') \frac{\vec{x} - \vec{x}'}{|\vec{x} - \vec{x}'|^2} = \frac{D_L D_{LS}}{\xi_0 D_S} \alpha(\xi_0 \vec{x}) \quad (1.25)$$

is the scaled deflection angle, and

$$\kappa(\vec{x}) = \frac{\Sigma(\xi_0 \vec{x})}{\Sigma_{cr}} \quad (1.26)$$

is the dimensionless surface mass density also known as the convergence. The term  $\Sigma_{cr}$  is the critical surface mass density expressed as

$$\Sigma_{cr} = \frac{c^2 D_S}{4\pi G D_L D_{LS}} \quad (1.27)$$

A key feature in terms of physical significance of critical surface mass density  $\Sigma_{cr}$  is that it yields the minimum value of  $\Sigma$  required to produce multiple images of the background source. Furthermore,  $\Sigma_{cr}$  can be used to setup a quantitative discriminating criteria for strong lensing ( $\Sigma \geq \Sigma_{cr}$ ) and weak lensing ( $\Sigma < \Sigma_{cr}$ ). This criterion can be easily qualified when we consider that strong lensing is generally observed in regions of high mass concentration, such as galaxy cluster cores or massive galaxies and as such the surface mass density usually surpasses the critical value. Similarly, for weak lensing where the observed effects are generally in the form of distortions and magnifications the surface mass density of the lensing structure is well within the critical value.

## 1.5 LENSING POTENTIAL AND LENSING DISTANCE

In case of an extended distribution of matter the deflection tendencies can be defined by its effective lensing potential.

$$\alpha(\vec{x}) = \nabla\psi(\vec{x}) \quad (1.28)$$

where

$$\psi(\vec{x}) = \frac{1}{\pi} \int_{\mathcal{R}^2} d^2x' \kappa(\vec{x}') \ln|\vec{x} - \vec{x}'| \quad (1.29)$$

The mapping of  $\vec{x} \rightarrow \vec{y}$  is a gradient mapping,

$$\vec{y} = \nabla \left( \frac{1}{2} \vec{x}^2 - \psi(\vec{x}) \right) \quad (1.30)$$

or,

$$\nabla\phi(\vec{x}, \vec{y}) = 0 \quad (1.31)$$

where,

$$\phi(\vec{x}, \vec{y}) = \frac{1}{2} (\vec{x} - \vec{y})^2 - \psi(\vec{x}) \quad (1.32)$$

is called the Fermat potential.

The Laplacian of the lensing potential, with the help of the identity  $\nabla^2 \ln|\vec{x} - \vec{x}'| = 2\pi\delta^2(\vec{x} - \vec{x}')$  is

$$\nabla^2\psi = 2\kappa \quad (1.33)$$

*i.e.*, twice the convergence.

Lensing potential, deflection angle, critical surface density all strongly depend on distances between the trinity of source, lens and observer. This in turn depends on the source and lens redshifts. The ratio of the distances as given below

$$\frac{D_L D_{LS}}{D_S} \quad (1.34)$$

is called lensing distance. The lensing distance reaches its maximum when the lens is placed between the observer and the source at an intermediate distance, and it increases with the source's redshift. It is evident that the effects produced by the lensing event increase with the lensing distance (and consequently the convergence).

It is instructive to notice the schematic panel given in [table. 1.1](#). As the relative position of the source keeps changing the corresponding nature and positioning of images also keeps changing. The following section will firmly establish the mathematical relation defining the exact image positioning based on the lens equation.

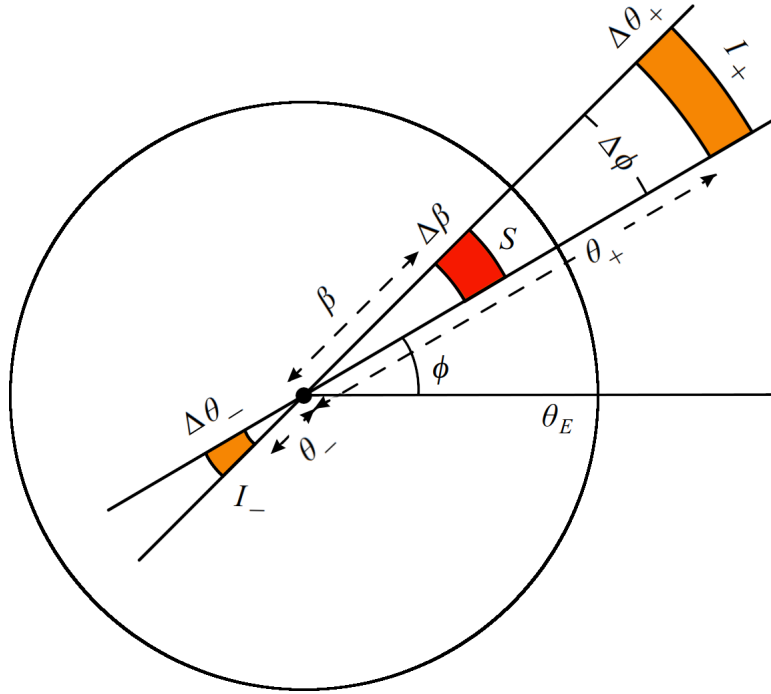
## 1.6 IMAGE LOCATION AND MAGNIFICATION FACTOR

Let us revisit the lens equation given by [eq. \(1.20\)](#), and ignore the vectorial notation for the present. Given the geometry depicted in [fig. \[1.5\]](#) we can assume from the geometry of small angles  $\theta = \frac{\xi}{D_L}$ . Also we know that  $\alpha = \frac{4GM}{\xi c^2}$ . Using this we can write [eq. \(1.20\)](#) as,

$$\beta = \theta - \left( \frac{4GM}{c^2} \frac{D_{LS}}{D_L D_S} \right) \frac{1}{\theta} \quad (1.35)$$

Based on this we can define a term called the characteristic bending angle,  $\alpha_0$  which depends only on the mass of the lens and the distances involved (discussed in [section 1.5](#)).

$$\alpha_0 = \left( \frac{4GM}{c^2} \frac{D_{LS}}{D_L D_S} \right)^{\frac{1}{2}} \quad (1.36)$$



**Figure 1.6:** Depiction of multiple images and magnification due to gravitational lensing. The source is given by the red patch whereas the images are depicted in orange.  $I_+$  represents the magnified image and  $I_-$  represents the diminished image. The magnified image lies outside the Einstein radius ( $\theta_E$ ) whereas the diminished image lies within the Einstein radius. The Einstein ring occurs for  $\beta = 0$  i.e., when the source lies exactly behind the lens.

This allows us to write eq. (1.35) as,

$$\beta = \theta - \frac{\alpha_0^2}{\theta} \quad (1.37)$$

For a point source lens  $\beta$  and  $\theta$  are co-planar and we can transform eq. (1.37) as

$$\begin{aligned} \theta^2 - \beta\theta - \alpha_0^2 &= 0 \\ \Rightarrow \theta &= \frac{1}{2} \left( \beta \pm \sqrt{4\alpha_0^2 + \beta^2} \right) \end{aligned} \quad (1.38)$$

This gives us the location of the images  $(\theta_+, \theta_-)$  formed due to gravitational lensing as a function of the true position of the source with respect to the lens [fig. [1.6]].

It is readily observable from eq. (1.38) which is quadratic in nature that one of the images will be magnified whereas another will be diminished. Let us consider a polar coordinate system with the lens at the centre. At the location of the lens the area element is  $dA = \beta \Delta\phi \Delta\beta$  which to the observer appears to be  $dA' = \theta \Delta\phi \Delta\theta$  as the angles get distorted due to the lens. The resulting magnification is therefore given by,

$$\mu = \frac{\theta \Delta\phi \Delta\theta}{\beta \Delta\phi \Delta\beta} = \frac{\theta \Delta\theta}{\beta \Delta\beta} = \frac{\theta}{\beta} \frac{\partial\theta}{\partial\beta} \quad (1.39)$$

where the derivative is of course,

$$\frac{\partial\theta}{\partial\beta} = \frac{1}{2} \left( 1 \pm \frac{\beta}{\sqrt{4\alpha_0^2 + \beta^2}} \right) \quad (1.40)$$

Substituting this in the expression for magnification we have,

$$\mu_{\pm} = \frac{1}{4} \left[ \frac{\beta}{\sqrt{4\alpha_0^2 + \beta^2}} + \frac{\sqrt{4\alpha_0^2 + \beta^2}}{\beta} \pm 2 \right] \quad (1.41)$$

As  $\beta \rightarrow \infty$ ,  $\mu_+ \rightarrow 1$  and  $\mu_- \rightarrow 0$ . This is clearly observed in the plot provided in fig. [1.7]. At the other end of the limit spectrum as  $\beta \rightarrow 0$  the magnification factors  $\mu_+$ ,  $\mu_- \rightarrow \infty$ . As such when  $\beta < \alpha_0$  the magnification due to gravitational lens becomes important.

Another important factor that should be considered is the brightness ratio of the two images,

$$\frac{\mu_+}{\mu_-} = \left[ \frac{(4\alpha_0^2 + \beta^2)^{\frac{1}{2}} + \beta}{(4\alpha_0^2 + \beta^2)^{\frac{1}{2}} - \beta} \right]^2 \quad (1.42)$$

which can be simplified to

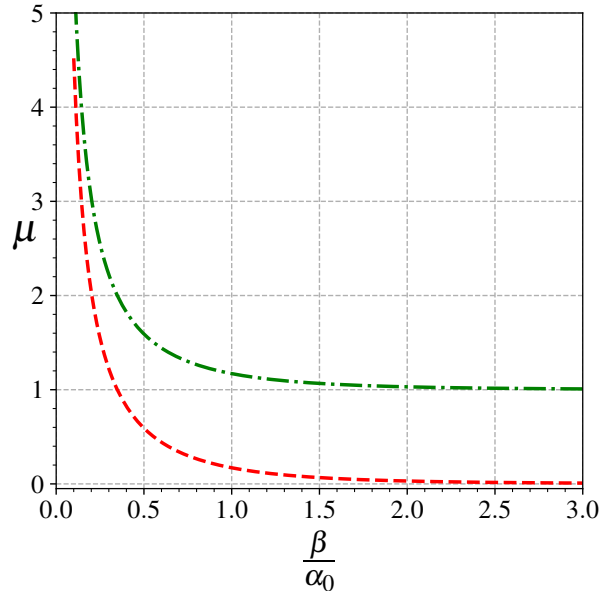
$$\frac{\mu_+}{\mu_-} = \left( \frac{\theta_+}{\theta_-} \right)^2 \quad (1.43)$$

Two other important relations are,

$$\theta_+ \theta_- = -\alpha_0^2 \quad (1.44)$$

and

$$\beta = \theta_+ - \theta_- \quad (1.45)$$



**Figure 1.7:** Plot depicting the variation of two different magnifications  $\mu_+$  and  $\mu_-$  vs  $\frac{\beta}{\alpha_0}$ . The green dash-dot line indicates  $\mu_+$  whereas the red dashed line indicates  $\mu_-$ .

## 1.7 LENS MAPPING AND DISTORTION

The actual shape of the images formed by gravitational lensing may be determined by solving the lens equation for all points within the extended source. It has already been established that deflection of the light ray is dictated by the lens properties which depend on the angular scales. For cases where the source is much smaller than the angular scales the lens mapping (*i.e.*, the correspondence between the image and the source positions) can be locally linearised. Consequently the distortion of the images can be described by the Jacobian matrix.

$$\mathcal{A}(\theta) = \frac{\partial \beta}{\partial \theta} = \left( \delta_{ij} - \frac{\partial \alpha_i(\theta)}{\partial \theta_j} \right) = \left( \delta_{ij} - \frac{\partial^2 \psi(\theta)}{\partial \theta_i \partial \theta_j} \right) \quad (1.46)$$

Note, that in terms involving  $\psi$  the variable is switched to  $\theta$  but this is not contrarian to the treatment presented earlier in [section 1.5](#) where the variable was  $x$ . The metric  $\mathcal{A}$  is a rank two symmetric tensor and is termed as the *lensing Jacobian*.  $\mathcal{A}$  can be divided into an isotropic and an anisotropic part,

$$\begin{aligned} \mathcal{A}^{\text{iso}} &= \frac{1}{2} \text{Tr} \mathcal{A}, I = \left[ 1 - \frac{1}{2} (\psi_{11} + \psi_{22}) \right] \delta_{ij} \\ &= \left( 1 - \frac{1}{2} \nabla^2 \psi \right) \delta_{ij} \\ &= (1 - \kappa) \delta_{ij} \end{aligned} \quad \begin{array}{l} \text{[where we have used eq. (1.33)]} \\ \end{array} \quad (1.47)$$

$$\begin{aligned}
 \mathcal{A}^{\text{aniso}} &= \mathcal{A} - \frac{1}{2} \text{Tr} \mathcal{A}, I \\
 &= \delta_{ij} - \psi_{ij} - \frac{1}{2} (1 - \psi_{11} + 1 - \psi_{22}) \delta_{ij} \\
 &= -\psi_{ij} + \frac{1}{2} (\psi_{11} + \psi_{22}) \delta_{ij} \\
 &= \begin{pmatrix} -\frac{1}{2} (\psi_{11} - \psi_{22}) & -\psi_{12} \\ -\psi_{12} & \frac{1}{2} (\psi_{11} - \psi_{22}) \end{pmatrix}
 \end{aligned} \tag{1.48}$$

The matrix described in eq. (1.48) is antisymmetric in nature and is called the *shear matrix*. The reasoning behind this nomenclature lies in the fact that it describes the distortion due to gravitational lensing, where the subscripts on  $\psi$  correspond to partial derivatives with respect to the vector components,  $(\theta_1, \theta_2)$  respectively. The *shear* can be described as a vector in the lens plane  $\vec{\gamma} = (\gamma_1, \gamma_2)$  [in complex notation –  $\gamma = \gamma_1 + i\gamma_2$ ] where

$$\gamma_1(\theta) = \frac{1}{2} (\psi_{11} - \psi_{22}) \tag{1.49}$$

$$\gamma_2(\theta) = \psi_{12} = \psi_{21} \tag{1.50}$$

The eigenvalues of the shear matrix are given by

$$\pm\gamma = \pm\sqrt{\gamma_1^2 + \gamma_2^2} \tag{1.51}$$

This implies that there exists an angle  $\rho$  for which the following relation holds good

$$\begin{pmatrix} \gamma_1 & \gamma_2 \\ \gamma_2 & -\gamma_1 \end{pmatrix} = \gamma \begin{pmatrix} \cos 2\rho & \sin 2\rho \\ \sin 2\rho & \cos 2\rho \end{pmatrix} \tag{1.52}$$

The angle  $\rho$  represents the direction of the eigenvector corresponding to the positive eigenvalue.

We can finally express the *lensing Jacobian*  $\mathcal{A}$  as

$$\begin{aligned}
 \mathcal{A} &= \mathcal{A}^{\text{iso}} + \mathcal{A}^{\text{aniso}} \\
 &= (1 - \kappa) \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \gamma \begin{pmatrix} \cos 2\rho & \sin 2\rho \\ \sin 2\rho & \cos 2\rho \end{pmatrix} \\
 &= \begin{pmatrix} 1 - \kappa - \gamma_1 & -\gamma_2 \\ -\gamma_2 & 1 - \kappa + \gamma_1 \end{pmatrix}
 \end{aligned} \tag{1.53}$$

This is a critical equation in gravitational lensing as it helps determine the distortion of the lensed images. Let us reiterate that two of the contributing ingredients in eq. (1.53) are  $\kappa$  (*the convergence*) and  $\gamma$  (*the shear*). The physical interpretation of eq. (1.53) is a culmination of *isotropic* and *anisotropic* source image transformations where

(a) any deformation in the image that results from convergence  $\kappa$ , stretches the lensed image uniformly in all directions and does not affect the actual shape.

(b) any deformation of the image affected by shear  $\gamma$  stretches the source along a specific direction  $\rho$

In other words the image size in comparison to the source is increased in the direction of the eigenvectors of  $\mathcal{A}$  with eigenvalue  $\gamma$ , and decreased in the perpendicular direction. Quantitatively speaking the magnitude of the magnifications and de-magnifications are given by the inverse of the tangential and radial eigenvalues, namely

$$\lambda_t = 1 - \kappa - \gamma; \quad \lambda_r = 1 - \kappa + \gamma \quad (1.54)$$

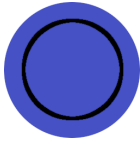
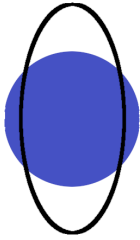
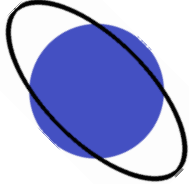
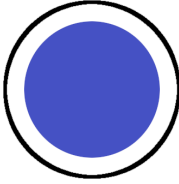
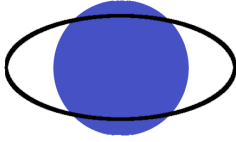
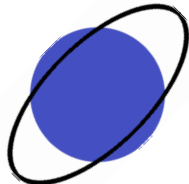
Therefore, for a circular source of radius  $r$  the convergence term affects the lensed image by mapping it onto a larger (or smaller) circle of radius

$$r_\kappa^I = \frac{r}{1 - \kappa} \quad (1.55)$$

On the other hand the shear term makes the circle elongated in the shape of an ellipse whose major and minor axes are

$$a = \frac{r}{1 - \kappa - \gamma}, \quad b = \frac{r}{1 - \kappa + \gamma} \quad (1.56)$$

**Table 1.4:** Effect of convergence and shear on source image as a consequence of gravitational lensing. The original source is depicted by the blue circle and the corresponding distorted/magnified/de-magnified image is depicted by the black outlined shape.

| Dependence on  | Convergence                                                                         | Re(Shear)                                                                           | Im(Shear)                                                                             |
|----------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Condition      | $\kappa < 0$                                                                        | $\gamma_1 < 0, \gamma_2 = 0$                                                        | $\gamma_2 < 0, \gamma_1 = 0$                                                          |
| Image geometry |  |  |  |
| Condition      | $\kappa > 0$                                                                        | $\gamma_1 > 0, \gamma_2 = 0$                                                        | $\gamma_2 > 0, \gamma_1 = 0$                                                          |
| Image geometry |  |  |  |

Of course [table. 1.4](#) is based on the simplistic assumption that the source is perfectly circular. However, in reality most sources are not circular but tend to have some ellipticity. Furthermore, sources like galaxies etc are not perfectly elliptical either. This merits a more robust definition of ellipticity than the ones derived in this section. A detailed and more complete discussion on the topic may be found in [Dodelson \(2017\)](#).

It is interesting to note in this context that the magnification tensor  $\mu$  is related to the lensing Jacobian as  $\mu = \mathcal{A}^{-1}$  *i.e.*, the inverse of the determinant of the Jacobian matrix gives the magnification. The same applies for tangential and radial components of magnification.

$$\mu \equiv \frac{1}{|\mathcal{A}|} = \frac{1}{(1 - \kappa)^2 - \gamma^2} \quad (1.57a)$$

$$\mu_t = \frac{1}{\lambda_t} = \frac{1}{1 - \kappa - \gamma} \quad (1.57b)$$

$$\mu_r = \frac{1}{\lambda_r} = \frac{1}{1 - \kappa + \gamma} \quad (1.57c)$$

It is worth noting in this context that the magnification ( $\mu_i$ ) of an image cannot be measured directly. Although the relative magnification  $\frac{\mu_i}{\mu_j}$  between two images  $i, j$  can be measured when they are resolved.

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## 1.8 CRITICAL CURVES AND CAUSTICS

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*Critical curves* are those that satisfy  $|\mathcal{A}| = 0$  on the lens plane, where the Jacobian determinant vanishes. Regions in the lens plane where the Jacobian determinant has the opposite sign are separated by these curves. The parity of images is shown by the sign of the Jacobian determinant. Images with negative parity (or negative Jacobian determinant) have inverted orientation, while images with positive parity (or positive Jacobian determinant) are said to have the same orientation as the unperturbed image. *Caustics* are critical curves mapped to the source plane using the lens equation. In a lensing system, the caustics and source position have a direct impact on the number of images. The number of images changes with the source position for a specific observer and lens position. Whenever the source crosses a *caustic* the number of images varies by two. [[Petters, Levine, and Wambsganss \(2001\)](#), [Congdon and Keeton \(2018\)](#)]

For instances where *caustics* are curves, the smooth arcs are called *folds* (having topology of a line). *Cusps* on the other hand are the singular points where two abutting fold arcs have coincident tangents with the folds arcs on opposite sides of the double tangent [[Narayan and Wallington \(1992\)](#)]. Sources on caustics are infinitely magnified. On the other hand, actual astrophysical scenarios do not involve infinite magnification. Initially, every source has a finite size and a finite magnification, as determined by the surface brightness-weighted point-source magnification across its solid angle. Another important point is that, geometrical optics approximations fail for point sources near critical curves, and wave optics descriptions should be used. We then observe that even point sources are magnified by a finite amount. [[Ohanian \(1983\)](#), [Wang and Turner \(1997\)](#), [Bozza, Pietroni, and Melchiorre \(2020\)](#), [Berry \(2021\)](#)]

## 1.9 TIME DELAY

When two or more images are formed in a gravitational lensing system, there tends to be different travel times along different photon trajectories [Cooke and Kantowski (1975), Kayser and Refsdal (1983)]. This happens primarily due to—

- (a) **geometrical time delay** – light rays take different amounts of time to reach the observer from the source because the path lengths are actually different.
- (b) **Shapiro (or potential) time delay** – this delay occurs as a consequence of the influence of gravitational field potential of the lens on the light ray [Shapiro (1964)].

The cumulative difference between the arrival times of two images is defined as their corresponding time delay.

In comparison to the undeviated light path the actual light path takes an extra amount of time given by the expression

$$\begin{aligned} \mathcal{T}(\vec{x}, \vec{y}) &= \frac{\xi_0^2}{c} \frac{D_S}{D_L D_{LS}} \phi(\vec{x}, \vec{y}) (1 + z_L) \\ &= \frac{\xi_0^2}{c} \frac{D_S}{D_L D_{LS}} \left[ \frac{(\vec{x} - \vec{y})^2}{2} - \psi(\vec{x}) \right] (1 + z_L) \end{aligned} \quad (1.58)$$

The term  $z_L$  refers to the redshift of the lens. Although  $\mathcal{T}(\vec{x}, \vec{y})$  cannot be measured the relative time delay between two images  $\Delta\mathcal{T} = \mathcal{T}_1 - \mathcal{T}_2$  can be determined. It is worth mentioning it in this context that the time delay is the only dimensional observable for a gravitational lensing system. As a result of this the time delay provides the overall length scale involved for the system.

### 1.9.1 Fermat's principle – A general relativistic adaptation

In the classical context the statement of Fermat's principle means that light always follows the quickest path between two points. Now, in general relativity the path of light curves are represented by null geodesics *i.e.*, the line element satisfies  $ds^2 = 0$  for any null curve between two events. As such it is tricky to lay down the Fermat's principle for null geodesics which are already a byproduct of extremalisation. Following in the footsteps of Schneider, Ehlers, and Falco (1992), we can lay down the theorem as

For any event  $S$  (emission of light from a source) let the worldline  $\mathcal{L}$  for an observer be connected by a null curve  $\gamma$ . Such a null curve will represent a photon trajectory *iff* its time of arrival  $\tau$  on  $\mathcal{L}$  is stationary under first order variations of  $\gamma$ , where the entire family of smooth null curves from  $S$  to  $\mathcal{L}$  are taken into account.

$$\delta\tau = 0 \quad (1.59)$$

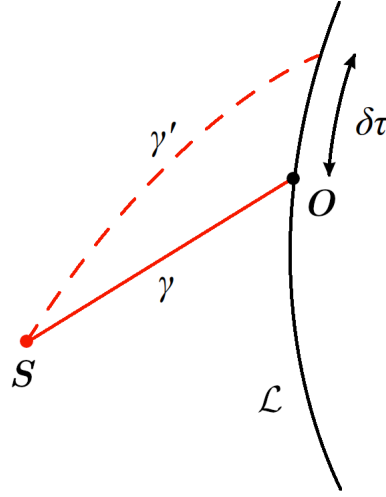


Figure 1.8: Illustration depicting the formulation of relativistic Fermat's principle

The path  $\gamma'$  is a slightly different path than  $\gamma$  so that the both of these paths emerging from  $S$  arrive at the worldline  $\mathcal{L}$  with a relative time delay of  $\delta\tau$  [fig. [1.8]]. Now, for any null curve like  $\gamma'$  slightly varying from  $\gamma$  we must have  $\delta\tau = 0$ . Comprehensive treatment of Fermat's principle in general relativity may be found in the following literature [Perlick (1990), Perlick (1990), Faraoni (1992), Perlick and Piccione (1998), Perlick (2000), Giannoni, Masiello, and Piccione (2002), Perlick (2006), Frolov (2013)].

Fermat's principle will have a strong bearing on the mathematical treatments in chapter 5. As a precursor it will be instructive to delve into the mathematical formulations regarding Fermat's principle for two specific spacetimes, namely (a) conformally stationary and (b) conformally static.

### Effective refractive index in conformally stationary spacetime through Fermat's principle

Any spacetime whose geometrical identity does not change with respect to time is termed as a stationary spacetime. The line element of a stationary spacetime has the form

$$ds^2 = e^{2\Phi} (dt - \omega_i dx^i)^2 - e^{-2\Phi} dl^2 \quad (1.60)$$

where  $dl^2 = \gamma_{ij} dx^i dx^j$ . Note that  $\Phi$ ,  $\omega_i$  and  $\gamma_{ij}$  depend on the spatial coordinates  $x_i$  alone. The term  $\omega_i$  determines rotational effects in the spacetime geometry. Abiding by the criteria for null geodesics,  $ds^2 = 0$  and applying it to eq. (1.60) we get

$$\begin{aligned} dt &= \omega_i dx^i + e^{-2\Phi} dl \\ \Rightarrow t &= \int_{\gamma} \omega_i dx^i + e^{-2\Phi} dl \end{aligned} \quad (1.61)$$

where we integrate over the null geodesic  $\gamma$ . Applying Fermat's principle to [eq. \(1.61\)](#) we get

$$\delta \int_{\gamma} \omega_i dx^i + e^{-2\Phi} dl = \delta \int_{\gamma} \left( \omega_i \frac{dx^i}{dl} + e^{-2\Phi} \right) dl = 0 \quad (1.62)$$

Comparing with the Fermat's principle in the classical domain we can then say from [eq. \(1.62\)](#) that the term

$$\left( \omega_i \frac{dx^i}{dl} + e^{-2\Phi} \right) = n_{\text{stationary}} \quad (1.63)$$

is analogous to the refractive index of the stationary spacetime. As [eq. \(1.63\)](#) reveals the effective refractive index is dependent on both position and direction. Thus we can conclude that the spatial paths of photons are geodesics for the Finsler metric  $(\omega_i dx^i + e^{-2\Phi} dl)$  which reduces to the Riemannian metric for  $\omega_i = 0$  [[section 5.1](#)].

### Effective refractive index in conformally static spacetime through Fermat's principle

For special cases where the spacetime geometry is irrotational we have  $\omega_i = 0$ . In instances such as this [eq. \(1.63\)](#) reduces down to

$$n_{\text{static}} = e^{-2\Phi} \quad (1.64)$$

Physically speaking we can attribute the effective refractive index as a property of isotropic, non-dispersive medium.

## 1.9.2 Determining the Shapiro and geometric time delay

Let us consider the weak field perturbation of the FLRW metric and express it as

$$ds^2 = \left( 1 + \frac{2\Phi}{c^2} \right) c^2 dt^2 - \left( 1 - \frac{2\Phi}{c^2} \right) d(x^i)^2 \quad (1.65)$$

Solving for the null geodesic we get,

$$t = \frac{1}{c} \int \left( 1 - \frac{2\Phi}{c^2} \right) dl = \frac{1}{c} l - \frac{2}{c^3} \int \Phi dl \quad (1.66)$$

With reference to [fig. \[1.5\]](#) the path SPO's euclidean length is given by  $l$  which can also be expressed as

$$l = SP + PO = \sqrt{\left( \vec{\xi} - \vec{\eta} \right)^2 + D_{LS}^2} + \sqrt{\vec{\xi}^2 + D_L^2} \approx D_{LS} + D_L + \frac{1}{2D_{LS}} \left( \vec{\xi} - \vec{\eta} \right)^2 + \frac{1}{2D_L} \vec{\xi}^2 \quad (1.67)$$

Let us evaluate the integral in [eq. \(1.66\)](#) from S to P which yields,

$$\int_S^P \Phi dl = GM \ln \frac{|\vec{\xi}|}{2D_{LS}} \quad (1.68)$$

where we have assumed a point mass potential  $\left(\Phi = -\frac{GM}{r}\right)$  and approximated by neglecting higher order terms. If we now consider an arbitrary length scale  $\xi_0$  for which  $\xi_0 < D_{LS}$  and  $|\vec{\xi}| < \xi_0$  then we can segment [eq. \(1.68\)](#) into two parts as shown below

$$\int_S^P \Phi dl = GM \left( \ln \frac{|\vec{\xi}|}{2\xi_0} + \ln \frac{\xi_0}{D_{LS}} \right) \quad (1.69)$$

Similarly for the portion PO we have

$$\int_P^O \Phi dl = GM \left( \ln \frac{|\vec{\xi}|}{2\xi_0} + \ln \frac{\xi_0}{D_L} \right) \quad (1.70)$$

Therefore, summing up the contribution from both portions of the approximated light ray path and expressing in terms of surface mass density we get,

$$\Delta t_{\text{Shapiro}} = -\frac{2}{c^3} \int_S^O \Phi dl = -\frac{4G}{c^3} \int d^2\xi' \Sigma \vec{\xi}' \ln \left( \frac{|\vec{\xi} - \vec{\xi}'|}{\xi_0} \right) + \text{const.} \quad (1.71)$$

Let us now consider the generic equation expressing the time delay for an arbitrary light path in comparison to the undeviated light path

$$\begin{aligned} c\Delta t &= \phi(\vec{\xi}, \vec{\eta}) + \text{const.} \\ &= \frac{D_L D_S}{2D_{LS}} \left( \frac{\vec{\xi}}{D_L} - \frac{\vec{\eta}}{D_S} \right)^2 - \psi(\vec{\xi}) + \text{const} \\ &= \frac{D_L D_S}{2D_{LS}} \left( \frac{\vec{\xi}}{D_L} - \frac{\vec{\eta}}{D_S} \right)^2 - \frac{4G}{c^2} \int d^2\xi' \Sigma \vec{\xi}' \ln \left( \frac{|\vec{\xi} - \vec{\xi}'|}{\xi_0} \right) \end{aligned} \quad (1.72)$$

where  $\phi$  is the Fermat potential [[compare with eq. \(1.32\)](#)] and  $\psi$  is the lensing potential. If we now apply the Fermat's principle to [eq. \(1.72\)](#) we get

$$\begin{aligned} \frac{\partial(\Delta t)}{\partial \vec{\xi}} &= 0 \\ \Rightarrow \frac{\partial}{\partial \xi} \left[ \frac{D_L D_S}{2D_{LS}} \left( \frac{\vec{\xi}}{D_L} - \frac{\vec{\eta}}{D_S} \right)^2 - \frac{4G}{c^2} \int d^2\xi' \Sigma(\xi) \ln \left( \frac{|\vec{\xi} - \vec{\xi}'|}{\xi_0} \right) \right] &= 0 \\ \Rightarrow -\vec{\eta} + \frac{D_S}{D_L} \vec{\xi} &= D_{LS} \times \frac{4G}{c^2} \int d^2\xi' \Sigma(\vec{\xi}') \frac{\vec{\xi} - \vec{\xi}'}{|\vec{\xi} - \vec{\xi}'|^2} \\ \Rightarrow \vec{\eta} &= \frac{D_S}{D_L} \vec{\xi} - D_{LS} \alpha(\vec{\xi}) \end{aligned} \quad (1.73)$$

This clearly shows that the lens equation [[eq. \(1.22\)](#)] can be obtained from the Fermat's principle as well.

Now, from the generic structure of eq. (1.72) the time delay *i.e.*, difference in time of arrival for two images indicated by  $\vec{\xi}_A$  and  $\vec{\xi}_B$  we have,

$$c(t_A - t_B) = \phi(\vec{\xi}_A, \vec{\eta}) - \phi(\vec{\xi}_B, \vec{\eta}) \quad (1.74)$$

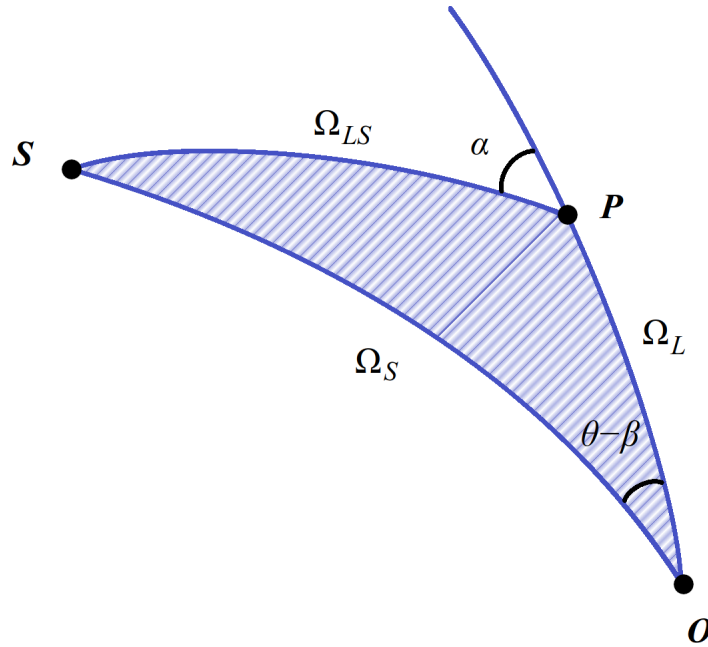
where  $t_A - t_B$  gives the time coordinate separation for the two different arrival times (at observer location) of the two images. Let us now substitute the expression for  $\phi$  in eq. (1.74)

$$\begin{aligned} c(t_A - t_B) &= \frac{D_L D_S}{2D_{LS}} \left[ \left( \frac{\vec{\xi}_A}{D_L} - \frac{\vec{\eta}}{D_S} \right)^2 - \left( \frac{\vec{\xi}_B}{D_L} - \frac{\vec{\eta}}{D_S} \right)^2 \right] + \frac{4G}{c^2} \int d^2 \xi' \Sigma(\vec{\xi}') \ln \left( \frac{|\vec{\xi}_B - \vec{\xi}'|}{|\vec{\xi}_A - \vec{\xi}'|} \right) \\ &= \frac{D_L D_{LS}}{2D_S} \left[ \left\{ \alpha(\vec{\xi}_A) \right\}^2 - \left\{ \alpha(\vec{\xi}_B) \right\}^2 \right] + \frac{4G}{c^2} \int d^2 \xi' \Sigma(\vec{\xi}') \ln \left( \frac{|\vec{\xi}_B - \vec{\xi}'|}{|\vec{\xi}_A - \vec{\xi}'|} \right) \end{aligned} \quad (1.75)$$

where we have substituted the lens equation (eq. (1.73)).

Before we proceed further let us first construct the geometrical structure describing the projection of the events/locations S, P and O into the 3-D comoving space of constant curvature (fig. [1.9]). This will be critical in conjunction with the metric representation of FLRW metric in eq. (1.76).

$$ds^2 = a^2(\zeta) [d\zeta^2 - d\Omega^2] \quad (1.76)$$



**Figure 1.9:** Structure of the geodesic triangle in 3-D Riemannian surface of constant curvature

In eq. (1.76) the *conformal time*  $\zeta$  is related to  $t$  and  $a(t)$  by the relation  $\zeta = c \int \frac{dt}{a(t)}$ . On the other hand  $d\Omega^2$  represents the metric for 3-D Riemannian space of constant curvature. In the case of null geodesics  $ds^2 = 0$  as usual and the geometrical time delay is given by

$$\Delta\zeta_{\text{geom}} = \Omega_{LS} + \Omega_L - \Omega_S \quad (1.77)$$

Note that in keeping with fig. [1.9] and eq. (1.76) the  $\Omega$  terms are the distances measured by the metric  $d\Omega^2$ . Owing to the fact that in comparison to the Hubble time  $a_0 = \frac{1}{H_0}$  the time delay scales are negligible we can assume the following approximate equation to hold good.

$$a_0 \Delta\zeta_{\text{geom}} = c \Delta t_{\text{geom}} \quad (1.78)$$

Since conformal transformation preserves angles the correspondence between angular distances and  $\Omega$ 's can be easily determined.

$$\frac{D_S}{a_S} = \sin \Omega_S \quad (1.79a)$$

$$\frac{D_L}{a_L} = \sin \Omega_L \quad (1.79b)$$

$$\frac{D_{LS}}{a_S} = \sin \Omega_{LS} \quad (1.79c)$$

From the geometry depicted in fig. [1.9] and eq. (1.77) we can write

$$\Delta\zeta_{\text{geom}} = \frac{\sin \Omega_L \sin \Omega_{LS}}{2 \sin \Omega_S} \alpha^2 \quad (1.80)$$

and

$$\alpha \sin \Omega_{LS} = (\vec{\theta} - \vec{\beta}) \sin \Omega_S \quad (1.81)$$

Substituting eq. (1.81) in eq. (1.80) we get

$$\Delta\zeta_{\text{geom}} = \frac{\sin \Omega_L \sin \Omega_S}{2 \sin \Omega_{LS}} (\vec{\theta} - \vec{\beta})^2 \quad (1.82)$$

Finally combining the result obtained above with eq. (1.79) we get

$$\begin{aligned} \Delta\zeta_{\text{geom}} &= \frac{1}{a_L} \frac{D_L D_S}{2 D_{LS}} (\vec{\theta} - \vec{\beta})^2 \\ \Rightarrow \frac{c}{a_0} \Delta t_{\text{geom}} &= \frac{1}{a_L} \frac{D_L D_S}{2 D_{LS}} (\vec{\theta} - \vec{\beta})^2 \\ \Rightarrow \Delta t_{\text{geom}} &= \frac{1}{c} \frac{a_0}{a_L} \frac{D_L D_S}{2 D_{LS}} (\vec{\theta} - \vec{\beta})^2 = \frac{1}{c} (1 + z_L) \frac{D_L D_S}{2 D_{LS}} (\vec{\theta} - \vec{\beta})^2 \end{aligned} \quad (1.83)$$

where we have used  $\frac{a_0}{a_L} = (1 + z_L)$ . Note that in eq. (1.71) where we have derived the Shapiro time delay the necessary correction due to redshift is not included. Taking this into account the cosmological Shapiro time delay is given by

$$\Delta t_{\text{Shapiro}} = -\frac{1}{c} (1 + z_L) \psi(\vec{\xi}) + \text{const.} \quad (1.84)$$

Therefore, the total time delay incurred by the deviated ray in comparison to the unperturbed ray is given by

$$\begin{aligned}\Delta t_{\text{tot}} &= \Delta t_{\text{Shapiro}} + \Delta t_{\text{geom}} \\ &= (1 + z_L) \left[ \frac{D_L D_S}{2D_{LS}} \left( \vec{\theta} - \vec{\beta} \right)^2 - \psi \left( \vec{\xi} \right) \right] + \text{const.}\end{aligned}$$

where the const. has the same value for all rays travelling from the source to the observer.

*She walks in beauty, like the night  
Of cloudless climes and starry skies;  
And all that's best of dark and bright  
Meet in her aspect and her eyes;  
Thus mellowed to that tender light  
Which heaven to gaudy day denies.*

—Lord Byron

# 2

## DEFLECTION OF LIGHT IN GENERAL RELATIVITY—CLASSICAL APPROACH

We present a detailed derivation of the deflection angle for a photon in a static isotropic spherically symmetric metric following the prescription outlined in [Weinberg \(1972\)](#). The descriptors of the metric “static” and “isotropic” in this case imply the existence of a set of *quasi-Minkowskian* coordinates  $[x^0 \equiv t, x^1, x^2, x^3]$  such that the invariant proper time  $d\tau^2 = g_{\mu\nu}dx^\mu dx^\nu$  is independent of  $t$ . The most general description of an isotropic spherically symmetric gravitational field is given by

$$ds^2 = -f(r)dt^2 + g(r)dr^2 + r^2d\theta^2 + r^2\sin^2\theta d\phi^2 \quad (2.1)$$

The generic expression for the deflection angle of a light ray under the influence of such a gravitational field may be derived by solving the geodesic equations given by

$$\frac{d^2x^\mu}{dp^2} + \Gamma_{\nu\lambda}^\mu \frac{dx^\nu}{dp} \frac{dx^\lambda}{dp} = 0 \quad (2.2)$$

In [eq. \(2.2\)](#)  $p$  denotes a parameter describing the trajectory. In order to write down the four equations resulting from the family of equations in [eq. \(2.2\)](#) we first need to calculate all the non-vanishing components

of the affine connection given by

$$\Gamma_{\nu\lambda}^{\mu} = \frac{1}{2}g^{\mu\eta} \left( \frac{\partial g_{\eta\nu}}{\partial x^{\lambda}} + \frac{\partial g_{\eta\lambda}}{\partial x^{\nu}} - \frac{\partial g_{\nu\lambda}}{\partial x^{\eta}} \right) \quad (2.3)$$

The non-vanishing components of the affine connection of the metric expressed in eq. (2.1) are tabulated below

$$\begin{aligned} \Gamma_{tt}^r &= \frac{1}{2g(r)} \frac{df(r)}{dr} & \Gamma_{\theta\theta}^r &= -\frac{r}{g(r)} & \Gamma_{\phi\phi}^r &= -\frac{r \sin^2 \theta}{g(r)} \\ \Gamma_{rr}^r &= \frac{1}{2g(r)} \frac{dg(r)}{dr} & \Gamma_{r\theta}^{\theta} &= \Gamma_{\theta r}^{\theta} = -\frac{1}{r} & \Gamma_{\phi\phi}^{\theta} &= -\sin \theta \cos \theta \\ \Gamma_{\phi r}^{\phi} &= \Gamma_{r\phi}^{\phi} = \frac{1}{r} & \Gamma_{\phi\theta}^{\phi} &= \Gamma_{\theta\phi}^{\phi} = \cot \theta & \Gamma_{tr}^t &= \Gamma_{rt}^t = \frac{1}{2f(r)} \frac{df(r)}{dr} \end{aligned} \quad (2.4)$$

Suitable substitution of these components in the geodesic equation yields the equations of motion for  $t$ ,  $r$ ,  $\theta$  and  $\phi$  in terms of the parameter  $p$ . The four different equations of motion are determined in the following subsection.

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## 2.1 EQUATIONS OF MOTION FOR THE GENERIC CASE OF A STATIC ISOTROPIC METRIC

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The equation of motion for  $t$ ,  $\theta$ ,  $\phi$  &  $r$  is derived from the geodesic equation and relevant non-vanishing affine connections as follows:–

**The equation of motion for  $t$**

$$\begin{aligned} \frac{d^2 t}{dp^2} + \Gamma_{tr}^t \frac{dt}{dp} \frac{dr}{dp} + \Gamma_{rt}^t \frac{dr}{dp} \frac{dt}{dp} &= 0 \\ \Rightarrow \frac{d^2 t}{dp^2} + \frac{f'(r)}{2f(r)} \frac{dt}{dp} \frac{dr}{dp} + \frac{f'(r)}{2f(r)} \frac{dr}{dp} \frac{dt}{dp} &= 0 \\ \Rightarrow \frac{d^2 t}{dp^2} + \frac{f'(r)}{f(r)} \frac{dt}{dp} \frac{dr}{dp} &= 0 \end{aligned} \quad (2.5)$$

**The equation of motion for  $\theta$**

$$\begin{aligned} \frac{d^2 \theta}{dp^2} + \Gamma_{r\theta}^{\theta} \frac{dr}{dp} \frac{d\theta}{dp} + \Gamma_{\theta r}^{\theta} \frac{d\theta}{dp} \frac{dr}{dp} + \Gamma_{\phi\phi}^{\theta} \frac{d\phi}{dp} \frac{d\phi}{dp} &= 0 \\ \Rightarrow \frac{d^2 \theta}{dp^2} + \frac{2}{r} \frac{dr}{dp} \frac{d\theta}{dp} - \sin \theta \cos \theta \left( \frac{d\phi}{dp} \right)^2 &= 0 \\ \Rightarrow \frac{d^2 \theta}{dp^2} + \frac{2}{r} \frac{dr}{dp} \frac{d\theta}{dp} - \sin \theta \cos \theta \left( \frac{d\phi}{dp} \right)^2 &= 0 \end{aligned} \quad (2.6)$$

### The equation of motion for $\phi$

$$\begin{aligned}
 & \frac{d^2\phi}{dp^2} + \Gamma_{\phi r}^{\phi} \frac{d\phi}{dp} \frac{dr}{dp} + \Gamma_{r\phi}^{\phi} \frac{dr}{dp} \frac{d\phi}{dp} + \Gamma_{\phi\theta}^{\phi} \frac{d\phi}{dp} \frac{d\theta}{dp} + \Gamma_{\theta\phi}^{\phi} \frac{d\theta}{dp} \frac{d\phi}{dp} = 0 \\
 \Rightarrow & \frac{d^2\phi}{dp^2} + 2\Gamma_{\phi r}^{\phi} \frac{d\phi}{dp} \frac{dr}{dp} + 2\Gamma_{\theta\phi}^{\phi} \frac{d\theta}{dp} \frac{d\phi}{dp} = 0 \\
 \Rightarrow & \frac{d^2\phi}{dp^2} + \frac{2}{r} \frac{d\phi}{dp} \frac{dr}{dp} + 2 \cot \theta \frac{d\theta}{dp} \frac{d\phi}{dp} = 0
 \end{aligned} \tag{2.7}$$

### The equation of motion for $r$

$$\begin{aligned}
 & \frac{d^2r}{dp^2} + \Gamma_{rr}^r \frac{dr}{dp} \frac{dr}{dp} + \Gamma_{tt}^r \frac{dt}{dp} \frac{dt}{dp} + \Gamma_{\theta\theta}^r \frac{d\theta}{dp} \frac{d\theta}{dp} + \Gamma_{\phi\phi}^r \frac{d\phi}{dp} \frac{d\phi}{dp} = 0 \\
 \Rightarrow & \frac{d^2r}{dp^2} + \frac{g'(r)}{2g(r)} \left( \frac{dr}{dp} \right)^2 + \frac{f'(r)}{2g(r)} \left( \frac{dt}{dp} \right)^2 - \frac{r}{g(r)} \left( \frac{d\theta}{dp} \right)^2 - \frac{r \sin^2 \theta}{g(r)} \left( \frac{d\phi}{dp} \right)^2 = 0
 \end{aligned} \tag{2.8}$$

The orbit of a particle governed by the equations [eq. \(2.5\)](#)–[eq. \(2.8\)](#) maybe considered to lie on the equatorial plane as a consequence of the gravitational field being isotropic. The immediate mathematical consequence of this physical scenario is that  $\theta = \frac{\pi}{2}$ . As a result of this the equation of motion for  $\theta$  described by [eq. \(2.6\)](#) is rendered satisfied and  $\theta$  can be excused as a dynamical variable. Armed with these insights, we can further simplify the three remaining equations of motion (namely the ones in terms of  $t$ ,  $r$  and  $\phi$ ) and recast them into a more approachable form with some standard mathematical manipulation.

Let us first approach the equation for  $t$ . Dividing both sides of the [eq. \(2.5\)](#) by  $\frac{dt}{dp}$  we get

$$\begin{aligned}
 & \frac{\frac{d^2t}{dp^2}}{\frac{dt}{dp}} + \frac{f'(r)}{f(r)} \frac{dr}{dp} = 0 \\
 \Rightarrow & \frac{d}{dp} \left[ \ln \frac{dt}{dp} + \ln f(r) \right] = 0 \\
 \Rightarrow & \frac{dt}{dp} = \frac{1}{f(r)}
 \end{aligned} \tag{2.9}$$

For the equation  $\phi$  given by [eq. \(2.6\)](#) the term  $2 \cot \theta \frac{d\theta}{dp}$  immediately vanishes as  $\theta = \frac{\pi}{2}$ , thus reducing the equation to

$$\begin{aligned}
 & \frac{d^2\phi}{dp^2} + \frac{2}{r} \frac{d\phi}{dp} \frac{dr}{dp} = 0 \\
 \Rightarrow & \frac{\frac{d^2\phi}{dp^2}}{\frac{d\phi}{dp}} + \frac{2}{r} \frac{dr}{dp} = 0 \quad \left[ \because \text{Dividing both sides by } \frac{d\phi}{dp} \right] \\
 \Rightarrow & \frac{d}{dp} \left[ \ln \frac{d\phi}{dp} + \ln r^2 \right] = 0
 \end{aligned}$$

$$\begin{aligned}
 &\Rightarrow \ln \frac{d\phi}{dp} = -\ln r^2 \\
 &\Rightarrow \ln \frac{d\phi}{dp} = \ln \frac{1}{r^2} + \ln J \\
 &\Rightarrow \frac{d\phi}{dp} = \frac{J}{r^2} \\
 &\Rightarrow r^2 \frac{d\phi}{dp} = J
 \end{aligned} \tag{2.10}$$

Here,  $J$  which occurred as the constant of integration represents the angular momentum per unit mass.

Upon substituting the results found in eq. (2.9) & eq. (2.10) in the eq. (2.8) along with  $\theta = \frac{\pi}{2}$  we get

$$\begin{aligned}
 &\frac{d^2 r}{dp^2} + \frac{g'(r)}{2g(r)} \left( \frac{dr}{dp} \right)^2 + \frac{f'(r)}{2g(r)} \left( \frac{dt}{dp} \right)^2 - \frac{r}{g(r)} \left( \frac{d\theta}{dp} \right)^2 - \frac{r \sin^2 \theta}{g(r)} \left( \frac{d\phi}{dp} \right)^2 = 0 \\
 &\Rightarrow \frac{d^2 r}{dp^2} + \frac{g'(r)}{2g(r)} \left( \frac{dr}{dp} \right)^2 + \frac{f'(r)}{2g(r)} \left( \frac{dt}{dp} \right)^2 - \frac{r}{g(r)} \left( \frac{d\phi}{dp} \right)^2 = 0 \\
 &\Rightarrow \frac{d^2 r}{dp^2} + \frac{g'(r)}{2g(r)} \left( \frac{dr}{dp} \right)^2 + \frac{f'(r)}{2g(r)} \frac{1}{f(r)^2} - \frac{r}{g(r)} \frac{J^2}{r^4} = 0 \\
 &\Rightarrow \frac{d^2 r}{dp^2} + \frac{g'(r)}{2g(r)} \left( \frac{dr}{dp} \right)^2 - \frac{J^2}{r^3 g(r)} + \frac{f'(r)}{2g(r)} \frac{1}{f(r)^2} = 0
 \end{aligned} \tag{2.11}$$

Multiplying both sides of eq. (2.11) with  $2g(r) \frac{dr}{dp}$  we get,

$$\begin{aligned}
 &2g(r) \frac{dr}{dp} + g'(r) \left( \frac{dr}{dp} \right)^3 - \frac{2J^2}{r^3} \frac{dr}{dp} + \frac{f'(r)}{f(r)^2} \frac{dr}{dp} = 0 \\
 &\Rightarrow \frac{d}{dp} \left[ g(r) \left( \frac{dr}{dp} \right)^2 \right] + \frac{d}{dp} \left[ \frac{J^2}{r^2} \right] - \frac{d}{dp} \left[ \frac{1}{f(r)} \right] = 0 \\
 &\Rightarrow \frac{d}{dp} \left[ g(r) \left( \frac{dr}{dp} \right)^2 + \frac{J^2}{r^2} - \frac{1}{f(r)} \right] = 0 \\
 &\Rightarrow g(r) \left( \frac{dr}{dp} \right)^2 + \frac{J^2}{r^2} - \frac{1}{f(r)} = -E
 \end{aligned} \tag{2.12}$$

Before proceeding further it is instructive to note that  $d\tau$  ( $\tau$  denotes the proper time) is proportional to  $dp$  which allows us to normalize the parameter  $p$  such that  $p = \tau$  for the case of massive particles. However, as we are interested in describing the motion of photons, we should take care to normalize  $p$  independently of  $\tau$  so that the constant of proportionality  $\frac{d\tau}{dp}$  vanishes for massless particles. These conditions help us to determine some necessary caveats regarding the determination of a particle orbit in the said static spacetime metric. In order to do that, let us begin with expressing the metric given in eq. (2.1) in terms of proper time which can be written as

$$-d\tau^2 = -f(r)dt^2 + g(r)dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2 \tag{2.13}$$

If we now substitute  $\theta = \frac{\pi}{2}$  and the results found in eq. (2.9), eq. (2.10) & eq. (2.12) in eq. (2.13) we get

$$-d\tau^2 = -f(r)dt^2 + g(r)dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2$$

$$\begin{aligned} \Rightarrow -d\tau^2 &= -f(r) \frac{dp^2}{f(r)^2} + g(r) \frac{1}{g(r)} \left( -E + \frac{1}{f(r)} - \frac{J^2}{r^2} \right) dp^2 + r^2 \frac{J^2}{r^4} dp^2 \\ \Rightarrow d\tau^2 &= E dp^2 \end{aligned} \quad (2.14)$$

eq. (2.14) mandates that in order for  $\frac{d\tau}{dp} = \text{constant}$  to be satisfied  $E$  must be positive for massive particles and zero for massless particles. Furthermore, eq. (2.12) dictates that in order for the orbit of a particle to assume a radius of  $r$  the following identity must hold, given that  $g(r)$  generally holds a positive value.

$$\frac{J^2}{r^2} + E \leq \frac{1}{f(r)} \quad (2.15)$$

Our equations of motion have been determined in terms of the parameter  $p$ . However, since we are concerned with the shape of the orbit, we can eliminate  $dp$  from eq. (2.10) & eq. (2.12). This allows us to express  $r$  as a function of  $\phi$ . Starting with the eq. (2.12) we proceed as follows,

$$\begin{aligned} g(r) \left( \frac{dr}{dp} \right)^2 + \frac{J^2}{r^2} - \frac{1}{f(r)} &= -E \\ \Rightarrow g(r) \left( \frac{dr}{d\phi} \right)^2 \left( \frac{d\phi}{dp} \right)^2 + \frac{J^2}{r^2} - \frac{1}{f(r)} &= -E \\ \Rightarrow g(r) \left( \frac{dr}{d\phi} \right)^2 \frac{J^2}{r^4} + \frac{J^2}{r^2} - \frac{1}{f(r)} &= -E \\ \Rightarrow \frac{g(r)}{r^4} \left( \frac{dr}{d\phi} \right)^2 + \frac{1}{r^2} - \frac{1}{J^2 f(r)} &= -\frac{E}{J^2} \end{aligned} \quad (2.16)$$

Furthermore, the eq. (2.10) and eq. (2.12) can be rewritten by eliminating  $p$  from them individually with the help of eq. (2.9),

$$r^2 \frac{d\phi}{dt} = J f(r) \quad (2.17)$$

$$\frac{g(r)}{f(r)^2} \left( \frac{dr}{dt} \right)^2 + \frac{J^2}{r^2} - \frac{1}{f(r)} = -E \quad (2.18)$$

Since we are interested in determining the deflection angle of a light ray in a gravitational field described by the metric in eq. (2.1) we reduce the eq. (2.16) to yield an expression for  $\phi$  as an integral in terms of  $r$ . The necessary steps are shown below.

$$\begin{aligned} \frac{g(r)}{r^4} \left( \frac{dr}{d\phi} \right)^2 + \frac{1}{r^2} - \frac{1}{J^2 f(r)} &= -\frac{E}{J^2} \\ \Rightarrow \frac{g(r)}{r^4} \left( \frac{dr}{d\phi} \right)^2 &= \frac{1}{J^2 f(r)} - \frac{E}{J^2} - \frac{1}{r^2} \\ \Rightarrow \left( \frac{dr}{d\phi} \right)^2 &= \frac{r^4}{J^2 f(r) g(r)} - \frac{E r^4}{J^2 g(r)} - \frac{r^2}{g(r)} \end{aligned}$$

$$\begin{aligned}
 \Rightarrow \left( \frac{dr}{d\phi} \right)^2 &= \frac{r^4}{g(r)} \left[ \frac{1}{J^2 f(r)} - \frac{E}{J^2} - \frac{1}{r^2} \right] \\
 \Rightarrow \frac{dr}{d\phi} &= \frac{r^2}{\sqrt{g(r)}} \sqrt{\frac{1}{J^2 f(r)} - \frac{E}{J^2} - \frac{1}{r^2}} \\
 \Rightarrow d\phi &= \frac{\sqrt{g(r)} dr}{r^2 \sqrt{\frac{1}{J^2 f(r)} - \frac{E}{J^2} - \frac{1}{r^2}}} \\
 \Rightarrow \phi &= \pm \int \frac{\sqrt{g(r)} dr}{r^2 \sqrt{\frac{1}{J^2 f(r)} - \frac{E}{J^2} - \frac{1}{r^2}}} \quad (2.19)
 \end{aligned}$$

In order to determine the deflection angle of any particle in a static isotropic gravitational field, we need to consider two other quantities, namely the velocity  $V$  of the particle and the so called impact parameter  $b$ . We also need to determine the interrelationship between these two quantities with  $E$  &  $J$  already introduced in our study earlier. For any particle entering the gravitational field from infinity *i.e.*,  $r \rightarrow \infty$  the metric becomes Minkowskian in nature, indicating that  $f(\infty) = g(\infty) = 1$ . Furthermore, we can consider the motion of the particle at infinity to be along a straight line with the constant velocity  $V$ .

The impact parameter  $b$  and the velocity  $V$  mentioned above can be mathematically expressed as

$$\begin{aligned}
 b &\simeq r \sin(\phi - \phi_\infty) \simeq r(\phi - \phi_\infty) \\
 V &\simeq \frac{d}{dt} (r \cos(\phi - \phi_\infty)) \simeq \frac{dr}{dt}
 \end{aligned}$$

We can insert these expressions for  $b$  and  $V$  in [eq. \(2.17\)](#) & [eq. \(2.18\)](#) to get the expressions  $J = bV$  and  $E = 1 - V^2$ . Of course the derivation of these expressions were carried out under the condition of motion at infinity. In the case of a photon, the velocity  $V = 1$  which makes the impact parameter  $b = J$  and  $E = 0$ . Since, in many cases the static isotropic metric represent objects like black holes we are also interested in calculating the closest distance of approach of a particle to the said object. Assuming that the closest distance of approach is  $r_0$ , let us work with [eq. \(2.16\)](#) to arrive at a suitable expression for  $J$  which depends only on the distance of closest approach and the velocity. [Since  $r_0$  will have a fixed value, we can claim  $\left( \frac{dr}{d\phi} \right)_{r=r_0} = 0$ ]

$$\begin{aligned}
 \frac{g(r)}{r^4} \left( \frac{dr}{d\phi} \right)^2 + \frac{1}{r^2} - \frac{1}{J^2 f(r)} &= -\frac{E}{J^2} \\
 \Rightarrow \frac{g(r_0)}{r_0^4} \left( \frac{dr}{d\phi} \right)_{r=r_0}^2 + \frac{1}{r_0^2} - \frac{1}{J^2 f(r_0)} &= -\frac{E}{J^2} \\
 \Rightarrow \frac{1}{r_0^2} - \frac{1}{J^2 f(r_0)} &= -\frac{E}{J^2} \\
 \Rightarrow \frac{1}{r_0^2} &= \frac{1}{J^2} \left[ \frac{1}{f(r_0)} - E \right]
 \end{aligned}$$

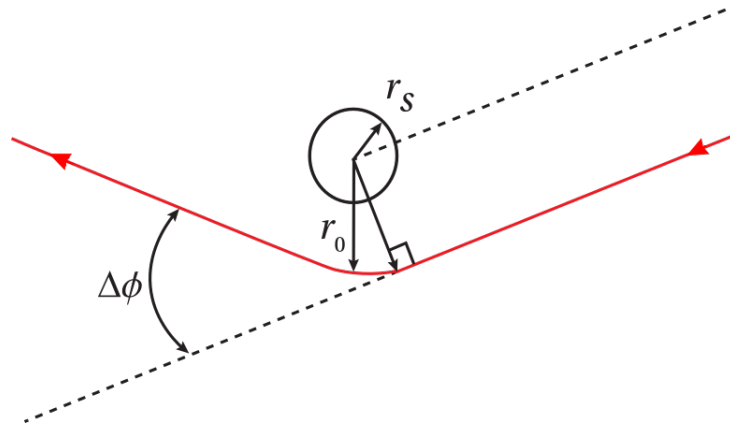


Figure 2.1: Classical depiction of bending of light under the influence of a gravitational field

$$\begin{aligned}
 \Rightarrow \frac{1}{r_0^2} &= \frac{1}{J^2} \left[ \frac{1}{f(r_0)} - 1 + V^2 \right] \\
 \Rightarrow J^2 &= r_0^2 \left[ \frac{1}{f(r_0)} - 1 + V^2 \right] \\
 \Rightarrow J &= r_0 \sqrt{\frac{1}{f(r_0)} - 1 + V^2}
 \end{aligned} \tag{2.20}$$

Substituting this value of  $J$  in eq. (2.19) we find the value of  $\phi$  purely as a function of  $r$ .

$$\begin{aligned}
 \phi &= \pm \int \frac{\sqrt{g(r)} dr}{r^2 \sqrt{\frac{1}{J^2 f(r)} - \frac{E}{J^2} - \frac{1}{r^2}}} \\
 \Rightarrow \phi(r) &= \phi_\infty + \int_r^\infty \frac{\sqrt{g(r)} dr}{\sqrt{\frac{1}{J^2} \left( \frac{1}{f(r)} - E \right) - \frac{1}{r^2}}} \\
 \Rightarrow \phi(r) &= \phi_\infty + \int_r^\infty \frac{\sqrt{g(r)} dr}{r^2 \left[ \frac{1}{r_0^2} \left( \frac{1}{f(r) - 1 + V^2} \right) \left( \frac{1}{f(r_0) - 1 + V^2} \right)^{-1} - \frac{1}{r^2} \right]^{\frac{1}{2}}}
 \end{aligned} \tag{2.21}$$

The cumulative change in the value of  $\phi$  as the particle approaches infinity to the closest distance of approach  $r_0$  before returning to infinity again is  $2|\phi(r_0) - \phi_\infty|$ . Now, if spacetime were flat then the trajectory would have been that of a straight line *i.e.*, the change in  $\phi$  would have been simply  $\pi$ . Therefore, the deflection of the orbit from a straight line would simply be  $\Delta\phi = 2|\phi(r_0) - \phi_\infty| - \pi$ . When  $\Delta\phi > 0$  the trajectory is bent towards the source of the gravitational field, whereas for  $\Delta\phi < 0$  the trajectory is bent away from the source of the gravitational field.

For the specific case of massless particles, [eq. \(2.21\)](#) reduces to

$$\begin{aligned}\phi(r) - \phi_\infty &= \int_r^\infty \frac{\sqrt{g(r)} dr}{r \left[ \frac{r^2}{r_0^2} \frac{f(r_0)}{f(r)} - 1 \right]^{\frac{1}{2}}} \\ \Rightarrow \phi(r) - \phi_\infty &= \int_r^\infty \frac{1}{r} \left[ \left( \frac{r}{r_0} \right)^2 \frac{f(r_0)}{f(r)} \frac{1}{g(r)} - \frac{1}{g(r)} \right]^{-\frac{1}{2}} dr\end{aligned}\quad (2.22)$$

Therefore, the most general expression for the bending angle of a light ray in a static isotropic metric is given as

$$\Delta\phi_{\text{photon}} = 2|\phi(r_0) - \phi_\infty| - \pi = 2 \int_{r_0}^\infty \frac{1}{r} \left[ \left( \frac{r}{r_0} \right)^2 \frac{f(r_0)}{f(r)} \frac{1}{g(r)} - \frac{1}{g(r)} \right]^{-\frac{1}{2}} dr - \pi \quad (2.23)$$

while the impact parameter in the case of a photon is given by the expression derived from [eq. \(2.20\)](#)

$$b_{\text{photon}} = r_0 f(r_0)^{-\frac{1}{2}} \quad (2.24)$$

To see a world in a grain of sand  
And a heaven in a wild flower,  
Hold infinity in the palm of your hand  
And eternity in an hour.

—William Blake, *Auguries of Innocence*

# 3

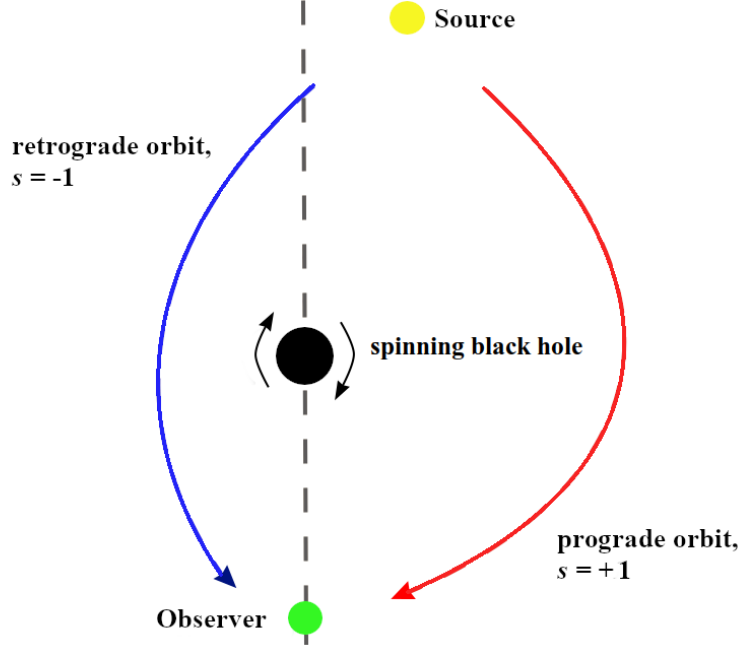
## DEFLECTION ANGLE OF LIGHT IN THE EQUATORIAL PLANE OF A KERR BLACK HOLE

Unlike the Schwarzschild black hole(*i.e.*, zero angular momentum case), the deflection angle of light(on the equatorial plane) in the vicinity of a spinning black hole(non-zero angular momentum case) is influenced differently depending on the trajectory of approach. This influence in question is dictated by how the direction of light ray is aligned with the spin of the black hole. When the photon trajectory is aligned in the same direction as the spin of the black hole we identify it as a *prograde orbit*. Whereas, when the photon trajectory is aligned opposite to the spin of the black hole we designate it as a *retrograde orbit*. Taking the Schwarzschild case as a reference point we find that the presence of angular momentum—

- (a) increases the deflection angle for the prograde case.
- (b) decreases the deflection angle for the retrograde case.

An interesting consequence of this non-uniform effect of black hole spin on the photon trajectory is the loss of symmetry about the axis of rotation. Furthermore, the greater the angular momentum of the black hole the more pronounced the departure of the bending angle from the Schwarzschild case. This section is

devoted to deriving the mathematical formulation of all these effects that are unique to the spinning black hole. We follow the same approach as — [Iyer and Hansen \(2009\)](#), [Iyer and Hansen \(2009\)](#) in this section and take the Kerr metric for our case study. From an evolutionary point of view of the subject, the foundations were laid out by — [Darwin \(1959\)](#), [Darwin \(1961\)](#), [Boyer and Lindquist \(1967\)](#), [Chandrasekhar \(1998\)](#) etc. The unique feature of the work [Iyer and Hansen \(2009\)](#) lies in the fact that they were the first to arrive at an explicit expression for the angle of deflection for both the prograde and retrograde cases.



**Figure 3.1:** Schematic diagram illustrating the prograde and retrograde orbits (on the equatorial plane) in the vicinity of a spinning black hole. In the above scenario the axis of rotation of the black hole points into the page.

The Kerr metric(in the equatorial plane) is expressed in the Boyer-Lindquist coordinates as

$$ds^2 = g_{tt}dt^2 + g_{rr}dr^2 + g_{\phi\phi}d\phi^2 + 2g_{t\phi}dtd\phi \quad (3.1)$$

where,

$$g_{tt} = -\left(1 - \frac{2M}{r}\right); \quad g_{rr} = \left(1 - \frac{2M}{r} + \frac{a^2}{r^2}\right)^{-1}; \quad g_{\phi\phi} = r^2 + a^2 + \frac{2a^2M}{r}; \quad g_{t\phi} = -\frac{2aM}{r}$$

Choosing the affine parameter  $\lambda$  along the photon trajectory we can write  $p^\mu = \dot{x}^\mu = \frac{dx^\mu}{d\lambda}$ . Based on our choice we can now write the conserved quantities– energy and azimuthal component of angular momentum as shown below,

$$E = -\xi_{(t)\mu}p^\mu = -g_{tt}\dot{t} - g_{t\phi}\dot{\phi}$$

$$\Rightarrow E = \left(1 - \frac{2M}{r}\right) \dot{t} + \frac{2aM}{r} \dot{\phi} \quad (3.2)$$

and

$$\begin{aligned} J_\phi &= \xi_{(\phi)\mu} p^\mu = g_{t\phi} \dot{t} + g_{\phi\phi} \dot{\phi} \\ &= -\frac{2aM}{r} \dot{t} + \left(r^2 + a^2 + \frac{2a^2M}{r}\right) \dot{\phi} \end{aligned} \quad (3.3)$$

By suitable algebraic manipulation of eq. (3.2) & eq. (3.3) we derive the expression for  $\dot{\phi}$  &  $\dot{t}$  as shown below. Multiplying eq. (3.2) with  $\left(1 - \frac{2M}{r}\right)^{-1}$  & eq. (3.3) with  $\left(\frac{2aM}{r}\right)^{-1}$  we get,

$$\begin{aligned} E \left(1 - \frac{2M}{r}\right)^{-1} - \frac{2aM}{r} \left(1 - \frac{2M}{r}\right)^{-1} \dot{\phi} &= \dot{t} \\ J_\phi \left(\frac{2aM}{r}\right)^{-1} - \left(r^2 + a^2 + \frac{2a^2M}{r}\right) \left(\frac{2aM}{r}\right)^{-1} \dot{\phi} &= -\dot{t} \end{aligned}$$

Adding the two equations above we get,

$$\begin{aligned} E \left(1 - \frac{2M}{r}\right)^{-1} + J_\phi \left(\frac{2aM}{r}\right)^{-1} - \left[ \frac{2aM}{r} \left(1 - \frac{2M}{r}\right)^{-1} + \left(r^2 + a^2 + \frac{2a^2M}{r}\right) \left(\frac{2aM}{r}\right)^{-1} \right] \dot{\phi} &= 0 \\ \Rightarrow \dot{\phi} \left[ \frac{2aM}{r} \left(1 - \frac{2M}{r}\right)^{-1} + \left(r^2 + a^2 + \frac{2a^2M}{r}\right) \left(\frac{2aM}{r}\right)^{-1} \right] &= E \left(1 - \frac{2M}{r}\right)^{-1} + J_\phi \left(\frac{2aM}{r}\right)^{-1} \end{aligned}$$

Multiplying both sides by  $\left(1 - \frac{2M}{r}\right) \frac{2aM}{r}$ ,

$$\begin{aligned} \Rightarrow \dot{\phi} \left[ \left(\frac{2aM}{r}\right)^2 + \left(r^2 + a^2 + \frac{2a^2M}{r}\right) \left(1 - \frac{2M}{r}\right) \right] &= E \frac{2aM}{r} + J_\phi \left(1 - \frac{2M}{r}\right) \\ \Rightarrow \dot{\phi} &= \frac{J_\phi \left[ 1 - \frac{2M}{r} + \frac{2aM}{r} \frac{E}{J_\phi} \right]}{r^2 + a^2 - 2Mr} \end{aligned} \quad (3.4)$$

For  $\dot{t}$  we multiply eq. (3.2) with  $\left(\frac{2aM}{r}\right)^{-1}$  & eq. (3.3) with  $\left(r^2 + a^2 + \frac{2a^2M}{r}\right)^{-1}$  thereby arriving at the following equations,

$$\begin{aligned} E \left(\frac{2aM}{r}\right)^{-1} &= \left(1 - \frac{2M}{r}\right) \left(\frac{2aM}{r}\right)^{-1} \dot{t} + \dot{\phi} \\ J_\phi \left(r^2 + a^2 + \frac{2a^2M}{r}\right)^{-1} &= -\frac{2aM}{r} \left(r^2 + a^2 + \frac{2a^2M}{r}\right)^{-1} \dot{t} + \dot{\phi} \end{aligned}$$

Upon subtracting the later equation from the former we get,

$$E \left(\frac{2aM}{r}\right)^{-1} - J_\phi \left(r^2 + a^2 + \frac{2a^2M}{r}\right)^{-1} = \left[ \left(1 - \frac{2M}{r}\right) \left(\frac{2aM}{r}\right)^{-1} + \frac{2aM}{r} \left(r^2 + a^2 + \frac{2a^2M}{r}\right)^{-1} \right] \dot{t}$$

Multiplying both sides by  $\frac{2aM}{r} \left( r^2 + a^2 + \frac{2a^2M}{r} \right)$  we get,

$$\begin{aligned}
 E \left( r^2 + a^2 + \frac{2a^2M}{r} \right) - J_\phi \frac{2aM}{r} &= \left[ \left( 1 - \frac{2M}{r} \right) \left( r^2 + a^2 + \frac{2a^2M}{r} \right) + \left( \frac{2aM}{r} \right)^2 \right] \dot{t} \\
 \Rightarrow \dot{t} &= \frac{E \left( r^2 + a^2 + \frac{2a^2M}{r} \right) - J_\phi \frac{2aM}{r}}{\left( 1 - \frac{2M}{r} \right) \left( r^2 + a^2 + \frac{2a^2M}{r} \right) + \left( \frac{2aM}{r} \right)^2} \\
 \Rightarrow \dot{t} &= \frac{Er(r^2 + a^2) - 2aM(J_\phi - aE)}{r(r^2 + a^2 - 2Mr)} \tag{3.5}
 \end{aligned}$$

We now employ the normalization condition  $p^2 = -m^2$  to find the relevant expression for the equation involving  $\dot{r}$ . For the Kerr metric the normalization condition takes the form

$$g_{tt}\dot{t}^2 + g_{\phi\phi}\dot{\phi}^2 + g_{rr}\dot{r}^2 + 2g_{t\phi}\dot{t}\dot{\phi} = -m^2 \tag{3.6}$$

For the sake of simplicity we assume,

$$\begin{aligned}
 1 - \frac{2M}{r} &\equiv A \\
 \frac{2aM}{r} &\equiv B \\
 r^2 + a^2 + \frac{2a^2M}{r} &\equiv C
 \end{aligned}$$

which helps us to rewrite eq. (3.2) & eq. (3.3) as

$$E = A\dot{t} + B\dot{\phi} \tag{3.7}$$

&

$$J_\phi = -B\dot{t} + C\dot{\phi} \tag{3.8}$$

From eq. (3.6) we proceed as follows,

$$\begin{aligned}
 g_{tt}\dot{t}^2 + g_{\phi\phi}\dot{\phi}^2 + g_{rr}\dot{r}^2 + 2g_{t\phi}\dot{t}\dot{\phi} &= -m^2 \\
 \Rightarrow - \left( 1 - \frac{2M}{r} \right) \dot{t}^2 + \left( r^2 + a^2 + \frac{2a^2M}{r} \right) \dot{\phi}^2 + \left( 1 - \frac{2M}{r} + \frac{a^2}{r^2} \right)^{-1} \dot{r}^2 - \frac{4aM}{r} \dot{t}\dot{\phi} &= -m^2 \\
 \Rightarrow - \left( 1 - \frac{2M}{r} \right) \dot{t}^2 + \left( r^2 + a^2 + \frac{2a^2M}{r} \right) \dot{\phi}^2 + r^2 (r^2 + a^2 - 2Mr)^{-1} \dot{r}^2 - \frac{4aM}{r} \dot{t}\dot{\phi} &= -m^2 \\
 \Rightarrow - A\dot{t}^2 + C\dot{\phi}^2 + \frac{r^2}{\Delta} \dot{r}^2 - \frac{4aM}{r} \dot{t}\dot{\phi} &= -m^2 \quad [\because \Delta \equiv (r^2 + a^2 - 2Mr)] \\
 \Rightarrow - A\dot{t}^2 + C\dot{\phi}^2 + \frac{r^2}{\Delta} \dot{r}^2 - 2B\dot{t}\dot{\phi} &= -m^2 \\
 \Rightarrow - A\dot{t}^2 - B\dot{t}\dot{\phi} - B\dot{t}\dot{\phi} + C\dot{\phi}^2 + \frac{r^2}{\Delta} \dot{r}^2 &= -m^2
 \end{aligned}$$

$$\begin{aligned}
 &\Rightarrow -\left(A\dot{t} + B\dot{\phi}\right)\dot{t} + \left(-B\dot{t} + C\dot{\phi}\right)\dot{\phi} + \frac{r^2}{\Delta}\dot{r}^2 = -m^2 \\
 &\Rightarrow -E\dot{t} + J_\phi\dot{\phi} + \frac{r^2}{\Delta}\dot{r}^2 = 0 \quad [\because m = 0 \text{ for null geodesics}] \\
 &\Rightarrow \dot{r}^2 r^2 = E\Delta\dot{t} - J_\phi\Delta\dot{\phi}
 \end{aligned} \tag{3.9}$$

In order to remove  $\dot{t}$  &  $\dot{\phi}$  from eq. (3.9) we restructure eq. (3.4) & eq. (3.5) as shown below. Let us first proceed with eq. (3.5) and get an expression for  $\Delta\dot{t}$ .

$$\begin{aligned}
 \dot{t} &= \frac{Er(r^2 + a^2) - 2aM(J_\phi - aE)}{r(r^2 + a^2 - 2Mr)} = \frac{Er(r^2 + a^2) - 2aM(J_\phi - aE)}{r\Delta} \\
 \Rightarrow \Delta\dot{t} &= E(r^2 + a^2) - \frac{2aM}{r}(J_\phi - aE) \\
 &= E(r^2 + a^2) - B(J_\phi - aE) \\
 &= E(r^2 + a^2 + Ba) - BJ_\phi \\
 &= E\left(r^2 + a^2 + \frac{2Ma^2}{r}\right) - BJ_\phi \\
 \Rightarrow \Delta\dot{t} &= CE - BJ_\phi
 \end{aligned} \tag{3.10}$$

Similarly, let us proceed with eq. (3.4) and get an expression for  $\Delta\dot{\phi}$ .

$$\begin{aligned}
 \dot{\phi} &= \frac{E\frac{2aM}{r} + J_\phi\left(1 - \frac{2M}{r}\right)}{r^2 + a^2 - 2Mr} \\
 &= \frac{BE + AJ_\phi}{\Delta} \\
 \Rightarrow \Delta\dot{\phi} &= BE + AJ_\phi
 \end{aligned} \tag{3.11}$$

Substituting eq. (3.10) & eq. (3.11) into eq. (3.9) we arrive at the expression for  $\dot{r}^2$ .

$$\begin{aligned}
 \dot{r}^2 r^2 &= E\Delta\dot{t} - J_\phi\Delta\dot{\phi} \\
 &= E(CE - BJ_\phi) - J_\phi(BE + AJ_\phi) \\
 &= CE^2 - 2BJ_\phi E - AJ_\phi^2 \\
 &= \left(r^2 + a^2 + \frac{2a^2M}{r}\right)E^2 - 2J_\phi E \frac{2aM}{r} - \left(1 - \frac{2M}{r}\right)J_\phi^2 \\
 \Rightarrow \dot{r}^2 &= E^2 + \frac{a^2}{r^2}E^2 + \frac{2Ma^2}{r^3}E^2 - \frac{4Ma}{r^3}J_\phi E - \frac{J_\phi^2}{r^2} + \frac{2M}{r^3}J_\phi^2
 \end{aligned} \tag{3.12}$$

We can incorporate the impact parameter  $b = \frac{J_\phi}{E}$  in eq. (3.12) as shown below. Moreover, in order to include the frame dragging effect of the rotating black hole we define the sign dependent impact parameter as  $b_s \equiv s \left| \frac{J_\phi}{E} \right| \equiv sb$  where  $s = -1$  for the retrograde orbits and  $s = +1$  for the prograde orbits.

$$\dot{r}^2 = J_\phi^2 \left[ \frac{1}{b^2} + \frac{a^2}{b^2 r^2} + \frac{2Ma^2}{r^3 b^2} - \frac{4Ma}{r^3 b_s} - \frac{1}{r^2} + \frac{2M}{r^3} \right] \tag{3.13}$$

Making the transition to reciprocal coordinates we get  $\frac{du}{d\phi} = -\frac{1}{r^2} \frac{\dot{r}}{\dot{\phi}}$ , which in turn yields  $\left(\frac{du}{d\phi}\right)^2 = \frac{1}{r^4} \frac{\dot{r}^2}{\dot{\phi}^2} = u^4 \frac{\dot{r}^2}{\dot{\phi}^2}$ . With the help of reciprocal coordinates [eq. \(3.4\)](#) assume the following form,

$$\dot{\phi}^2 = J_\phi^2 \left[ \frac{1 - 2Mu + \frac{2Ma u}{b_s}}{a^2 - \frac{2M}{u} + \frac{1}{u^2}} \right]^2 \quad (3.14)$$

Similarly, the expression for  $\dot{r}^2$  becomes,

$$\begin{aligned} \dot{r}^2 &= J_\phi^2 \left[ \frac{1}{b^2} + \frac{a^2 u^2}{b^2} + \frac{2Ma^2 u^3}{b^2} - \frac{4Ma u^3}{b_s} - u^3 + 2Mu^3 \right] \\ &= J_\phi^2 \left[ 2Mu^3 \left( 1 - \frac{2a}{b_s} + \frac{a^2}{b^2} \right) - u^2 \left( 1 - \frac{a^2}{b^2} \right) + \frac{1}{b^2} \right] \\ &= J_\phi^2 \left[ 2Mu^3 \left( 1 - \frac{a}{b_s} \right)^2 - u^2 \left( 1 - \frac{a^2}{b^2} \right) + \frac{1}{b^2} \right] \end{aligned} \quad (3.15)$$

Combining [eq. \(3.14\)](#) & [eq. \(3.15\)](#) we can derive the expression for  $\left(\frac{du}{d\phi}\right)^2$ ,

$$\begin{aligned} \left(\frac{du}{d\phi}\right)^2 &= \frac{u^4 J_\phi^2 \left[ 2Mu^3 \left( 1 - \frac{a}{b_s} \right)^2 - u^2 \left( 1 - \frac{a^2}{b^2} \right) + \frac{1}{b^2} \right]}{J_\phi^2 \left[ \frac{1 - 2Mu + \frac{2Ma u}{b_s}}{a^2 - \frac{2M}{u} + \frac{1}{u^2}} \right]^2} \\ &= \frac{[1 - 2Mu + a^2 u^2]^2}{\left[ 1 - 2Mu \left( 1 - \frac{a}{b_s} \right) \right]^2} \left\{ 2M \left( 1 - \frac{a}{b_s} \right)^2 u^3 - \left( 1 - \frac{a^2}{b^2} \right) u^2 + \frac{1}{b^2} \right\} \end{aligned} \quad (3.16)$$

Let us consider that the photon trajectory originates in the asymptotic region and approaches the black hole before reaching the viewer located in the asymptotic region. The closest the photon can get to the black hole while tracing out the path described above can be  $r_0$  (say). Integrating the equation of motion for  $\phi$  and subtracting  $\pi$  from it yields the bending angle. Therefore, the deflection angle for the Kerr black hole can now be expressed as,

$$\hat{\alpha}_{\text{Kerr}} = -\pi + 2 \int_0^{\frac{1}{r_0}} \frac{\left\{ 1 - 2Mu \left( 1 - \frac{a}{b_s} \right) \right\}}{\left\{ 1 - 2Mu + a^2 u^2 \right\} \left\{ 2M \left( 1 - \frac{a}{b_s} \right)^2 u^3 - \left( 1 - \frac{a^2}{b^2} \right) u^2 + \frac{1}{b^2} \right\}^{\frac{1}{2}}} du \quad (3.17)$$

In order to derive the exact solution for [eq. \(3.17\)](#) we need to solve the polynomials within the curly brackets. Let us first tackle the polynomial  $\left\{ 2M \left( 1 - \frac{a}{b_s} \right)^2 u^3 - \left( 1 - \frac{a^2}{b^2} \right) u^2 + \frac{1}{b^2} \right\}$ . As such we identify this cubic

polynomial with  $B(u)$  (say) and note that it has a maximum of two positive roots and one negative root. For our purposes we assume the three roots as  $u_1, u_2$  &  $u_3$  respectively where  $u_1$  is the negative root and  $u_2$  &  $u_3$  are the two real distinct positive roots. In order to find these solutions we assume two intermediate constants  $P$  &  $Q$  [Chandrasekhar (1998)] which facilitates expressing the roots as,

$$\left. \begin{aligned} u_1 &= \frac{P - 2M - Q}{4Mr_0} \\ u_2 &= \frac{1}{r_0} \\ u_3 &= \frac{P - 2M + Q}{4Mr_0} \end{aligned} \right\} \quad (3.18)$$

The dependency of the variable  $P$  on  $a, b, s$  &  $r_0$  may be determined by comparing the coefficients in  $B(u)$  with the relevant counterparts of our targeted cubic polynomial in eq. (3.17).

$$\begin{aligned} B(u) &= 2M \left(1 - \frac{a}{b_s}\right)^2 u^3 - \left(1 - \frac{a^2}{b^2}\right) u^2 + \frac{1}{b^2} \\ &= 2M \left(1 - \frac{a}{b_s}\right)^2 \left[ u^3 - \frac{1}{2M} \frac{\left(1 - \frac{a^2}{b^2}\right)}{\left(1 - \frac{a}{b_s}\right)^2} u^2 + \frac{1}{b^2} \frac{1}{2M} \frac{1}{\left(1 - \frac{a}{b_s}\right)^2} \right] \\ &\equiv 2M \left(1 - \frac{a}{b_s}\right)^2 (u - u_1)(u - u_2)(u - u_3) \\ &= 2M \left(1 - \frac{a}{b_s}\right)^2 [u^3 - (u_1 + u_2 + u_3) u^2 + (u_1 u_2 + u_1 u_3 + u_2 u_3) u - u_1 u_2 u_3] \end{aligned} \quad (3.19)$$

Taking the coefficient of  $u^2$  we get,

$$\begin{aligned} \frac{1}{2M} \frac{\left(1 - \frac{a^2}{b^2}\right)}{\left(1 - \frac{a}{b_s}\right)^2} &= u_1 + u_2 + u_3 \\ \Rightarrow \frac{1}{2M} \frac{\left(1 - \frac{a^2}{b^2}\right)}{\left(1 - \frac{a}{b_s}\right)^2} &= \frac{P - 2M - Q}{4Mr_0} + \frac{1}{r_0} + \frac{P - 2M + Q}{4Mr_0} = \frac{P}{2Mr_0} \\ \Rightarrow P &= r_0 \frac{\left(1 + \frac{a}{b_s}\right)}{\left(1 - \frac{a}{b_s}\right)} \end{aligned} \quad (3.20)$$

Before proceeding further with deriving other critical equations we concentrate on arriving at some working expressions for the terms like  $r_0 \rightarrow$  distance of closest approach,  $r_{sc} \rightarrow$  radius of the circular photon orbit, etc.

Let us revisit the expression for  $\dot{r}^2$  and mould it into a more accommodating form.

$$\begin{aligned}\dot{r}^2 &= J_\phi^2 \left[ 2Mu^3 \left( 1 - \frac{a}{b_s} \right)^2 - u^2 \left( 1 - \frac{a^2}{b^2} \right) + \frac{1}{b^2} \right] \\ \Rightarrow \frac{\dot{r}^2}{J_\phi^2} &= 2Mu^3 \left( 1 - \frac{a}{b_s} \right)^2 - u^2 \left( 1 - \frac{a^2}{b^2} \right) + \frac{1}{b^2}\end{aligned}\quad (3.21)$$

Now we know that,  $\frac{1}{b^2}$  can be expressed as,

$$\frac{1}{b^2} = \frac{\dot{r}^2}{J_\phi^2} + W_{\text{eff}}(r) \quad (3.22)$$

Let us now compare eq. (3.21) with eq. (3.22) and express it such that we have our expression for  $W_{\text{eff}}$  in relation to the Kerr metric.

$$\frac{\dot{r}^2}{J_\phi^2} = \frac{1}{b^2} - \left[ u^2 \left( 1 - \frac{a^2}{b^2} \right) - 2Mu^3 \left( 1 - \frac{a}{b_s} \right)^2 \right]$$

where,

$$W_{\text{eff}}(r) = \left[ \frac{1}{r^2} \left( 1 - \frac{a^2}{b^2} \right) - 2M \frac{1}{r^3} \left( 1 - \frac{a}{b_s} \right)^2 \right] \quad (3.23)$$

Now, if we consider the trajectory of a photon approaching from infinity, *i.e.*, commencing its journey from an asymptotic region towards the black hole and then finally turning back. The turning points in cases such as this is a very well defined location designated by the distance of closest approach,  $r_0$ . This  $r_0$  in turn depends on the impact parameter  $b$  given by,

$$\begin{aligned}\left. \frac{\dot{r}^2}{J_\phi^2} \right|_{r=r_0} &= \frac{1}{b^2} - \left[ u_0^2 \left( 1 - \frac{a^2}{b^2} \right) - 2Mu_0^3 \left( 1 - \frac{a}{b_s} \right)^2 \right] = 0 \\ \Rightarrow \frac{r_0^3}{b^2} - r_0 \left( 1 - \frac{a^2}{b^2} \right) + 2M \left( 1 - \frac{a}{b_s} \right)^2 &= 0 \\ \Rightarrow r_0^3 - b^2 \left( 1 - \frac{a^2}{b^2} \right) r_0 + 2Mb^2 \left( 1 - \frac{a}{b_s} \right)^2 &= 0\end{aligned}\quad (3.24)$$

Clearly eq. (3.24) is in the form of a cubic equation whose general form is,

$$a_1 x^3 + a_2 x^2 + a_3 x + a_4 = 0 \quad (3.25)$$

In order to extract the relevant solutions of the cubic equation we need some key terms as given below,

$$\alpha = \frac{3a_1 a_3 - a_2^2}{9a_1^2}; \quad \beta = \frac{9a_1 a_2 a_3 - 27a_1^2 a_4 - 2a_2^3}{54a_1^3}; \quad \cos \theta = \frac{\beta}{\sqrt{-\alpha^3}}$$

Comparing the generic cubic equation described by eq. (3.25) with eq. (3.24) we find that,

$$a_1 = 1; \quad a_2 = 0; \quad a_3 = -b^2 \left( 1 - \frac{a^2}{b^2} \right); \quad a_4 = 2Mb^2 \left( 1 - \frac{a}{b_s} \right)^2$$

utilising which we find,

$$\alpha = \frac{3 \times 1 \times \left\{ -b^2 \left( 1 - \frac{a^2}{b^2} \right) \right\} - 0}{9} = -\frac{b^2 \left( 1 - \frac{a^2}{b^2} \right)}{3}$$

and

$$\beta = \frac{-27 \times 2Mb^2 \left( 1 - \frac{a}{b_s} \right)^2}{54} = -Mb^2 \left( 1 - \frac{a}{b_s} \right)^2$$

Therefore,

$$\begin{aligned} \cos \theta &= \frac{\beta}{\sqrt{-\alpha^3}} = \frac{-Mb^2 \left( 1 - \frac{a}{b_s} \right)^2}{\sqrt{\left\{ \frac{b^2 \left( 1 - \frac{a^2}{b^2} \right)}{3} \right\}^3}} = -\frac{Mb^2 \left( 1 - \frac{a}{b_s} \right)^2}{\frac{b^3 \left( 1 - \frac{a^2}{b^2} \right)^{\frac{3}{2}}}{3\sqrt{3}}} = -\frac{3\sqrt{3}M \left( 1 - \frac{a}{b_s} \right)^2}{b \left( 1 - \frac{a^2}{b^2} \right)^{\frac{3}{2}}} \\ \Rightarrow \theta &= \cos^{-1} \left\{ -\frac{3\sqrt{3}M \left( 1 - \frac{a}{b_s} \right)^2}{b \left( 1 - \frac{a^2}{b^2} \right)^{\frac{3}{2}}} \right\} \end{aligned} \quad (3.26)$$

The solution of the cubic equation [eq. \(3.25\)](#) is actually given in terms of  $\theta$  as,

$$x = 2\sqrt{-\alpha} \cos \left( \frac{1}{3}\theta \right) - \frac{1}{3}a_2 \quad (3.27)$$

which for our case helps us determine the expression for the closest approach as,

$$\begin{aligned} r_0 &= 2\sqrt{\frac{b^2 \left( 1 - \frac{a^2}{b^2} \right)}{3}} \cos \left[ \frac{1}{3} \cos^{-1} \left\{ -\frac{3\sqrt{3}M \left( 1 - \frac{a}{b_s} \right)^2}{b \left( 1 - \frac{a^2}{b^2} \right)^{\frac{3}{2}}} \right\} \right] \\ \Rightarrow r_0 &= \frac{2b}{\sqrt{3}} \sqrt{1 - \frac{a^2}{b^2}} \cos^{-1} \left\{ -\frac{3\sqrt{3}M \left( 1 - \frac{a}{b_s} \right)^2}{b \left( 1 - \frac{a^2}{b^2} \right)^{\frac{3}{2}}} \right\} \end{aligned} \quad (3.28)$$

It is worth reiterating in this context that since [eq. \(3.28\)](#) is presented in terms of  $b_s$  the expression for  $r_0$  automatically yields results for both the prograde as well as retrograde cases.

Now, in order to find out  $r_{sc}$ , we invoke the condition,  $\left. \frac{dW_{\text{eff}}}{dr} \right|_{r=r_{sc}} = 0$ .

$$\left. \frac{dW_{\text{eff}}}{dr} \right|_{r=r_{sc}} = 0$$

$$\begin{aligned}
 &\Rightarrow \frac{d}{dr} \left[ \frac{1}{r^2} \left( 1 - \frac{a^2}{b^2} \right) - \frac{2M}{r^3} \left( 1 - \frac{a}{b_{sc}} \right)^2 \right] \bigg|_{r=r_{sc}} = 0 \\
 &\Rightarrow - \left( 1 - \frac{a^2}{b^2} \right) \frac{2}{r_{sc}^3} + \frac{6M}{r_{sc}^4} \left( 1 - \frac{a}{b_{sc}} \right)^2 = 0 \\
 &\Rightarrow r_{sc} \left( 1 - \frac{a^2}{b^2} \right) = 3M \left( 1 - \frac{a}{b_{sc}} \right)^2 \\
 &\Rightarrow r_{sc} = 3M \frac{\left( 1 - \frac{a}{b_{sc}} \right)}{\left( 1 + \frac{a}{b_{sc}} \right)} = 3M \frac{b_{sc} - a}{b_{sc} + a} \tag{3.29}
 \end{aligned}$$

Comparing eq. (3.29) with eq. (3.20) we can say that  $r_0 = r_{sc}$  &  $b_s = b_{sc}$  the variable  $P = 3M$ . Now, when the spin of the black hole is non-existent, *i.e.*,  $a = 0$  we have  $r_{sc} \rightarrow r_c = 3M$ . Since we are concerned with the Kerr metric currently we note that for the prograde case the critical value for closest approach is  $r_0 < 3M$  while for the retrograde case it is  $r_0 > 3M$ .

Going back to eq. (3.19) and comparing the coefficients of  $u$ ,

$$\begin{aligned}
 &u_1 u_2 + u_2 u_3 + u_1 u_3 = 0 \\
 &\Rightarrow \frac{P - 2M - Q}{4Mr_0} \frac{1}{r_0} + \frac{P - 2M - Q}{4Mr_0} \frac{P - 2M + Q}{4Mr_0} + \frac{P - 2M + Q}{4Mr_0} \frac{1}{r_0} = 0 \\
 &\Rightarrow P - 2M - Q + P - 2M + Q + \frac{(P - 2M - Q)(P - 2M + Q)}{4M} = 0 \\
 &\Rightarrow 8PM - 16M^2 + P^2 + 4M^2 - 4PM - Q^2 = 0 \\
 &\Rightarrow Q^2 = (P + 6M)(P - 2M) \tag{3.30}
 \end{aligned}$$

And finally with our last comparison we have,

$$\begin{aligned}
 &-\frac{P - 2M - Q}{4Mr_0} \frac{1}{r_0} \frac{P - 2M + Q}{4Mr_0} = \frac{1}{2M} \frac{1}{b^2 \left( 1 - \frac{a}{b_s} \right)^2} \\
 &\Rightarrow \frac{Q^2 - (P - 2M)^2}{8Mr_0^3} = \frac{1}{b^2 \left( 1 - \frac{a}{b_s} \right)^2} \tag{3.31}
 \end{aligned}$$

The remaining critical equations involving the key terms may be derived by substituting the expression for  $r_{sc}$  from eq. (3.29) in the equation  $\frac{\dot{r}^2}{J_\phi^2} = 0$ ,

$$\begin{aligned}
 &\frac{\dot{r}^2}{J_\phi^2} \bigg|_{r=r_{sc}} = 0 \\
 &\Rightarrow \frac{1}{b^2} - \left[ \frac{1}{r_{sc}^2} \left( 1 - \frac{a^2}{b^2} \right) - \frac{2M}{r_{sc}^3} \left( 1 - \frac{a}{b_{sc}} \right)^2 \right] = 0
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow \frac{r_{\text{sc}}^3}{b_{\text{sc}}^2} &= r_{\text{sc}} \left(1 - \frac{a^2}{b^2}\right) - 2M \left(1 - \frac{a}{b_{\text{sc}}}\right)^2 \\
 \Rightarrow \frac{r_{\text{sc}}^3}{b_{\text{sc}}^2} &= 3M \frac{1 - \frac{a}{b_{\text{sc}}}}{1 + \frac{a}{b_{\text{sc}}}} \left(1 - \frac{a}{b_{\text{sc}}}\right) \left(1 + \frac{a}{b_{\text{sc}}}\right) - 2M \left(1 - \frac{a}{b_{\text{sc}}}\right)^2 \\
 \Rightarrow \frac{r_{\text{sc}}^3}{b_{\text{sc}}^2} &= 3M \left(1 - \frac{a}{b_{\text{sc}}}\right)^2 - 2M \left(1 - \frac{a}{b_{\text{sc}}}\right)^2 = M \left(1 - \frac{a}{b_{\text{sc}}}\right)^2 \\
 \Rightarrow b_{\text{sc}}^2 &= \frac{r_{\text{sc}}^3}{M \left(1 - \frac{a}{b_{\text{sc}}}\right)^2} = \frac{27M^3 \left(1 - \frac{a}{b_{\text{sc}}}\right)^3}{M \left(1 + \frac{a}{b_{\text{sc}}}\right)^3 \left(1 - \frac{a}{b_{\text{sc}}}\right)^2} = \frac{27M^2 \left(1 - \frac{a}{b_{\text{sc}}}\right)}{\left(1 + \frac{a}{b_{\text{sc}}}\right)^3} = \frac{27M^2 (b_{\text{sc}} - a) b_{\text{sc}}^3}{(b_{\text{sc}} + a)^3 b_{\text{sc}}} \\
 \Rightarrow (b_{\text{sc}} + a)^3 &= 27M^2 (b_{\text{sc}} - a)
 \end{aligned} \tag{3.32}$$

Rewriting eq. (3.29) as  $M^2 = \frac{r_{\text{sc}}^2 (b_{\text{sc}} + a)^2}{9 (b_{\text{sc}} - a)^2}$  with eq. (3.32) we get,

$$\begin{aligned}
 (b_{\text{sc}} + a)^3 &= 27 \times \frac{r_{\text{sc}}^2 (b_{\text{sc}} + a)^2}{9 (b_{\text{sc}} - a)^2} \times (b_{\text{sc}} - a) \\
 \Rightarrow (b_{\text{sc}} + a) &= 3r_{\text{sc}}^2 \frac{1}{(b_{\text{sc}} - a)} \\
 \Rightarrow b_{\text{sc}}^2 &= 3r_{\text{sc}}^2 + a^2
 \end{aligned} \tag{3.33}$$

In order to solve eq. (3.32) for both the prograde( $s = +1$ ) and retrograde( $s = -1$ ) cases we need to treat them separately. For the prograde case we denote the associated impact parameter with  $sb_{+c}$  whereas for the retrograde case we denote it with  $sb_{-c}$ .

**For prograde orbits**—  $a > 0, s = +1$ . The cubic equation[eq. (3.32)] can be expressed in the form

$$(sb_{+c} + a)^3 - 27M^2 (sb_{+c} + a) + 54M^2 a = 0 \tag{3.34}$$

which is a cubic equation of  $(sb_{+c} + a)$ . Comparing this equation with eq. (3.25) we find,

$$\begin{aligned}
 \alpha &= \frac{3a_1 a_3 - a_2^2}{9a_1^2} = \frac{3 \times (-27M^2)}{9} = -9M^2; \\
 \beta &= \frac{9a_1 a_2 a_3 - 27a_1^2 a_4 - 2a_2^3}{54a_1^3} = \frac{-27 \times (54M^2 a)}{54} = -27M^2 a; \\
 \theta &= \cos^{-1} \left( \frac{\beta}{\sqrt{-\alpha^3}} \right) = \cos^{-1} \left( \frac{-27M^2 a}{\sqrt{9^3 M^6}} \right) = \cos^{-1} \left( -\frac{a}{M} \right)
 \end{aligned}$$

Therefore, the root of eq. (3.34) will be,

$$sb_{+c} + a = 2\sqrt{-\alpha} \cos \left( \frac{1}{3}\theta \right) - \frac{1}{3}a_2 = 2\sqrt{9M^2} \cos \left[ \frac{1}{3} \cos^{-1} \left( -\frac{a}{M} \right) \right] = 6M \cos \left[ \frac{1}{3} \cos^{-1} \left( -\frac{a}{M} \right) \right]$$

Since, for the prograde case  $s = +1$ , we have

$$b_{+c} = -a + 6M \cos \left[ \frac{1}{3} \cos^{-1} \left( -\frac{a}{M} \right) \right] \tag{3.35}$$

**For retrograde orbits**–  $a < 0$ ,  $s = -1$ . The cubic equation[eq. (3.32)] can be expressed in the form

$$(sb_{-c} - a)^3 - 27M^2 (sb_{-c} - a) - 54M^2 a = 0 \quad (3.36)$$

which is a cubic equation of  $(sb_{-c} - a)$ . Comparing this equation with eq. (3.25) we find,

$$\begin{aligned} \alpha &= \frac{3a_1a_3 - a_2^2}{9a_1^2} = \frac{3 \times (-27M^2)}{9} = -9M^2; \\ \beta &= \frac{9a_1a_2a_3 - 27a_1^2a_4 - 2a_2^3}{54a_1^3} = \frac{-27 \times (-54M^2a)}{54} = 27M^2a; \\ \theta &= \cos^{-1} \left( \frac{\beta}{\sqrt{-\alpha^3}} \right) = \cos^{-1} \left( \frac{27M^2a}{\sqrt{9^3 M^6}} \right) = \cos^{-1} \left( \frac{a}{M} \right) \end{aligned}$$

Therefore, the root of eq. (3.36) will be,

$$sb_{-c} - a = 2\sqrt{-\alpha} \cos \left( \frac{1}{3}\theta \right) - \frac{1}{3}a_2 = 2\sqrt{9M^2} \cos \left[ \frac{1}{3} \cos^{-1} \left( \frac{a}{M} \right) \right] = 6M \cos \left[ \frac{1}{3} \cos^{-1} \left( \frac{a}{M} \right) \right]$$

Since, for the retrograde case  $s = -1$ , we have

$$\begin{aligned} -b_{-c} &= a + 6M \cos \left[ \frac{1}{3} \cos^{-1} \left( \frac{a}{M} \right) \right] \\ \Rightarrow b_{-c} &= -a - 6M \cos \left[ \frac{1}{3} \cos^{-1} \left( \frac{a}{M} \right) \right] \end{aligned} \quad (3.37)$$

We can now combine the two solutions expressed in eq. (3.35) & eq. (3.37) to find the general solution for  $b_{sc}$

$$b_{sc} = -a + s6M \cos \left[ \frac{1}{3} \cos^{-1} \left( -s \frac{a}{M} \right) \right] \quad (3.38)$$

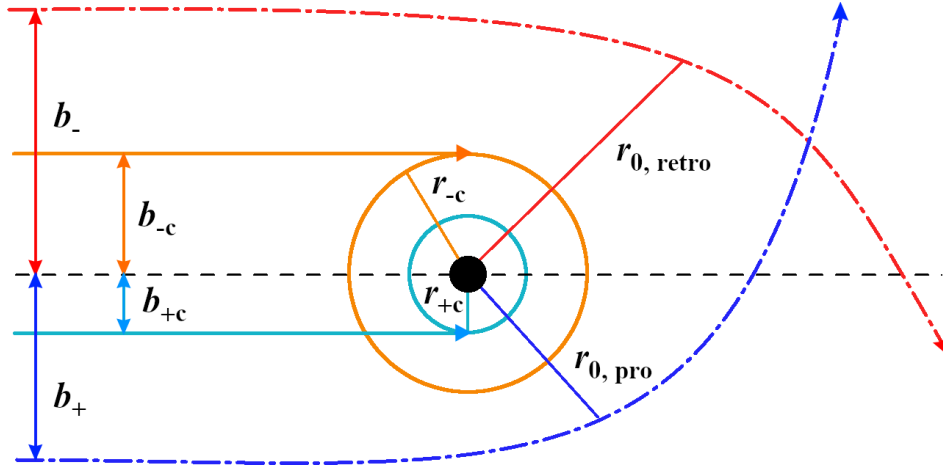
In order to determine  $r_{sc}$  we first restructure eq. (3.32) and substitute  $b_{sc}$  from eq. (3.38).

$$\begin{aligned} (b_{sc} + a)^3 &= 27M^2 (b_{sc} - a) \\ \Rightarrow \frac{b_{sc} - a}{b_{sc} + a} &= \frac{(b_{sc} + a)^2}{27M^2} = \frac{1}{27M^2} \left[ s6M \cos \left\{ \frac{1}{3} \cos^{-1} \left( -s \frac{a}{M} \right) \right\} \right]^2 \\ \Rightarrow \frac{b_{sc} - a}{b_{sc} + a} &= \frac{36}{27} \cos^2 \left\{ \frac{1}{3} \cos^{-1} \left( -s \frac{a}{M} \right) \right\} \end{aligned} \quad (3.39)$$

Finally we can substitute this in eq. (3.29) to find the expression for  $r_{sc}$ ,

$$\begin{aligned} r_{sc} &= 3M \frac{b_{sc} - a}{b_{sc} + a} = 3M \times \frac{36}{27} \cos^2 \left\{ \frac{1}{3} \cos^{-1} \left( -s \frac{a}{M} \right) \right\} = 2M \left[ 2 \cos^2 \left\{ \frac{1}{3} \cos^{-1} \left( -s \frac{a}{M} \right) \right\} \right] \\ \Rightarrow r_{sc} &= 2M \left[ 1 + \cos \left\{ \frac{2}{3} \cos^{-1} \left( -s \frac{a}{M} \right) \right\} \right] \end{aligned} \quad (3.40)$$

Substituting,  $s = -1$  &  $+1$  in eq. (3.38) & eq. (3.40) we can derive the expressions for  $b_{+c}$ ,  $b_{-c}$ ,  $r_{+c}$ ,  $r_{-c}$  respectively. The pictorial representation provided in fig. [3.2] presents a detailed depiction of these quantities for both the prograde and retrograde orbits under the influence of a spinning black hole.



**Figure 3.2:** Overhead view of the prograde and retrograde orbits (for the equatorial plane) around a Kerr black hole with a detailed depiction of the critical parameters like radii of innermost circular orbits, distances of closest approach and impact parameters for these corresponding cases. The spin orientation is the same as in [fig. \[3.1\]](#). The dashed-dotted line indicates the trajectory of closest approach, red for retrograde case and blue for prograde case. The solid line indicates the innermost circular orbit, orange for retrograde case and cyan for prograde case.

This is in good agreement with the results obtained in [Chandrasekhar \(1998\)](#)(equation 87, pg-330). The key factor that separates the approach of [Iyer and Hansen \(2009\)](#) is that the information regarding prograde and retrograde orbits is inbuilt in the expression by default. As such this makes for an extremely potent tool in studying the phenomenon of gravitational lensing in the vicinity of a rotating black hole. It is also instructive to study the plots of  $\frac{b_{sc}}{M}$  vs  $\frac{a}{M}$  ([fig. \[3.3\]](#)) and  $\frac{r_{sc}}{M}$  vs  $\frac{a}{M}$  ([fig. \[3.4\]](#)) for three distinct cases—(a) Schwarzschild, (b) Kerr prograde and (c) Kerr retrograde.

Let us now return to [eq. \(3.17\)](#) and examine the remaining polynomials. Starting off with the quadratic polynomial  $\{1 - 2Mu + a^2u^2\}$  and denoting its roots as  $u_p$  &  $u_n$  we have,

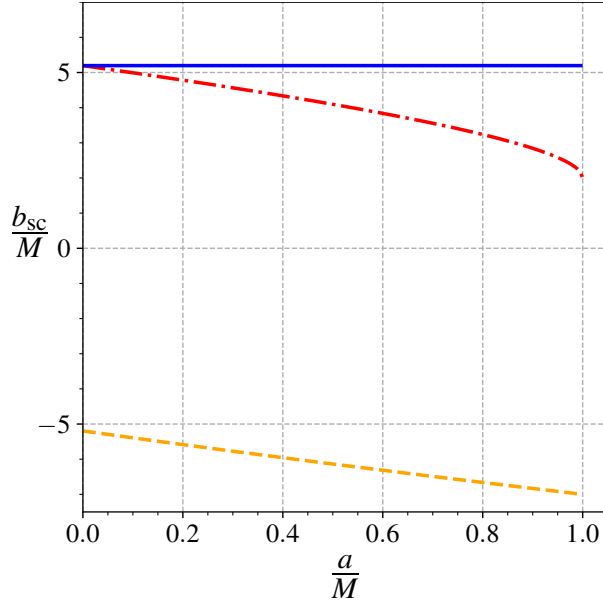
$$u_p = \frac{M + \sqrt{M^2 - a^2}}{a^2}; u_n = \frac{M - \sqrt{M^2 - a^2}}{a^2} \quad (3.41)$$

In order to facilitate the integration in [eq. \(3.17\)](#) we need to express the ratio of the two remaining polynomials as a combination of two partial fractions.

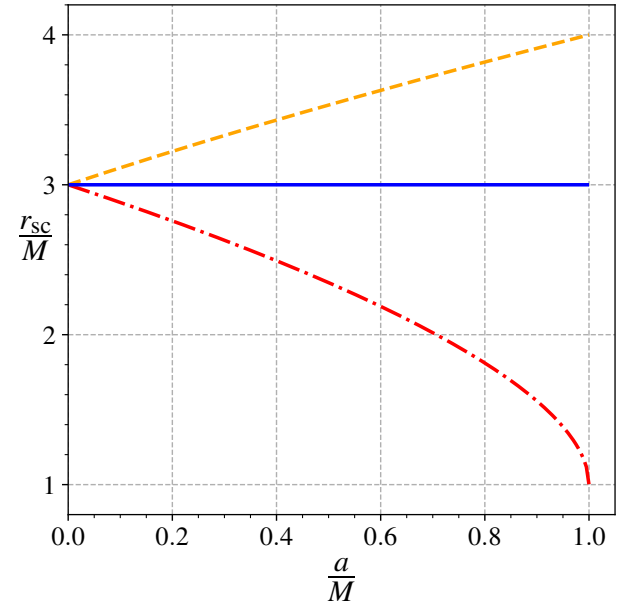
$$\frac{1 - 2Mu(1 - \omega_s)}{1 - 2Mu + a^2u^2} = \frac{C_p}{u_p - u} + \frac{C_n}{u_n - u} \quad (3.42)$$

Here  $\omega_s = \frac{a}{b_s}$ . Simplifying the sum of partial fractions we have,

$$\frac{C_p}{u_p - u} + \frac{C_n}{u_n - u} = \frac{C_p(u_n - u) + C_n(u_p - u)}{(u_p - u)(u_n - u)} = \frac{(C_p u_n + C_n u_p) - (C_p + C_n)u}{u^2 - (u_p + u_n)u + u_p u_n} \quad (3.43)$$



**Figure 3.3:** Plotting the critical impact parameter  $\frac{b_{sc}}{M}$  as a function of the spin parameter  $\frac{a}{M}$ . The lines are represented as follows:- Schwarzschild - Blue solid line, Kerr prograde - Red dash-dot line, Kerr retrograde - Orange dashed line



**Figure 3.4:** Plotting the critical radius  $\frac{r_{sc}}{M}$  as a function of the spin parameter  $\frac{a}{M}$ . The lines are represented as follows:- Schwarzschild - Blue solid line, Kerr prograde - Red dash-dot line, Kerr retrograde - Orange dashed line

Similarly, the ratio of the polynomials can be expressed as,

$$\frac{1 - 2Mu(1 - \omega_s)}{1 - 2Mu + a^2u^2} = \frac{1 - 2Mu(1 - \omega_s)}{a^2 \left( u^2 - \frac{2M}{a^2}u + \frac{1}{a^2} \right)} = \frac{\frac{1}{a^2} - \frac{2M}{a^2}(1 - \omega_s)u}{u^2 - \frac{2M}{a^2}u + \frac{1}{a^2}} \quad (3.44)$$

Comparing eq. (3.43) & eq. (3.44) we get,

$$C_p + C_n = \frac{2M}{a^2} (1 - \omega_s) \quad (3.45)$$

$$C_p u_n + C_n u_p = \frac{1}{a^2} \quad (3.46)$$

Solving eq. (3.45) & eq. (3.46) we find, the relevant expressions for  $C_p$  &  $C_n$ ,

$$C_n = \frac{a^2 - 2M(1 - \omega_s)(M - \sqrt{M^2 - a^2})}{2a^2\sqrt{M^2 - a^2}} \quad (3.47)$$

$$C_p = \frac{2M(1 - \omega_s)(M + \sqrt{M^2 - a^2}) - a^2}{2a^2\sqrt{M^2 - a^2}} \quad (3.48)$$

Let us now rewrite [eq. \(3.17\)](#) in terms of  $u_1, u_2, u_3, u_p, u_n, C_p, C_n$ .

$$\begin{aligned}
 \hat{\alpha}_{\text{Kerr}} &= -\pi + 2 \int_0^{\frac{1}{r_0}} \frac{\left\{ 1 - 2Mu \left( 1 - \frac{a}{b_s} \right) \right\}}{\{ 1 - 2Mu + a^2 u^2 \} \left\{ 2M \left( 1 - \frac{a}{b_s} \right)^2 u^3 - \left( 1 - \frac{a^2}{b^2} \right) u^2 + \frac{1}{b^2} \right\}^{\frac{1}{2}}} du \\
 \Rightarrow \hat{\alpha}_{\text{Kerr}} &= -\pi + 2 \int_0^{\frac{1}{r_0}} \left[ \frac{C_p}{u_p - u} + \frac{C_n}{u_n - u} \right] \left[ \frac{1}{\sqrt{B(u)}} \right] du \\
 \Rightarrow \hat{\alpha}_{\text{Kerr}} &= -\pi + 2 \int_0^{\frac{1}{r_0}} \left[ \frac{C_p}{(u_p - u) \sqrt{B(u)}} + \frac{C_n}{(u_n - u) \sqrt{B(u)}} \right] du \\
 \Rightarrow \hat{\alpha}_{\text{Kerr}} &= -\pi + \frac{2}{\sqrt{2M}(1 - \omega_s)} \int_0^{\frac{1}{r_0}} \frac{C_p}{(u_p - u) \sqrt{(u - u_1)(u - u_2)(u - u_3)}} du \\
 &\quad + \frac{2}{\sqrt{2M}(1 - \omega_s)} \int_0^{\frac{1}{r_0}} \frac{C_n}{(u_n - u) \sqrt{(u - u_1)(u - u_2)(u - u_3)}} du \tag{3.49}
 \end{aligned}$$

In order to account for the two different phases of the photon trajectory we separate the original limits of the integral  $\int_0^{u_2}$  into  $\int_0^{u_1} + \int_{u_1}^{u_2} \rightarrow -\int_{u_1}^0 + \int_{u_1}^{u_2}$  where we use  $u_2 = \frac{1}{r_0}$  [from [eq. \(3.18\)](#)]. This allows us to express [eq. \(3.49\)](#) as

$$\begin{aligned}
 \hat{\alpha}_{\text{Kerr}} &= -\pi + \sqrt{\frac{2}{M}} \frac{C_p}{1 - \omega_s} \left[ \int_{u_1}^{u_2} \frac{du}{(u_p - u) \sqrt{(u_3 - u)(u_2 - u)(u - u_1)}} \right. \\
 &\quad \left. - \int_{u_1}^0 \frac{du}{(u_p - u) \sqrt{(u_3 - u)(u_2 - u)(u - u_1)}} \right] \\
 &\quad + \sqrt{\frac{2}{M}} \frac{C_n}{1 - \omega_s} \left[ \int_{u_1}^{u_2} \frac{du}{(u_n - u) \sqrt{(u_3 - u)(u_2 - u)(u - u_1)}} \right. \\
 &\quad \left. - \int_{u_1}^0 \frac{du}{(u_n - u) \sqrt{(u_3 - u)(u_2 - u)(u - u_1)}} \right] \tag{3.50}
 \end{aligned}$$

The integrals expressed in [eq. \(3.50\)](#) are of the form as that of §233.02 in [Gradshteyn and Ryzhik \(2007\)](#). It is worth mentioning in this context that [Byrd and Friedman \(1971\)](#) is also an indispensable resource for complicated integrals. The general form of the integrals under consideration is

$$\int_c^T \frac{dx}{(r - x) \sqrt{(a - x)(b - x)(x - c)}} = \frac{2}{(r - c) \sqrt{a - c}} \Pi(\gamma, \zeta, q)$$

$$[\text{when } a > b \geq T > c, r \neq c] \quad (3.51)$$

where,

$$\gamma = \sin^{-1} \sqrt{\frac{T-c}{b-c}}; \quad q = \sqrt{\frac{b-c}{a-c}}; \quad \zeta = \frac{b-c}{r-c} \quad (3.52)$$

In the case of the integrals with the limits  $\int_{u_1}^{u_2}$  we have,

$$\begin{aligned} \gamma &= \sin^{-1} \sqrt{\frac{T-c}{b-c}} = \sin^{-1} \sqrt{\frac{u_2-u_1}{u_2-u_1}} = \sin^{-1}(1) = \frac{\pi}{2}; \\ q &= \sqrt{\frac{b-c}{a-c}} = \sqrt{\frac{u_2-u_1}{u_3-u_1}} = \sqrt{\frac{Q+6M-P}{2Q}}; \\ \zeta_{p,n} &= \frac{b-c}{r-c} = \frac{u_2-u_1}{u_{p,n}-u_1} = \frac{Q+6M-P}{4Mr_0 \left( \frac{M \pm \sqrt{M^2-a^2}}{a^2} \right) - P + 2M + Q} \end{aligned}$$

Whereas, for the integrals with the limits  $\int_{u_1}^0$  we have,

$$\gamma = \sin^{-1} \sqrt{\frac{T-c}{b-c}} = \sin^{-1} \sqrt{\frac{0-u_1}{u_2-u_1}} = \sin^{-1} \sqrt{\frac{Q+2M-P}{Q+6M-P}}$$

The terms  $\zeta_{p,n}$  &  $q$  remain unchanged, irrespective of the limits of integration.

Finally the coefficients of the integrals combine together as follows,

$$\begin{aligned} &\frac{2}{(r-c)\sqrt{a-c}} \times \sqrt{\frac{2}{M} \frac{C_{p,n}}{1-\omega_s}} = \frac{2}{(u_{p,n}-u_1)\sqrt{u_3-u_1}} \sqrt{\frac{2}{M} \frac{C_{p,n}}{1-\omega_s}} \\ &= \frac{2}{(u_{p,n}-u_1) \sqrt{\frac{P-2M+Q}{4Mr_0} - \frac{P-2M-Q}{4Mr_0}}} \sqrt{\frac{2}{M} \frac{C_{p,n}}{1-\omega_s}} = \frac{2\sqrt{2}}{(u_{p,n}-u_1) \sqrt{M} \sqrt{\frac{Q}{2Mr_0}}} \frac{C_{p,n}}{1-\omega_s} \\ &= \frac{4}{1-\omega_s} \sqrt{\frac{r_0}{Q}} \frac{C_{p,n}}{u_{p,n}-u_1} = \frac{4}{1-\omega_s} \sqrt{\frac{r_0}{Q}} \Omega_{p,n} \end{aligned} \quad (3.53)$$

where  $\Omega_{p,n} \equiv \frac{C_{p,n}}{u_{p,n}-u_1}$ . Finally, [eq. \(3.50\)](#) has the solution,

$$\hat{\alpha}_{\text{Kerr}} = -\pi + \frac{4}{1-\omega_s} \sqrt{\frac{r_0}{Q}} \left[ \Omega_p \left\{ \Pi\left(\frac{\pi}{2}, \zeta_p, q\right) - \Pi(\gamma, \zeta_p, q) \right\} + \Omega_n \left\{ \Pi\left(\frac{\pi}{2}, \zeta_n, q\right) - \Pi(\gamma, \zeta_n, q) \right\} \right] \quad (3.54)$$

The term  $q$  in the elliptic integral is defined in the domain  $0 \leq q^2 \leq 1$ . In [eq. \(3.54\)](#) the term  $\Pi\left(\frac{\pi}{2}, \zeta_{p,n}, q\right)$  is referred to as the complete elliptical integral of the third kind whereas the term  $\Pi(\gamma, \zeta_{p,n}, q)$  is identified as incomplete elliptical integral of the third kind. For the Schwarzschild case *i.e.*, when  $a = 0$ , we have

(with the help of some straightforward substitutions)  $\Omega_p = 1$ ,  $\Omega_n = 0$  &  $\zeta_p = 0$  which allows us to reduce eq. (3.54) to the form

$$\hat{\alpha} = -\pi + 4\sqrt{\frac{r_0}{Q}} \left[ \Pi\left(\frac{\pi}{2}, 0, q\right) - (\gamma, 0, q) \right] \quad (3.55)$$

where the terms within parentheses are the complete and incomplete integrals of the first kind. This is in agreement with the results derived in [Iyer and Petters \(2007\)](#) for the Schwarzschild case.

Finally, in order to have a functional expression of bending angle of a photon trajectory in the vicinity of a Kerr black hole we need to express eq. (3.54) in terms of some convenient notation. We start off with the terms,

$$h = \frac{M}{r_0}; \quad \omega_s = \frac{a}{b_s}; \quad \omega_0 = \frac{a^2}{M^2}; \quad h_{sc} = \frac{1 + \omega_s}{1 - \omega_s}; \quad r_{sc} = \frac{3M}{h_{sc}}$$

where the sign associated with  $\omega_s$  determines the nature of the orbit being prograde or retrograde. It is instructive to note that for the zero spin case, *i.e.*,  $a = 0$  we have  $[\omega_s, \omega_0 \rightarrow 0, h_{sc} \rightarrow 1]$  and our expression reduces to the Schwarzschild case. Furthermore, in the absence of any mass *i.e.*,  $M = 0$  we have  $h \rightarrow 0$  and our expression reduces to the scenario for a flat spacetime which means zero deflection of the light ray. Also for the distance of closest approach  $r_0 \rightarrow \infty$  we have  $h \rightarrow 1$  as expected. We also define the variables,

$$h' = 1 - \frac{3h}{h_{sc}} \equiv 1 - 3\frac{M}{r_0} \left( \frac{1 - \omega_s}{1 + \omega_s} \right) \quad \& \quad b' = 1 - \frac{sb_{sc}}{b}$$

which for the zero spin case reduces to  $h' \rightarrow 1 - \frac{3M}{r_0}$  which is the same as the Schwarzschild case as shown in [Iyer and Petters \(2007\)](#). The term  $b'$  is a dimensionless quantity which enables us to define an impact parameter that extends from the strong deflection limit to the weak deflection limit. The quantity  $s$  ensures that  $b'$  is confined within the domain  $0 \rightarrow 1$ . For  $a \rightarrow 0$  eq. (3.32) yields  $b_{sc} = 3\sqrt{3}M$  and we have  $b' = 1 - \frac{3\sqrt{3}M}{b}$ .

Now, we can express the intermediate variables in eq. (3.54),  $\frac{r_0}{Q}$ ,  $q$ ,  $\gamma$ ,  $\zeta_{p,n}$  &  $\Omega_{p,n}$  in terms of  $h$ ,  $h_{sc}$ ,  $\omega_0$  &  $\omega_s$  as shown below,

$$\frac{r_0}{Q} = \frac{1}{h_{sc} \sqrt{\left(1 - \frac{2h}{h_{sc}}\right) \left(1 + \frac{6h}{h_{sc}}\right)}} \quad (3.56)$$

where we use eq. (3.20) & eq. (3.30).

$$q = \sqrt{\frac{Q + 6M - P}{2Q}} = \frac{\sqrt{\left(1 - \frac{2h}{h_{sc}}\right) \left(1 + \frac{6h}{h_{sc}}\right) + \frac{6h}{h_{sc}} - 1}}{2\sqrt{\left(1 - \frac{2h}{h_{sc}}\right) \left(1 + \frac{6h}{h_{sc}}\right)}} \quad (3.57)$$

$$\gamma = \sin^{-1} \sqrt{\frac{Q + 2M - P}{Q + 6M - P}} = \sin^{-1} \sqrt{\frac{1 - \frac{2h}{h_{sc}} - \sqrt{\left(1 - \frac{2h}{h_{sc}}\right) \left(1 + \frac{6h}{h_{sc}}\right)}}{1 - \frac{6h}{h_{sc}} - \sqrt{\left(1 - \frac{2h}{h_{sc}}\right) \left(1 + \frac{6h}{h_{sc}}\right)}}} \quad (3.58)$$

$$\begin{aligned} \zeta_{p, n} &= \frac{Q + 6M - P}{4Mr_0 \left( \frac{M \pm \sqrt{M^2 - a^2}}{a^2} \right) - P + 2M + Q} \\ &= \frac{1 - \frac{6h}{h_{sc}} - \sqrt{\left(1 - \frac{2h}{h_{sc}}\right) \left(1 + \frac{6h}{h_{sc}}\right)}}{1 - \frac{2h}{h_{sc}} - \sqrt{\left(1 - \frac{2h}{h_{sc}}\right) \left(1 + \frac{6h}{h_{sc}}\right)} - \frac{4}{\omega_0 h_{sc}} (1 \pm \sqrt{1 - \omega_0})} \end{aligned} \quad (3.59)$$

$$\Omega_{p, n} = \frac{C_{p, n}}{u_{p, n} - u_1} = \frac{\pm (1 \pm \sqrt{1 - \omega_0}) (1 - \omega_s) \mp \frac{\omega_0}{2}}{\sqrt{1 - \omega_0} \left[ 1 \pm \sqrt{1 - \omega_0} - \frac{\omega_0 h_{sc}}{4} \left\{ 1 - \frac{2h}{h_{sc}} - \sqrt{\left(1 - \frac{2h}{h_{sc}}\right) \left(1 + \frac{6h}{h_{sc}}\right)} \right\} \right]} \quad (3.60)$$

Substituting eq. (3.56), eq. (3.57), eq. (3.58), eq. (3.59) & eq. (3.60) in eq. (3.54) we arrive at the desired solution for the bending angle. Out of the quantities  $r_0$ ,  $h$ ,  $h_{sc}$ ,  $\omega_0$  and  $\omega_s$  only  $\omega_0$  is independent of the impact parameter  $b$ . As mentioned earlier the quantity  $s$  assumes a  $-ve$  sign for retrograde orbits. A striking quality of the photon orbits around spinning black holes is that the frame dragging effect influences the photon trajectories differently depending on whether the spin is aligned along or against the direction of the photon. The degree of deflection from the initial trajectory is significantly larger for prograde orbits than for retrograde orbits. As a consequence of this with increase in  $a$  the relativistic images [Virbhadra and Ellis (2000), Virbhadra (2009), Virbhadra, Narasimha, and Chitre (1998)] gradually shift “outwards” for the retrograde case and “inwards” for the prograde case Iyer and Hansen (2009). Investigations regarding higher order relativistic images may be carried out by equating the expression for deflection angle  $\hat{\alpha} = 2n\pi$  and substituting it in the lens equation [Schneider, Ehlers, and Falco (1992), Wambsganss (1998)]. The analytical solution derived thus provide a critical tool in the study of gravitational lensing.

*Fairy tales do not give the child his first idea of bogey. What fairy tales give the child is his first clear idea of the possible defeat of bogey. The baby has known the dragon intimately ever since he had an imagination. What the fairy tale provides for him is a St. George to kill the dragon.*

—G. K. Chesterton

# 4

## BENDING ANGLE OF LIGHT AND THE GAUSS-BONNET THEOREM

A general tendency in the literature while defining the bending angle of particle trajectories has been to assume the source to be located at infinity. However, in reality the lens objects responsible for the deviation in particle trajectories are very much within a finite range from the source. In fact for cases of non-asymptotically flat spacetime there exist strong caveats against considering an infinite distance between the lensing object and the source. As such a more precise and accurate definition of the bending angle is required which takes into account modifications resulting from corrections due to finite-distance between the lens object and the source. [Gibbons and Werner \(2008\)](#) have employed the Gauss-Bonnet theorem to the so-called optical metric (alternatively Fermat metric) of the lensing object for the spherically symmetric cases to determine the deflection of a light ray. In this work they considered two domains defined as—

- $\mathcal{D}_1$  — The domain enclosed by two light rays with the lensing object at the centre. This basically establishes the connection between multiple images and the topology of the spacetime.
- $\mathcal{D}_2$  — The domain enclosed on one end by a non-geodesic circular arc and by a light ray on the other end. This serves the purpose of determining the deflection angle in the purely asymptotic case.

The deflection angle in the asymptotic case was defined by them as the surface integral of the Gaussian curvature over the domain  $\mathcal{D}_2$ . However, the observer and source were considered to be in the asymptotically Euclidean region during the computation of deflection angle. Following this approach [Ishihara et al. \(2016\)](#) & [Ono et al. \(2017\)](#) extended the treatment to include the finite distance corrections to compute the bending angle for the non asymptotically flat spacetimes. The regimen prescribed by [Ono et al. \(2017\)](#) for calculating the bending angle can be outlined as follows—

- (a) Expressing the given metric in the Rander-Finsler form to extract the Fermat metric which is essentially a three dimensional space describing the particle trajectory.
- (b) The particle trajectory on the equatorial plane is defined by extracting the Riemannian part of the Randers-Finsler metric and isolating the equatorial part. This essentially defines the two-dimensional Riemannian manifold describing the trajectory of the particle.
- (c) Defining the integral region on the two dimensional Riemannian manifold. The integral region is enclosed by the following curves
  - (i) the trajectory of the particle
  - (ii) a radial outward curve passing through the lensing object and the source.
  - (iii) a radial outward curve passing through the lensing object and the receiver(observer).
  - (iv) an auxiliary circular arc constructed with the lensing object as the centre.
- (d) Applying the Gauss-Bonnet theorem to the integral region as defined above in order to get the deflection angle in terms of integrals of the Gaussian and geodesic curvature.
- (e) Establishing that the above expression is equivalent to an alternative definition involving only the jump angles and the azimuthal coordinate separation of the viewer and source.

Developing on the regimen described above [Ono et al. \(2017\)](#) have successfully constructed an alternative framework for evaluating the deflection angle where they circumvent the tedious computation of integrals involving the Gaussian and geodesic curvatures. Following their work a flurry of investigations [[Ishihara et al. \(2017\)](#), [Ono et al. \(2018\)](#), [Övgün \(2018\)](#), [Ono and Asada \(2019\)](#), [Övgün et al. \(2019\)](#)] involving their methodology was successfully applied to a number of other cases (in some cases extending to the strong deflection limit).

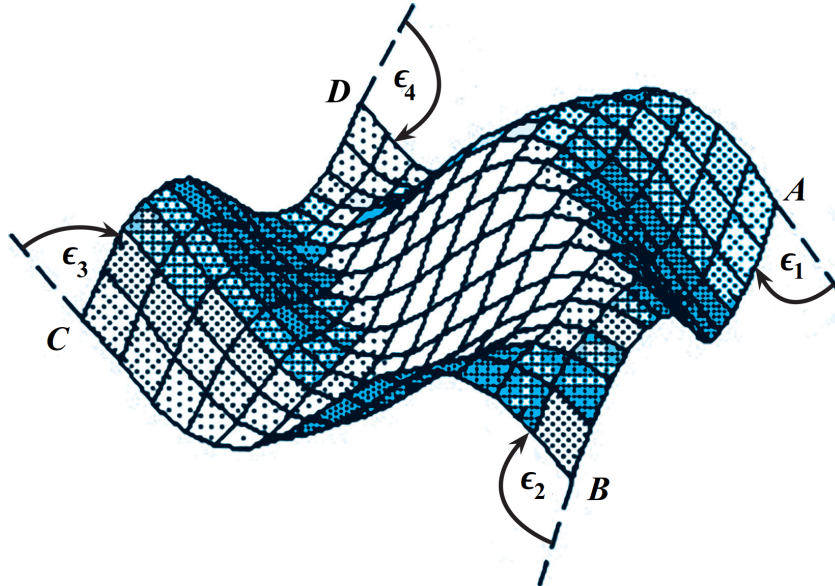
In the following sub-sections we present an elementary account of the Gauss-Bonnet theorem (to the extent required for the purpose of determining the deflection angle in a curved spacetime) followed by a detailed description of the *modus operandi* of [Ishihara et al. \(2016\)](#) & [Ono et al. \(2017\)](#) outlined above, as applied to the cases of both spherically symmetric and axis-symmetric spacetimes.

## 4.1 THE GAUSS-BONNET THEOREM

The inter-relationship between the topology of a surface to its intrinsic differential geometry is one of the many areas of interest while studying the motion of particles in General Relativity. The Gauss-Bonnet theorem [do Carmo (1976), Oprea (2019)] provides us with an extremely potent tool to serve this very purpose. In fact it is one of the most profound theorems of differential geometry that links local and global geometry. Let us consider there exists a patch of surface  $\mathcal{D}$  (fig. [4.1]) enclosed by a set of  $n$  boundary curves denoted by  $\partial\mathcal{D}_i$ . Now, if  $\epsilon_i$  denote the exterior angles formed where the boundary curves  $\partial\mathcal{D}_i$  meet and they have the geodesic curvature  $\kappa_g(s_i)$  with  $s_i$  representing a point on  $\partial\mathcal{D}_i$  then the mathematical formulation of the theorem reads

$$\iint_{\mathcal{D}} K dS + \sum_{i=1}^n \int_{\partial\mathcal{D}_i} \kappa_g d\ell + \sum_{i=1}^n \epsilon_i = 2\pi\chi(\mathcal{D}) \quad (4.1)$$

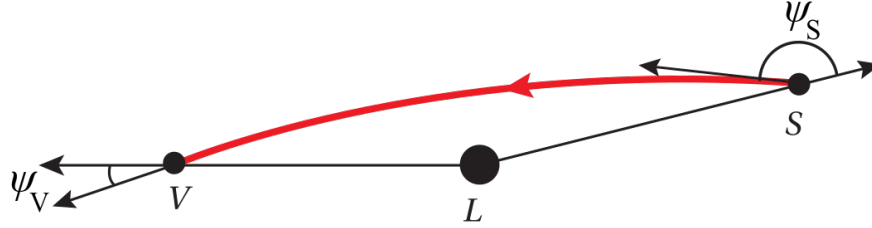
where  $K$ ,  $dS$  and  $d\ell$  are the Gaussian curvature, area element of  $\mathcal{D}$  and line element of  $\partial\mathcal{D}_i$  respectively. The term  $\chi(\mathcal{D})$  is called the Euler characteristic number of  $\mathcal{D}$ . As we are considering the motion of particles in the given spacetime the consequent manifold under consideration is a compact connected smooth surface the corresponding Euler characteristic is equal to 1. It should be noted that the sign of the line element  $\ell$  should be chosen such that its compatibility with the orientation of the surface is maintained.



**Figure 4.1:** Illustrating the Gauss-Bonnet theorem. A patch of surface created by the intersection of several curves thereby subtending the exterior angle  $\epsilon_i$  at each point of intersection.

## 4.2 GRAVITATIONAL LENS, PHOTON TRAJECTORY AND GEOMETRY

The geometrical depiction of the manifold as chalked out by the presence of a lensing object, source of light, viewer and the photon trajectory lies at the heart of the current topic. Let us, therefore, briefly spend some time describing the geometry of the said manifold that provides the integrable region as required by the Gauss-Bonnet theorem.



**Figure 4.2:** Two dimensional Riemannian manifold  ${}^2\mathcal{R}$  where  $LS$ ,  $LV$  &  $SV$  are the spatial projections of the geodesics in the equatorial plane.

Let us concentrate on [fig. \[4.2\]](#) which represents the triangle  ${}^V\nabla_L^S$  enclosed by the photon trajectory traced out from the source to the viewer after suffering deflection due to the presence of the lensing object  $L$  and the radial geodesics from  $L$  through the viewer( $V$ ) & source( $S$ ). It is this very space that will be described by the Fermat metric extracted from the metric defining the spacetime under consideration. It must be specified in this context that as we are interested in the trajectory of particles in the equatorial plane the [fig. \[4.2\]](#) actually depicts a two dimensional Riemannian manifold  ${}^2\mathcal{R}$  where  $LS$ ,  $LV$  &  $SV$  are the spatial projections of the geodesics in the equatorial plane. A critical point to note in this context is that photon trajectory as depicted in [fig. \[4.2\]](#) may or may not have a non-vanishing geodesic curvature depending on the original spacetime metric. However, this does not interfere with the fact that the photon trajectory in four-dimensional spacetime is represented by a null geodesic.

Let us now concentrate on the angles between the photon trajectory and the radial direction as well as the coordinate separation angle between the source & viewer. As can be clearly seen from [fig. \[4.2\]](#)  $\psi_V$  &  $\psi_S$  are the angles at the viewer and source positions respectively. Now, if  $\phi_V$  and  $\phi_S$  are taken to represent the longitudes of the viewer and source respectively then  $\phi_V - \phi_S = \phi_{VS}$ (say) gives the coordinate separation angle. Furthermore, let us define the internal angle subtended at the center *i.e.*, the lensing object  $L$  as  $\psi_L$ . Since we will be applying the Gauss-Bonnet theorem to this surface we would like to specify that  $\psi_V$  is the opposite angle of the interior angle at  $V$  (*both the interior angle and exterior angles are same in this case*) and  $\psi_S$  is the exterior angle at  $S$  ( $\pi - \psi_S$  is the interior angle). Let us assume that  $\theta_i$  ( $i = 1, 2, 3$ ) denote the interior angles of the triangle  ${}^V\nabla_L^S$ . This allows us to write the sum of the interior angles as

$$\begin{aligned}
 \sum_{i=1}^3 \theta_i &= \psi_V + \psi_L + (\pi - \psi_S) = \psi_V + \psi_L - \psi_S + \pi = \alpha_\psi + \pi \\
 \Rightarrow \alpha_\psi &= \sum_{i=1}^3 \theta_i - \pi
 \end{aligned} \tag{4.2}$$

where we have defined

$$\alpha_\psi = \psi_V + \psi_L - \psi_S \tag{4.3}$$

A very important observation in this context is that if  ${}^V\nabla_L^S$  is flat then obviously  $\alpha_\psi = 0$  on account of  $\sum_{i=1}^3 \theta_i = \pi$ . This automatically leads us to draw the inference that  $\alpha_\psi$  provides us with a measure of the deviation from the Euclidean space.

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### 4.3 MATHEMATICAL FORMULATION OF DEFLECTION ANGLE FROM THE GAUSS-BONNET THEOREM

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In our attempt to determine the deflection angle with the help of Gauss-Bonnet theorem the choice of an appropriate integrable region is of critical importance. As will become evident during the course of this subsection a suitable choice for our purposes is quite tricky. The two regions that lend themselves readily for the purposes of our investigation are the embedded triangle  ${}^V\nabla_L^S$  and the embedded quadrilateral  ${}^\infty\nabla_S^\infty$ .

#### 4.3.1 Gauss-Bonnet theorem in the region ${}^V\nabla_L^S$

Let us now apply the Gauss-Bonnet theorem to domain represented by the embedded triangle  ${}^V\nabla_L^S$  allowing us to write the [eq. \(4.1\)](#) as

$$\iint_{{}^V\nabla_L^S} K dS + \sum_{i=1}^3 \int_{\partial\mathcal{D}_i} \kappa_g d\ell + \sum_{i=1}^3 \epsilon_i = 2\pi\chi({}^V\nabla_L^S) \tag{4.4}$$

Now, since  $\epsilon_i$  and  $\theta_i$  represent the exterior and interior angles we have  $\sum_{i=1}^3 \epsilon_i + \sum_{i=1}^3 \theta_i = 3\pi$ . This coupled with [eq. \(4.2\)](#) allows us to rewrite [eq. \(4.4\)](#) as

$$\begin{aligned}
 \iint_{{}^V\nabla_L^S} K dS + \int_L^S \kappa_{g(LS)} d\ell + \int_S^V \kappa_{g(SV)} d\ell + \int_V^L \kappa_{g(VL)} d\ell + 2\pi - \alpha_\psi &= 2\pi\chi({}^V\nabla_L^S) \\
 \Rightarrow \alpha_\psi &= \iint_{{}^V\nabla_L^S} K dS + \int_L^S \kappa_{g(LS)} d\ell + \int_S^V \kappa_{g(SV)} d\ell + \int_V^L \kappa_{g(VL)} d\ell + 2\pi - 2\pi\chi({}^V\nabla_L^S)
 \end{aligned}$$

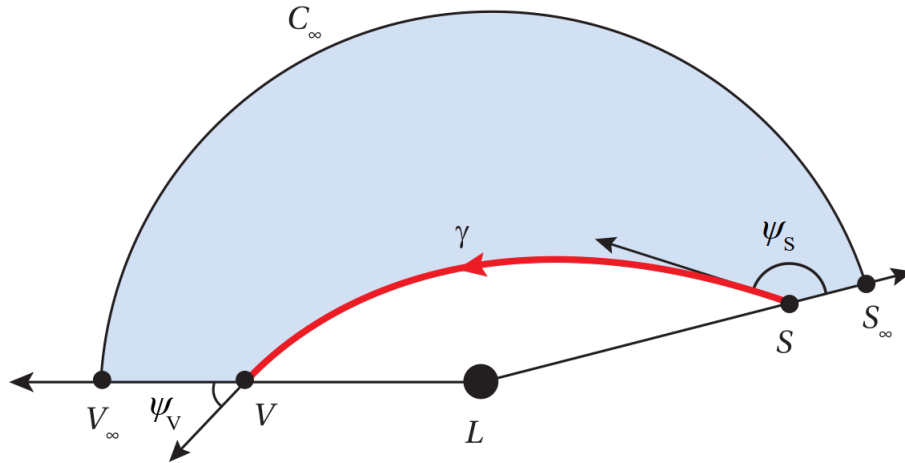
$$\Rightarrow \alpha_\psi = \iint_{{}^V\nabla_L^S} K dS + 0 + \int_S^V \kappa_{g(SV)} d\ell + 0 + 2\pi - 2\pi\chi({}^V\nabla_L^S) \quad (4.5)$$

The two geodesic curvatures along the radial geodesics **LS** & **VL** vanish. On the other hand the geodesic curvature of **SV** may or may not have a non-zero value.

However, there are two critical drawbacks in the formulation as it stands in eq. (4.5). Firstly,  $\psi_L$  included in  $\alpha_\psi$  by definition is the interior angle subtended at the lensing object **L**. Now, in studies associated with general relativity the lensing object is a black hole singularity more often than not. As such it becomes impossible to construct any relevant definition of  $\psi_L$ . Secondly, as **L** may be a singularity a portion of the surface patch of the embedded triangle  ${}^V\nabla_L^S$  that includes the black hole horizon is no longer smooth. An immediate consequence of this is that we cannot now assign the Euler characteristic  $\chi({}^V\nabla_L^S) = 1$  which prevents us from getting a workable expression for  $\alpha_\psi$  from eq. (4.5).

In order to avoid these shortcomings it becomes necessary to switch out the integrable region from  ${}^V\nabla_L^S$  to a more appropriate one where the manifold under consideration would qualify as a smooth uniform patch and we do not have to encounter  $\psi_L$ .

### 4.3.2 Gauss-Bonnet theorem in the region ${}^\infty\nabla_S^\infty$



**Figure 4.3:** Extracting the integrable surface region depicted as the quadrilateral  ${}^\infty\nabla_S^\infty$ . The said region is indicated by the portion shaded in cyan

First let us consider another embedded triangle fig. [4.3]  ${}^\infty\nabla_L^\infty$  which encloses the embedded triangle  ${}^V\nabla_L^S$ . The extremal boundary of the triangle  ${}^\infty\nabla_L^\infty$  is a circular arc of radius  $r_c$  with **L** at its centre and intersecting with the projections of the radial geodesics through the viewer and source. In fact these points of intersections might very well be taken as the location for the viewer and source located at infinity (for

$r_c \rightarrow \infty$ ). For the asymptotically flat spacetime, at  $r_c \rightarrow \infty$  the geodesic curvature of the auxiliary circular arc can be taken as  $\kappa_c \rightarrow \frac{1}{r_c}$  along with  $d\ell \rightarrow r_c d\phi$ <sup>1</sup>.

Let us now apply the Gauss-Bonnet theorem to the integrable region given by the embedded quadrilateral  ${}^\infty_V \square_S^\infty$ . The exterior angles at the vertices R,  $V_\infty$ ,  $S_\infty$  & S are  $\psi_V$ ,  $\frac{\pi}{2}$ ,  $\frac{\pi}{2}$  &  $\pi - \psi_S$  respectively. Also, geodesic curvatures associated with the radial geodesics will vanish of course. And since the embedded square  ${}^\infty_V \square_S^\infty = {}^\infty_V \nabla_L^\infty - {}^V \nabla_L^S$  we do not have to take into consideration whether the lensing object is a black hole singularity or not, thereby allowing us to write  $\chi({}^\infty_V \square_S^\infty) = 1$ .

$$\begin{aligned}
 & \iint_{{}^\infty_V \square_S^\infty} K dS + \sum_{i=1}^4 \int_{\partial \mathcal{D}_i} \kappa_g d\ell + \sum_{i=1}^4 \epsilon_i = 2\pi \chi({}^\infty_V \square_S^\infty) \\
 & \Rightarrow \iint_{{}^\infty_V \square_S^\infty} K dS + \int_S^{S_\infty} \kappa_g^{(SS_\infty)} d\ell + \int_{S_\infty}^{V_\infty} \kappa_g^{(S_\infty V_\infty)} d\ell + \int_{V_\infty}^V \kappa_g^{(V_\infty V)} d\ell + \int_V^S \kappa_g^{(VS)} d\ell \\
 & \quad + \epsilon_V + \epsilon_{V_\infty} + \epsilon_{S_\infty} + \epsilon_S = 2\pi \chi({}^\infty_V \square_S^\infty) \\
 & \Rightarrow \iint_{{}^\infty_V \square_S^\infty} K dS + 0 + \int_{\phi_S}^{\phi_V} \frac{1}{r_c} r_c d\phi + 0 + \int_V^S \kappa_g^{(VS)} d\ell + \psi_V + \frac{\pi}{2} + \frac{\pi}{2} + (\pi - \psi_S) = 2\pi \chi({}^\infty_V \square_S^\infty) \\
 & \Rightarrow \iint_{{}^\infty_V \square_S^\infty} K dS + \phi_{VS} + \int_V^S \kappa_g^{(VS)} d\ell + \psi_V - \psi_S = 0 \\
 & \therefore \alpha_D = - \iint_{{}^\infty_V \square_S^\infty} K dS + \int_S^V \kappa_g^{(VS)} d\ell \tag{4.6a} \\
 & = \psi_V - \psi_S + \phi_{VS} \tag{4.6b}
 \end{aligned}$$

Needless to say [eq. \(4.6a\)](#) & [eq. \(4.6b\)](#) actually depict the generic expression for the deflection angle  $\alpha_D$  of a photon. A quick check of the formula can be performed by testing it out in the asymptotically flat case, where the expression for deflection angle should take the same form as the standard form generally given in textbooks. In the conventional approach for deriving the expression of bending angle, as **S** & **V** are taken to be located at infinity we can safely assume that  $\psi_S = \pi$  and  $\psi_V = 0$ . This effectively reduces [eq. \(4.6b\)](#) to the form  $\alpha_D^{\text{AF}} = \phi_{VS} - \pi$  which is nothing but the traditional textbook definition of deflection angle. A more rigorous and detailed discussion on the justification for accepting [eq. \(4.6b\)](#) as a consistent definition of the bending angle maybe found in [Takizawa et al. \(2020\)](#).

However, the definition of bending angle involving the integrals of the Gaussian curvature and the geodesic curvature as expressed in [eq. \(4.6a\)](#) demands the assumption of asymptotic flatness ( $r \rightarrow \infty$ ).

<sup>1</sup>Since for  $r_c \rightarrow \infty$  the spacetime can be considered asymptotically flat we can express the metric describing the line element involved with the integral of geodesic curvature as  $d\ell^2 = dr^2 + r^2 (d\theta^2 + \sin^2 \theta d\phi^2)$ . For the equatorial plane  $\theta = \frac{\pi}{2}$  which allows us to express  $d\ell$  as  $r_c d\phi$

Takizawa et al. (2020), followed by Huang et al. (2024) have addressed this issue in detail. Huang et al. (2024) have in fact firmly established a generalized GW formalism of determining the deflection angle which does not require any form of asymptotic flatness.

Depending on the spacetime we are working with we can either use the form involving the integrals of the Gaussian curvature and geodesic curvature or the form involving the jump angles and coordinate angular separation between the source and observer. Although, in general the second form nearly always turns up complicated elliptic integrals. In spite of that a thorough consistency check between the results obtained by both the forms have been presented in Ono et al. (2017). Using the results as derived in this section we can now utilise them for computing the deflection angle for different types of spacetimes. Furthermore, following the developments in Huang et al. (2024) & Takahashi et al. (2023) it was firmly established that the GW, OIA & HC method are all equivalent and eq. (4.7) gives us a consistent definition of the deflection angle without requiring any special consideration of properties associated with the asymptotic limit of the spacetime.

$$\alpha_D = \psi_V - \psi_S + \phi_{VS} = \int_S^V \left( \frac{d\psi}{d\phi} + 1 \right) d\phi \quad (4.7)$$

One of the most critical points to note in this context is that provided the photon trajectory remains fixed the deflection angle as evaluated by the Gauss-Bonnet based approach is not dependent on the domains of integration.

In the following section we mainly devote ourselves to reviewing the development of mathematical techniques essential for determining the bending angle of light rays in—

- (a) Spherically symmetric spacetimes
- (b) Axisymmetric spacetimes

using the formalism developed here. In particular the determination of the angles  $\psi$  and  $\phi$  are the main objectives in chapter 5. On the other hand determination of the bending angle by evaluating integrals of  $K$  and  $\kappa_g$  forms the main subject matter of chapter 6.

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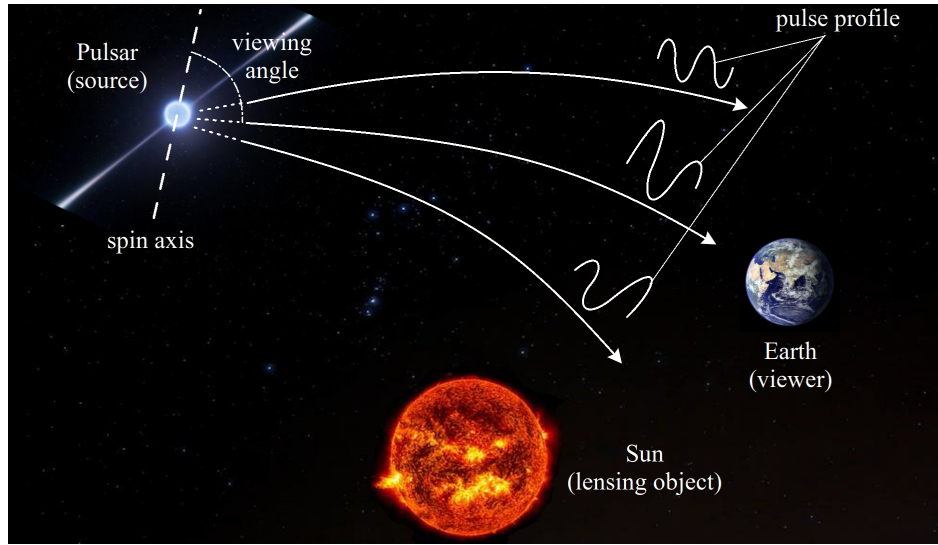
## 4.4 IMPLICATIONS FOR DIRECT ASTRONOMICAL OBSERVATIONS

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In terms of actual observations that can be carried out with the assistance of modern astronomical apparatus eq. (4.6b) is immensely relevant. Given that the location of the source and the observer are known the angles described in eq. (4.6b) can be easily determined in principle. Considering that the Very Long Baseline Interferometry(VLBI) is able to provide extremely precise radio observations the relative positions of a pulsar, the Earth and the lensing object Sun(say) can be determined from the ephemeris [Madison et al. (2013), Wang et al. (2017), Deller et al. (2019), Liu et al. (2023), Ding et al. (2024), Pearlman (2024)]. Since there

is a multitude of pulsars whose spin period, pulse signal, pulse profiles etc features are well documented they are ideal as potential source candidates for verifying weak lensing predictions [Walker (1996), Lai and Rafikov (2005), Hobill et al. (2008)]. As such we can consider an arbitrary pulsar emitting radio signals with a constant period in an anisotropic manner as a source. Given these considerations we can easily determine the following:—

- (a) The angle  $\psi_V$  at the Earth between the solar direction and the pulsar direction.
- (b) The angle  $\phi_{VS}$ .
- (c) In conjunction with information regarding the direction of pulse radiation and pulsar position the angle  $\psi_S$  can be easily determined. Since the viewing angle of the pulsar as seen by the viewer is accessible from the pulse profiles the direction of radiating pulses that reach the viewer can also be determined in principle. It is worth noting in this context that as a consequence of the Earth's motion around the Sun the viewing keeps continually changing with respect to time as illustrated in fig. [4.4].



**Figure 4.4:** The figure presents a scenario where we have the Sun as the lensing object, Earth as the location of the viewer and a pulsar as the source. The pulsar emits radio signals at periodic intervals in a specific anisotropic manner. The pulse profile associated with the emissions allows us to determine the radiation direction at the actual location of the pulsar. The relative positions of the Earth, Sun & Pulsar may be determined with the help of the ephemeris which in turn allows us to determine  $\phi_{VS}$  and  $\psi_S$ . The angle  $\psi_V$  can be ascertained by observing the pulsar as described in the text. This illustrates a scenario wherein the accuracy of eq. (4.6b) describing  $\alpha_D$  can be tested.

Determination of the angle  $\psi_S$  can be a very involved process and merits a somewhat more elaborate description. The pulse profile and shape are dependent on the viewing angle with respect to the spin axis of the pulsar. This provides us with a strong motivation for considering pulsars whose spin axis is known

to us from some astronomical observations. Furthermore, a significant issue in this context is that the spin axis of any isolated pulsar remains unchanged with respect to time. As the Earth keeps changing position due to its motion around the Sun the viewer records different viewing angles subtended by the pulsar with passage of time. As a result of this the corresponding pulse shape also keeps changing. Now by recording these changes in pulse shape the intrinsic direction of the radio emissions can be determined, at least in principle. This in turn allows us to determine the angle between the spin axis and the direction of the emitted light from the pulsar to the viewer. An added benefit of this approach is that we can glean information like the radial direction from the lensing object or the intrinsic position of any such known pulsar from the ephemeris. While from a technological point of view such measurements are still subject to handicaps the scientific principle behind measuring  $\psi_S$  is quite sound. Thus all the angles  $\psi_V$ ,  $\psi_S$  &  $\phi_{VS}$  can be determined from astronomical observations thereby determining  $\alpha_D$  as given by [eq. \(4.6b\)](#). Thus it can be claimed that the theoretical framework developed here has a strong potential for experimental verification based on astronomical observations.

*These are the sort of things people ought to look at. Things without pretensions, satisfied to be merely themselves.*

*—Aldous Huxley, The Doors of Perception and Heaven and Hell*

# 5

## BENDING ANGLE OF LIGHT IN AXISYMMETRIC SPACETIME BY OIA FORMALISM

Let us first consider the case of stationary axisymmetric spacetime followed by the spherically symmetric case. The generic form of the metric in the axisymmetric case in polar coordinates is given as

$$\begin{aligned} ds^2 &= g_{\mu\nu} dx^\mu dx^\nu \\ &= -A(r, \theta) dt^2 - 2H(r, \theta) dt d\phi + B(r, \theta) dr^2 + C(r, \theta) d\theta^2 + D(r, \theta) d\phi^2 \end{aligned} \quad (5.1)$$

For a photon orbit restricted to the equatorial plane we work with the assumption a local reflection symmetry with respect to  $\theta = \frac{\pi}{2}$  is observed. Furthermore,  $A\left(r, \frac{\pi}{2}\right), B\left(r, \frac{\pi}{2}\right), D\left(r, \frac{\pi}{2}\right) > 0$  is also an implicit assumption [Ono et al. (2017), Huang et al. (2024)]. As outlined in [chapter 4](#) in order to employ the OIA method we first need to reduce the metric to the Randers-Finsler form.

## 5.1 EXTRACTING THE FERMAT METRIC FROM THE RANDERS-FINSLER FORM

The Randers-Finsler form [Chanda et al. (2019), Werner (2012), Gibbons et al. (2009)] of any metric is an invaluable tool in the arsenal of any researcher investigating the geometrodynamics of any given spacetime. In general it is structured as

$$ds^2 = -h (dt - \omega_i dx^i)^2 + h^{-1} \xi_{ij} dx^i dx^j \quad (5.2)$$

Detailed accounts of this is available in the literature—Frolov (2013), Landau and Lifshitz (1980). With a suitable rescaling eq. (5.2) can be written as

$$dS^2 = - (dt - \beta_i dx^i)^2 + {}^{(F)}\gamma_{ij} dx^i dx^j \quad (5.3)$$

where  $ds^2 = h dS^2$  and  $\gamma_{ij}^{(F)}$  is the spatial metric Asada and Kasai (2000) commonly referred to as the optical metric or Fermat metric. Note in spite of its appearance the Fermat metric is not the same as the induced metric encountered during 3 + 1 splitting of a 4-d metric. The Fermat metric &  $\beta_i$  can be expressed in terms of the components of the given spacetime metric as

$$\gamma_{ij}^{(F)} = \frac{g_{ij}}{h} + \frac{g_{ti}}{h} \frac{g_{tj}}{h}, \quad \beta_i = \frac{g_{ti}}{h} \quad (5.4)$$

The term  $\beta_i$  gives us a measure of the departure of the Fermat metric from the geodesic. Invoking the null condition  $ds^2 = 0$  we can write from eq. (5.3)

$$dt = \sqrt{{}^{(F)}\gamma_{ij} dx^i dx^j} + \beta_i dx^i \quad (5.5)$$

$\gamma_{ij}^{(F)}$  is a Riemannian metric whereas  $\beta_i dx^i$  represents an one-form. In cases of metric describing spherically symmetric spacetime  $\beta_i dx^i = 0$  whereas for axisymmetric cases  $\beta_i \neq 0$ . A significant observation can be inferred from this, namely that  $\gamma_{ij}^{(F)}$  which is a Riemannian metric is sufficient for structuring the optical geometry for spherically symmetric cases. Whereas, for axisymmetric cases the inclusion of a non-zero  $\beta_i$  ensures that the optical geometry is described by a Randers-Finsler manifold Bao et al. (2012) which essentially reduces to Riemannian geometry iff  $\beta_i = 0$ . In fact one of the reasons we favour the terminology Fermat metric over optical metric as a descriptor of  $\gamma_{ij}^{(F)}$  is specifically because it does not universally lay out the optical geometry.

The axisymmetric metric expressed by eq. (5.1) can be reduced to the Randers-Finsler form as shown below.

$$\begin{aligned} ds^2 &= -A(r, \theta) dt^2 - 2H(r, \theta) dt d\phi + B(r, \theta) dr^2 + C(r, \theta) d\theta^2 + D(r, \theta) d\phi^2 \\ &= A \left[ - \left( dt^2 + 2 \frac{H}{A} dt d\phi + \frac{H^2}{A^2} d\phi^2 \right) \right] + A^{-1} \left[ \frac{B}{A} dr^2 + \frac{C}{A} d\theta^2 + \frac{D}{A} d\phi^2 + \frac{H^2}{A^2} \phi^2 \right] \end{aligned}$$

$$= -A \left( dt + \frac{H}{A} d\phi \right)^2 + A^{-1} \left[ \frac{B}{A} dr^2 + \frac{C}{A} d\theta^2 + \frac{AD + H^2}{A^2} d\phi^2 \right] \quad (5.6)$$

A comparison with the form presented in [eq. \(5.3\)](#) with [eq. \(5.6\)](#) allows us to write

$$\gamma_{ij}^{(F)} dx^i dx^j = \frac{B}{A} dr^2 + \frac{C}{A} d\theta^2 + \frac{AD + H^2}{A^2} d\phi^2 \quad (5.7)$$

$$\beta_i dx^i = -\frac{H}{A} d\phi \quad (5.8)$$

The steps leading upto [eq. \(5.6\)](#) is undoubtedly a pedestrian approach *en route* to deriving the Fermat metric and we have simply provided it as a easy way for someone new to the literature to verify the accuracy of [eq. \(5.4\)](#). In any case it is needless to say that preference must be given to [eq. \(5.4\)](#) while deriving  $\gamma_{ij}^{(F)}$  &  $\beta_i$ . Thus the matrix form of  $\gamma_{ij}^{(F)}$ , for axisymmetric spacetime is

$$\gamma_{ij}^{(F)} = \begin{pmatrix} \frac{B}{A} & 0 & 0 \\ 0 & \frac{C}{A} & 0 \\ 0 & 0 & \frac{AD + H^2}{A^2} \end{pmatrix} \quad (5.9)$$

### 5.1.1 Fermat metric— Why?

Determining the jump angles at the viewer and source locations is central to the methodology of GW-OIA formalism as is evident from our previous discussions in [section 4.2](#) & [section 4.3](#). Now, in order to accomplish this we need to determine two unit vectors namely —

- (a) the **unit tangential vector**  $\tau^i$  (say) along the photon trajectory
- (b) the **unit radial vector**  $\rho^i$  (say) along the outgoing direction

This is exactly where the Fermat metric comes in. The Fermat metric is critical for deriving a working expression of these two unit vectors. The beauty and simplicity of the Fermat metric lies in the fact that by the very nature of its definition it represents a Riemannian space( ${}^3\mathcal{R}$ ) wherein the trajectory of a photon(or any massless particle for that matter) can be represented by a simple spatial curve. In fact in case of any stationary spacetime the arc length along the photon trajectory, as described by the Fermat metric, is directly linked with the time associated with the time-like Killing vector. Now, if we take  $\ell_{ph}$  as the arc length along the photon trajectory then we can structure its definition with the aid of the Fermat metric as

$$d\ell_{ph}^2 = \gamma_{ij}^{(F)} dx^i dx^j \quad (5.10)$$

provided of course that  $\gamma^{ij(F)}$  obeys the condition

$$\gamma^{ij(F)} \gamma_{jk}^{(F)} = \delta_k^i \quad (5.11)$$

A detailed proof that  $\ell_{\text{ph}}$  as defined above is indeed an affine parameter along the photon trajectory is provided below [Asada and Kasai (2000)].

### 5.1.2 Proof of $\ell_{\text{ph}}$ is an affine parameter

Let us assume that the tangent vector of the photon trajectory is given by  $\tau^i = \frac{dx^i}{d\lambda}$  where  $\lambda$  is an affine parameter<sup>1</sup>. This allows us to write the geodesic equation for  $\tau^i$  as

$$\frac{d\tau_i}{d\lambda} = \frac{1}{2}g_{\mu\nu,i}\tau^\mu\tau^\nu \quad (5.12)$$

Since the spacetime under consideration is stationary there is no dependency on the time coordinate  $x^0$  which in turn implies that  $\tau_0$  remains constant along the geodesic *i.e.*,  $\frac{d\tau_0}{d\lambda} = 0$ . Now the null condition expressed as  $\tau^i\tau_i = 0$  which leads to<sup>2</sup>

$$\begin{aligned} 0 &= (\tau_0)^2 \\ \Rightarrow 0 &= \frac{dx^i}{d\lambda} \frac{dx_i}{d\lambda} = \frac{dx^i}{d\lambda} \gamma_{ij}^{(\text{F})} \frac{dx^j}{d\lambda} = \gamma_{ij}^{(\text{F})} \frac{dx^i}{d\lambda} \frac{dx^j}{d\lambda} \\ \Rightarrow 0 &= \left( \frac{d\ell_{\text{ph}}}{d\lambda} \right)^2 \\ \therefore \frac{d}{d\lambda} \left( \frac{d\ell_{\text{ph}}}{d\lambda} \right) &= 0 \end{aligned} \quad (5.13)$$

We arrive at eq. (5.13) with the aid of eq. (5.10). Having firmly established that  $\ell_{\text{ph}}$  as defined here must represent an affine parameter we can now express the unit tangential vector along the spatial curve as

$$\tau^i = \frac{dx^i}{d\ell_{\text{ph}}} \quad (5.14)$$

It is curious to notice that the stationary time coordinate  $t$  in the context of spacetime metric assumes the identity of the ratio between the arc length parameter to the spatial distance parameter in Fermat metric geometry.

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## 5.2 DETERMINATION OF THE ANGLES $\psi_V$ , $\psi_S$ & $\phi_{VS}$

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The angle  $\phi_{VS}$  is the one ever-present term in any mathematical formulation of the deflection angle and something which has been dealt with extensively in chapter 4. The only difference in this case is that we are no longer entertaining the assumption of an infinite separation between the source and viewer. How the

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<sup>1</sup>the present section does not discuss proper time and the use of the symbol  $\tau$  is restricted to its representation of tangent vector alone

<sup>2</sup>Note that for the purposes of raising and lowering indices the matrix  $\gamma_{ij}^{(\text{F})}$  is used instead of  $g_{ij}$ .

consideration of finite distance affects the representation of  $\phi_{VS}$  is discussed in [section 5.2.4](#). In order to find the angles  $\psi_V$  &  $\psi_S$  we first need to define the unit tangent vector and unit radial vector as depicted in the [fig. \[4.2\]](#) & [fig. \[4.3\]](#).

### 5.2.1 Defining the unit tangent vector and unit radial vector

From [eq. \(5.14\)](#) we can extract the components of the unit tangent vector as

$$\tau^i = \left( \frac{dr}{d\ell_{ph}}, 0, \frac{d\phi}{d\ell_{ph}} \right) \quad (5.15)$$

where there is no contribution from the  $\theta$  part as we restrict ourselves to the equatorial plane wherein  $\theta = \frac{\pi}{2}$ . However, instead of keeping  $\tau^i$  in the form expressed in [eq. \(5.15\)](#) we can restructure it to our benefit as

$$\begin{aligned} \tau^i &= \left( \frac{dr}{d\phi} \frac{d\phi}{d\ell_{ph}}, 0, \frac{d\phi}{d\ell_{ph}} \right) \\ \Rightarrow \tau^i &= \frac{1}{\zeta} \left( \frac{dr}{d\phi}, 0, 1 \right) \end{aligned} \quad (5.16)$$

where we can easily determine the value of  $\frac{1}{\zeta}$  by enforcing the condition  $\gamma_{ij}^{(F)} \tau^i \tau^j = 1$  and using [eq. \(5.7\)](#) in conjunction with [eq. \(5.16\)](#). Although, since it will be helpful to express our values involving the impact parameter we first derive the orbit equations using the properties of killing vectors as shown in [section 5.2.2](#). [The derivation of the final form of unit tangential vector is resumed in [section 5.2.3](#).]

The unit radial vector is simply defined as an unit tangent vector along the radial line on the equatorial plane in  ${}^3\mathcal{R}$ . Mathematically speaking the unit radial vector may be represented by

$$\rho^i = (\rho^r, 0, 0) \quad (5.17)$$

Now given the specifications of the Fermat metric the unit radial vector should abide by the stipulation

$$\gamma_{ij}^{(F)} \rho^i \rho^j = 1 \quad (5.18)$$

which allows us to write

$$\begin{aligned} \gamma_{ij}^{(F)} \rho^i \rho^j &= 1 \\ \Rightarrow \gamma_{rr}^{(F)} (\rho^r)^2 + \gamma_{\theta\theta}^{(F)} (\rho^\theta)^2 + \gamma_{\phi\phi}^{(F)} (\rho^\phi)^2 &= 1 \\ \Rightarrow \gamma_{rr}^{(F)} (\rho^r)^2 &= 1 \\ \Rightarrow \rho^r &= \frac{1}{\sqrt{\gamma_{rr}^{(F)}}} \\ \therefore \rho^i &= \left( \frac{1}{\sqrt{\gamma_{rr}^{(F)}}}, 0, 0 \right) \end{aligned} \quad (5.19)$$

### 5.2.2 Orbit equation in terms of impact parameter

Energy  $E$  and azimuthal component of angular momentum  $J$  are conserved along the geodesics (since the metric has no explicit dependence on  $t$  and  $\phi$ ). If we express the killing vectors with  $\Xi_{(t)}$  &  $\Xi_{(\phi)}$  then we can utilise it to construct two equations of motion in the spacetime eq. (5.1) involving  $E$  and  $J$  as follows

$$E = -\Xi_{(t)\mu} p^\mu = -g_{tt}\dot{t} - g_{t\phi}\dot{\phi} = A\dot{t} + H\dot{\phi} \quad (5.20)$$

and

$$J = -\Xi_{(\phi)\mu} p^\mu = g_{\phi t}\dot{t} + g_{\phi\phi}\dot{\phi} = -H\dot{t} + D\dot{\phi} \quad (5.21)$$

One of the other equation motion involving the radial part can be constructed from the null condition  $p^\mu p_\mu = 0$  as applied to the spacetime metric eq. (5.1). Now, it is well known that  $u \cdot u = g_{\alpha\beta} \frac{dx^\alpha}{d\lambda} \frac{dx^\beta}{d\lambda}$  which coupled with the null condition as applied to eq. (5.1) allows us to write

$$0 = -A\dot{t}^2 + B\dot{r}^2 + C\dot{\theta}^2 + D\dot{\phi}^2 - 2H\dot{t}\dot{\phi} \quad (5.22)$$

For the equatorial plane this reduces to the form

$$0 = -A\dot{t}^2 + B\dot{r}^2 + D\dot{\phi}^2 - 2H\dot{t}\dot{\phi} \quad (5.23)$$

$\lambda$  represents an affine parameter and  $\dot{\cdot}$  represents any derivative w.r.t to the affine parameter.

The impact parameter is defined as  $b = \frac{J}{E}$  as usual. We can now use this to express the impact parameter in terms of the components of the metric eq. (5.1).

$$\begin{aligned} b &= \frac{J}{E} \\ &= \frac{-H\dot{t} + D\dot{\phi}}{A\dot{t} + H\dot{\phi}} \\ &= \frac{-H + D\frac{\dot{\phi}}{\dot{t}}}{A + H\frac{\dot{\phi}}{\dot{t}}} \\ &= \frac{-H + D\frac{d\phi}{d\lambda} \frac{d\lambda}{dt}}{A + H\frac{d\phi}{d\lambda} \frac{d\lambda}{dt}} \\ &= \frac{-H + D\frac{d\phi}{dt}}{A + H\frac{d\phi}{dt}} \end{aligned} \quad (5.24)$$

Let us now resume our treatment of the eq. (5.23) to mould it into a more suitable form

$$0 = -A\dot{t}^2 + B\dot{r}^2 + D\dot{\phi}^2 - 2H\dot{t}\dot{\phi}$$

$$\begin{aligned}
 &= -A\dot{t}^2 - H\dot{t}\dot{\phi} + D\dot{\phi}^2 + -H\dot{t}\dot{\phi} + B\dot{r}^2 \\
 &= -\dot{t} \left( A\dot{t} + H\dot{\phi} \right) + \dot{\phi} \left( D\dot{\phi} - H\dot{t} \right) + B\dot{r}^2 \\
 &= -\dot{t}E + \dot{\phi}J + B\dot{r}^2 \\
 \Rightarrow B\dot{r}^2 &= E\dot{t} - J\dot{\phi} \\
 \Rightarrow \dot{r}^2 &= \frac{E}{B}\dot{t} - \frac{J}{B}\dot{\phi} \\
 \Rightarrow \left( \frac{dr}{d\phi} \frac{d\phi}{d\lambda} \right)^2 &= \frac{E}{B}\dot{t} - \frac{J}{B}\dot{\phi} \\
 \Rightarrow \left( \frac{dr}{d\phi} \right)^2 &= \frac{E\dot{t} - J\dot{\phi}}{B\dot{\phi}^2} \tag{5.25}
 \end{aligned}$$

Before proceeding further we need to express  $\dot{t}$  and  $\dot{\phi}$  in terms of  $E$ ,  $J$  and the components of the spacetime metric. A very simple algebraic manipulation of [eq. \(5.20\)](#) & [eq. \(5.21\)](#) allows us to write

$$\dot{\phi} = \frac{EH + AJ}{AD + H^2} \tag{5.26}$$

and

$$\dot{t} = \frac{ED - HJ}{AD + H^2} \tag{5.27}$$

Substituting the expressions in [eq. \(5.26\)](#) & [eq. \(5.27\)](#) in [eq. \(5.25\)](#) we get the radial equation in terms of the impact parameter and components of the spacetime metric as follows

$$\begin{aligned}
 \left( \frac{dr}{d\phi} \right)^2 &= \frac{E \frac{ED - HJ}{AD + H^2} - J \frac{EH + AJ}{AD + H^2}}{B \left( \frac{EH + AJ}{AD + H^2} \right)^2} \\
 &= \frac{E^2 \left( D - \frac{HJ}{E} \right) - JE \left( H + \frac{AJ}{E} \right)}{BE^2 \left( H + \frac{AJ}{E} \right)^2} (AD + H^2) \\
 &= \frac{\left[ (D - Hb) - \frac{J}{E} (H + Ab) \right]}{B (H + Ab)^2} (AD + H^2) \\
 &= \frac{(AD + H^2) (D - 2Hb - Ab^2)}{B (H + Ab)^2} \tag{5.28}
 \end{aligned}$$

[eq. \(5.28\)](#) in its present form can be immediately plugged into [eq. \(5.19\)](#) to get the expression for the unit tangential vector discussed in [section 5.2.1](#).

In terms of reciprocal coordinate  $u = \frac{1}{r}$  we have

$$\frac{du}{d\phi} = -\frac{1}{r^2} \frac{dr}{d\phi} \Rightarrow \left( \frac{du}{d\phi} \right)^2 = u^4 \left( \frac{dr}{d\phi} \right)^2 \tag{5.29}$$

With the aid of eq. (5.29), eq. (5.28) can be expressed in terms of the reciprocal coordinate as

$$\left(\frac{du}{d\phi}\right)^2 = u^4 \frac{(AD + H^2)(D - 2Hb - Ab^2)}{B(H + Ab)^2} = F(u) \quad (5.30)$$

### 5.2.3 Deriving the expression of the angles $\psi_V$ & $\psi_S$

A careful study of the fig. [4.2] establishes that the angles  $\psi_V$  and  $\psi_S$  are subtended between the unit tangential and unit radial vectors at the points **V** and **S** respectively. Now any angle  $\psi$  measured from the radial direction to the tangent drawn on the spatial curve representing the photon trajectory can be mathematically expressed as

$$\cos \psi = \gamma_{ij}^{(F)} \tau^i \rho^j \quad (5.31)$$

The only existing component of unit radial vector has already been derived in eq. (5.19). The expression for the unit tangential vector eq. (5.16) is now engineered to its final form as shown below. First we proceed with evaluating the expression for  $\frac{1}{\zeta}$  by utilising  $\gamma_{ij}^{(F)} \tau^i \tau^j = 1$  where we plug in the expression of  $\left(\frac{dr}{d\phi}\right)^2$  from eq. (5.28).

$$\begin{aligned} \gamma_{ij}^{(F)} \tau^i \tau^j &= 1 \\ \Rightarrow \gamma_{rr}^{(F)} \tau^r \tau^r + \gamma_{\theta\theta}^{(F)} \tau^\theta \tau^\theta + \gamma_{\phi\phi}^{(F)} \tau^\phi \tau^\phi &= 1 \\ \Rightarrow \frac{B}{A} \left(\frac{1}{\zeta} \frac{dr}{d\phi}\right)^2 + 0 + \frac{AD + H^2}{A^2} \frac{1}{\zeta^2} &= 1 \\ \Rightarrow \frac{1}{\zeta^2} \left[ \frac{B}{A} \frac{(AD + H^2)(D - 2Hb - Ab^2)}{B(H + Ab)^2} + \frac{AD + H^2}{A^2} \right] &= 1 \\ \Rightarrow \frac{1}{\zeta^2} \left[ \frac{(AD + H^2)(AD - 2HAb - A^2b^2) + (AD + H^2)(H + Ab)^2}{A^2(H + Ab)^2} \right] &= 1 \\ \Rightarrow \frac{1}{\zeta^2} \left[ \frac{(AD + H^2)(AD - 2HAb - A^2b^2 + H^2 + A^2b^2 + 2HAb)}{A^2(H + Ab)^2} \right] &= 1 \\ \Rightarrow \frac{1}{\zeta^2} \frac{(AD + H^2)^2}{A^2(H + Ab)^2} &= 1 \\ \Rightarrow \frac{1}{\zeta} = \frac{A(H + Ab)}{AD + H^2} \end{aligned} \quad (5.32)$$

Therefore, the components of the unit tangent vector can now be written as

$$\tau^i = \frac{A(H + Ab)}{AD + H^2} \left( \frac{dr}{d\phi}, 0, 1 \right) \quad (5.33)$$

Now, finally we can substitute the unit tangential and radial vectors given by eq. (5.33) & eq. (5.19) respectively in eq. (5.31).

$$\cos \psi = \gamma_{rr}^{(F)} \tau^r \rho^r$$

$$\begin{aligned}
 &= \gamma_{rr}^{(F)} \frac{A(H + Ab)}{AD + H^2} \frac{dr}{d\phi} \frac{1}{\sqrt{\gamma_{rr}^{(F)}}} \\
 &= \sqrt{\gamma_{rr}^{(F)}} \frac{A(H + Ab)}{AD + H^2} \frac{dr}{d\phi}
 \end{aligned} \tag{5.34}$$

However, there is a dependency on  $\frac{dr}{d\phi}$  in eq. (5.34), which does not lend itself to yield a simple user friendly expression of  $\psi$ . This issue can be easily resolved by transforming the expression in terms of  $\sin \psi$  so that  $\psi$  can be expressed only in terms of local dependencies. Squaring eq. (5.34) and substituting the expression of  $\left(\frac{dr}{d\phi}\right)^2$  from eq. (5.28) and  $\gamma_{rr}^{(F)}$  from eq. (5.9) we get

$$\begin{aligned}
 \cos^2 \psi &= \gamma_{rr}^{(F)} \frac{A^2 (H + Ab)^2}{(AD + H^2)^2} \left(\frac{dr}{d\phi}\right)^2 \\
 &= \gamma_{rr}^{(F)} \frac{A^2 (H + Ab)^2}{(AD + H^2)^2} \frac{(AD + H^2)(D - 2Hb - Ab^2)}{B(H + Ab)^2} \\
 &= \frac{B}{A} \frac{A^2}{B} \frac{D - 2Hb - Ab^2}{AD + H^2} \\
 &= \frac{AD - 2HAb - A^2b^2}{AD + H^2} \\
 \Rightarrow 1 - \cos^2 \psi &= 1 - \frac{AD - 2HAb - A^2b^2}{AD + H^2} \\
 &= \frac{AD + H^2 - AD + 2HAb + A^2b^2}{AD + H^2} \\
 \therefore \sin \psi &= \frac{H + Ab}{\sqrt{AD + H^2}}
 \end{aligned} \tag{5.35}$$

Therefore, the angle  $\psi$  can now be expressed as

$$\psi = \sin^{-1} \frac{H + Ab}{\sqrt{AD + H^2}} \tag{5.36}$$

Now, since we have restricted ourselves to the equatorial plane,  $A$ ,  $D$  &  $H$  are all functions of  $r$  or  $\frac{1}{u}$ . Clearly at the points **V** & **S**,  $r$  assumes the value  $r_V = \frac{1}{u_V}$  and  $r_S = \frac{1}{u_S}$  respectively. Therefore, the expressions for  $\psi_V$  &  $\psi_S$  can be written as

$$\psi_V = \sin^{-1} \frac{H\left(\frac{1}{u_V}\right) + A\left(\frac{1}{u_V}\right)b}{\sqrt{A\left(\frac{1}{u_V}\right)D\left(\frac{1}{u_V}\right) + H\left(\frac{1}{u_V}\right)^2}} \tag{5.37}$$

and

$$\psi_S = \pi - \sin^{-1} \frac{H\left(\frac{1}{u_S}\right) + A\left(\frac{1}{u_S}\right)b}{\sqrt{A\left(\frac{1}{u_S}\right)D\left(\frac{1}{u_S}\right) + H\left(\frac{1}{u_S}\right)^2}} \tag{5.38}$$

respectively. [eq. \(5.37\)](#) & [eq. \(5.38\)](#) are in their respective forms as  $\psi_V$  is acute and  $\psi_S$  is obtuse([fig. \[4.3\]](#)). It is worth noticing in this context that expressing  $\psi$  in terms of *sine* has the added benefit that  $\sin \psi$  always assumes a positive value as the domain of  $\psi$  is  $0 \leq \psi \leq \pi$ .

#### 5.2.4 Deriving the expression of the angle $\phi_{VS}$

The azimuthal coordinate separation  $\phi_{VS}$  is simply expressed as

$$\phi_{VS} = \int_S^V d\phi \quad (5.39)$$

Now we have already derived  $\psi_V$  &  $\psi_S$  in terms of  $u$  we are motivated to express  $\phi_{VS}$  in the same fashion. To this end we can utilise [eq. \(5.30\)](#) to express the integral in [eq. \(5.39\)](#) in terms of  $u$ . The only tricky bit in this context is determining the limits of integration after transforming the integral. In the conventional case we were working with the definition describing a photon arriving from infinity to the distance of closest approach  $r_0 = \frac{1}{u_0}$  before traveling back to infinity. This would automatically mandate the lower limit of  $u$  as 0. However, since the very essence of the Gauss-Bonnet theorem-inspired approach relies on the realistic premise that the source and viewer are at finite distances, the limits should be determined accordingly. As such we simply segment the integral into two distinct portions namely

- the photon trajectory from the source(S) to the turning point marking the distance of closest approach ( $r_0$ )
- the photon trajectory from the turning point to the viewer(V)

Therefore, [eq. \(5.39\)](#) can now be expressed as

$$\phi_{VS} = \int_{u_S}^{u_0} \frac{1}{\sqrt{F(u)}} du + \int_{u_V}^{u_0} \frac{1}{\sqrt{F(u)}} du \quad (5.40)$$

Substituting [eq. \(5.37\)](#), [eq. \(5.38\)](#) & [eq. \(5.40\)](#) in [eq. \(4.6\)](#) we can now express the bending angle of a photon trajectory in axisymmetric spacetimes as

$$\begin{aligned} \alpha_D = \sin^{-1} \frac{H\left(\frac{1}{u_V}\right) + A\left(\frac{1}{u_V}\right)b}{\sqrt{A\left(\frac{1}{u_V}\right)D\left(\frac{1}{u_V}\right) + H\left(\frac{1}{u_V}\right)^2}} - \pi - \sin^{-1} \frac{H\left(\frac{1}{u_S}\right) + A\left(\frac{1}{u_S}\right)b}{\sqrt{A\left(\frac{1}{u_S}\right)D\left(\frac{1}{u_S}\right) + H\left(\frac{1}{u_S}\right)^2}} \\ + \int_{u_S}^{u_0} \frac{1}{\sqrt{F(u)}} du + \int_{u_V}^{u_0} \frac{1}{\sqrt{F(u)}} du \end{aligned} \quad (5.41)$$

Using this generic expression we can now evaluate the bending angle for any axisymmetric spacetime wherein the source and viewer are located at finite distances. We now apply [eq. \(5.41\)](#) to the specific case of Kerr metric [[Teukolsky \(2015\)](#)].

### 5.3 APPLYING OIA FORMALISM TO KERR METRIC

The Kerr metric can be expressed in terms of Boyer-Lindquist coordinates as

$$ds^2 = - \left( 1 - \frac{2Mr}{\Sigma} \right) dt^2 - \frac{4aMr \sin^2 \theta}{\Sigma} d\phi dt + \frac{\Sigma}{\Delta} dr^2 + \Sigma d\theta^2 + \left( r^2 + a^2 + \frac{2a^2 Mr \sin^2 \theta}{\Sigma} \right) \sin^2 \theta d\phi^2 \quad (5.42)$$

where

$$\Sigma = r^2 + a^2 \cos^2 \theta \quad (5.43)$$

and

$$\Delta = r^2 + a^2 - 2Mr \quad (5.44)$$

#### 5.3.1 Evaluating $\psi_V$ & $\psi_S$ for Kerr spacetime

Substituting the respective values of  $A$ ,  $D$  &  $H$  for the Kerr case in [eq. \(5.36\)](#) and assuming  $\theta = \frac{\pi}{2}$  (since we are only considering the equatorial case) we can express  $\sin \psi$  as

$$\begin{aligned} \sin \psi &= \frac{H + Ab}{\sqrt{AD + H^2}} \\ &= \frac{\frac{2aMr}{r^2} + \left( 1 - \frac{2Mr}{r^2} \right) b}{\sqrt{\left( 1 - \frac{2Mr}{r^2} \right) \left( r^2 + a^2 + \frac{2a^2 Mr}{r^2} \right) + \frac{4a^2 M^2 r^2}{r^4}}} \\ &= \frac{\frac{2aM}{r} + b - \frac{2Mb}{r}}{\sqrt{\left( 1 - \frac{2M}{r} \right) \left( r^2 + a^2 + \frac{2a^2 M}{r} \right) + \frac{4a^2 M^2}{r^2}}} \\ &= \frac{b + \frac{2M}{r} (a - b)}{\sqrt{r^2 + a^2 + \frac{2a^2 M}{r} - 2Mr - \frac{2a^2 M}{r} - \frac{4a^2 M}{r^2} + \frac{4a^2 M}{r^2}}} \\ &= \frac{b + \frac{2M}{r} (a - b)}{\sqrt{r^2 + a^2 - 2Mr}} \\ &= \frac{b \left[ 1 + \frac{2M}{r} \left( \frac{a}{b} - 1 \right) \right]}{r \sqrt{1 + \frac{a^2}{r^2} - \frac{2M}{r}}} \\ &= \frac{b}{r} \times \frac{1 - \frac{2M}{r} + \frac{2aM}{br}}{\sqrt{1 - \frac{2M}{r} + \frac{a^2}{r^2}}} \end{aligned}$$

$$\begin{aligned}
 &= \frac{b}{r} \times \left(1 - \frac{2M}{r} + \frac{2aM}{br}\right) \times \left(1 - \frac{2M}{r} + \frac{a^2}{r^2}\right)^{-\frac{1}{2}} \\
 &= \frac{b}{r} \times \left(1 - \frac{2M}{r} + \frac{2aM}{br}\right) \times \left[1 + \frac{M}{r} - \frac{a^2}{2r^2} + \mathcal{O}\left\{\left(\frac{2M}{r} - \frac{a^2}{r^2}\right)^2\right\}\right] \\
 &= \frac{b}{r} \times \left[1 + \frac{M}{r} - \frac{a^2}{2r^2} - \frac{2M}{r} - \frac{2M^2}{r^2} + \frac{Ma^2}{r^3} + \frac{2aM}{br} + \frac{2aM^2}{br^2} - \frac{a^3M}{br^3} + \mathcal{O}\left\{\left(\frac{2M}{r} - \frac{a^2}{r^2}\right)^2\right\}\right] \\
 &= \frac{b}{r} \times \left[1 - \frac{M}{r} + \frac{2aM}{br}\right] + \mathcal{O}\left(\frac{M^2}{r^2}, \frac{a^2}{r^2}, \frac{aM^2}{r^3}\right) \\
 &= bu \times \left(1 - Mu + \frac{2aM}{b}u\right) + \mathcal{O}(M^2u^2, a^2u^2, aM^2u^3)
 \end{aligned} \tag{5.45}$$

In the above expression we only retain terms of the order  $\frac{M}{r}$  in the leading term and terms upto  $\frac{M^2}{r^3}$  in the  $\mathcal{O}()$  while completely dismissing terms involving  $r^{-4}$  and beyond. We can now solve for  $\psi$  as

$$\begin{aligned}
 \psi &= \sin^{-1}(bu - bMu^2 + 2aMu^2) + \mathcal{O}(M^2u^2, a^2u^2, aM^2u^3) \\
 &= \sin^{-1}(bu) - bMu^2 \frac{d}{d(bu)} [\sin^{-1}(bu)] + 2aMu^2 \frac{d}{d(bu)} [\sin^{-1}(bu)] + \mathcal{O}(M^2u^2, a^2u^2, aM^2u^3) \\
 &= \sin^{-1}(bu) - \frac{bMu^2}{\sqrt{1-b^2u^2}} + \frac{2aMu^2}{\sqrt{1-b^2u^2}} + \mathcal{O}(M^2u^2, a^2u^2, aM^2u^3)
 \end{aligned} \tag{5.46}$$

We have applied Taylor expansion<sup>3</sup> to the  $\sin^{-1}$  term with  $bu$  as the variable  $x$  and  $(-bMu^2 + 2aMu^2)$  as  $h$  where  $h \ll x$ . Terms are only retained upto the first order. From the generic expression of  $\psi$  in Kerr spacetime we can write the expressions of the terms  $\psi_V$  and  $\psi_S$  as

$$\psi_V = \sin^{-1}(bu_V) - \frac{bMu_V^2}{\sqrt{1-b^2u_V^2}} + \frac{2aMu_V^2}{\sqrt{1-b^2u_V^2}} + \mathcal{O}(M^2u_V^2, a^2u_V^2, aM^2u_V^3) \tag{5.47}$$

and

$$\begin{aligned}
 \pi - \psi_S &= \sin^{-1}(bu_S) - \frac{bMu_S^2}{\sqrt{1-b^2u_S^2}} + \frac{2aMu_S^2}{\sqrt{1-b^2u_S^2}} + \mathcal{O}(M^2u_S^2, a^2u_S^2, aM^2u_S^3) \\
 \Rightarrow \psi_S &= \pi - \sin^{-1}(bu_S) + \frac{bMu_S^2}{\sqrt{1-b^2u_S^2}} - \frac{2aMu_S^2}{\sqrt{1-b^2u_S^2}} + \mathcal{O}(M^2u_S^2, a^2u_S^2, aM^2u_S^3)
 \end{aligned} \tag{5.48}$$

### 5.3.2 Evaluating $\phi_{VS}$ for Kerr spacetime

The angle  $\phi_{VS}$  for the Kerr spacetime can be derived from the expression presented in eq. (5.40). But first we need to evaluate  $F(u)$  by substituting the appropriate values of  $A$ ,  $B$ ,  $D$  &  $H$  for the Kerr case in eq. (5.30). Let us start off with the form in eq. (5.28) before we derive the expression of  $F(u)$ .

$$\left(\frac{dr}{d\phi}\right)^2 = \frac{AD + H^2}{B} \frac{D - 2Hb - Ab^2}{(H + Ab)^2}$$

<sup>3</sup> $f(x \pm h) = f(x) \pm hf'(x) + \frac{h^2}{2}f''(x) \pm \frac{h^3}{6}f'''(x) + \dots$

$$\begin{aligned}
 &= \frac{\left(1 - \frac{2Mr}{r^2}\right) \left(r^2 + a^2 + \frac{2a^2Mr}{r^2}\right) + \left(\frac{2aMr}{r^2}\right)^2}{\frac{r^2}{r^2 - 2Mr + a^2}} \\
 &\quad \times \frac{\left(r^2 + a^2 + \frac{2a^2Mr}{r^2}\right) - \frac{4aMr}{r^2}b - \left(1 - \frac{2Mr}{r^2}b^2\right)}{\left[\frac{2amr}{r^2} + \left(1 - \frac{2Mr}{r^2}\right)b\right]^2} \\
 &= \frac{\left(1 - \frac{2M}{r}\right) \left(r^2 + a^2 + \frac{2a^2M}{r}\right) + \frac{4a^2M^2}{r^2}}{r^2} \times (r^2 - 2Mr + a^2) \\
 &\quad \times \frac{\left(r^2 + a^2 + \frac{2a^2M}{r}\right) - \frac{4aM}{r}b - \left(1 - \frac{2M}{r}b^2\right)}{\left[\frac{2am}{r} + \left(1 - \frac{2M}{r}\right)b\right]^2} \\
 &= \frac{r^2 + a^2 + \frac{2a^2M}{r} - 2Mr - \frac{2a^2M}{r} - \frac{4a^2M^2}{r^2} + \frac{4a^2M^2}{r^2}}{r^2} \times (r^2 - 2Mr + a^2) \\
 &\quad \times \frac{r^2 + a^2 + \frac{2a^2M}{r} - \frac{4abM}{r} - b^2 + \frac{2Mb^2}{r}}{\left[\frac{b^2}{r} \left(\frac{2aM}{b^2} + \frac{br}{b^2} - \frac{2Mb}{b^2}\right)\right]^2} \\
 &= \frac{(r^2 + a^2 - 2Mr)^2}{r^2} \times \frac{r^2 + a^2 + \frac{2a^2M}{r} - \frac{4abM}{r} - b^2 + \frac{2Mb^2}{r}}{\frac{b^4}{r^2} \left(\frac{2aM}{b^2} + \frac{r}{b} - \frac{2M}{b}\right)^2} \\
 &= \frac{(r^2 + a^2 - 2Mr)^2 \times \left(r^2 + a^2 + \frac{2a^2M}{r} - \frac{4abM}{r} - b^2 + \frac{2Mb^2}{r}\right)}{b^4 \left(\frac{2aM}{b^2} + \frac{r}{b} - \frac{2M}{b}\right)^2} \\
 &= \frac{b^4 \left(\frac{r^2}{b^2} + \frac{a^2}{b^2} - \frac{2Mr}{b^2}\right)^2 \times \frac{b^3}{r} \left(\frac{r^3}{b^3} + \frac{a^2r}{b^3} + \frac{2a^2M}{b^3} - \frac{4aM}{b^2} - \frac{r}{b} + \frac{2M}{b}\right)}{b^4 \left(\frac{2aM}{b^2} + \frac{r}{b} - \frac{2M}{b}\right)^2} \\
 &= \frac{b^2 \left[\frac{r}{b} \left(\frac{r}{b} - \frac{2M}{b}\right) + \frac{a^2}{b^2}\right]^2 \left[\frac{a^2}{b^2} \left(\frac{2M}{b} + \frac{r}{b}\right) - \frac{4aM}{b^2} + \frac{2M}{b} - \frac{r}{b} + \frac{r^3}{b^3}\right]}{\frac{r}{b} \left(\frac{2aM}{b^2} + \frac{r}{b} - \frac{2M}{b}\right)^2} \\
 &\simeq \frac{b^2 \left[\frac{r}{b} \left(\frac{r}{b} - \frac{2M}{b}\right)\right]^2 \left[-\frac{4aM}{b^2} + \frac{2M}{b} - \frac{r}{b} + \frac{r^3}{b^3}\right]}{\frac{r}{b} \left(\frac{2aM}{b^2} + \frac{r}{b} - \frac{2M}{b}\right)^2}
 \end{aligned}$$

$$\begin{aligned}
 & [\because \text{Neglecting terms } \geq a^2 \text{ for slow rotation approximation}] \\
 &= \frac{b^2 \left[ x \left( x - \frac{2M}{b} \right) \right]^2 \left[ -\frac{4aM}{b^2} + \frac{2M}{b} - x + x^3 \right]}{x \left( x + \frac{2aM}{b^2} - \frac{2M}{b} \right)^2} \\
 & \quad [\because \text{Substituting } \frac{r}{b} = x] \\
 &= \frac{b^2 x^2 \left( x^2 + \frac{4M^2}{b^2} - \frac{4Mx}{b} \right) \left( -\frac{4aM}{b^2} + \frac{2M}{b} - x + x^3 \right)}{x^3 \left[ 1 - \left( \frac{2M}{bx} - \frac{2aM}{b^2 x} \right) \right]^2} \\
 &\simeq \frac{\frac{b^2 x^2}{x^3} \left( x^2 - \frac{4M}{b} x \right) \left( -\frac{4aM}{b^2} + \frac{2M}{b} - x + x^3 \right)}{\left[ 1 - \frac{2M}{bx} \left( 1 - \frac{a}{b} \right) \right]^2} \\
 & \quad [\because \text{Neglecting terms } \geq M^2] \\
 &= \frac{b^2 \left( x - \frac{4M}{b} \right) \left( -\frac{4aM}{b^2} + \frac{2M}{b} - x + x^3 \right)}{\left[ 1 + \frac{4M^2}{b^2 x^2} \left( 1 - \frac{a}{b} \right)^2 - \frac{4M}{bx} \left( 1 - \frac{a}{b} \right) \right]} \\
 &= \frac{b^2 \left( x - \frac{4M}{b} \right) \left( -\frac{4aM}{b^2} + \frac{2M}{b} - x + x^3 \right)}{1 - \frac{4M}{bx} \left( 1 - \frac{a}{b} \right)} \\
 &= b^2 \left( x - \frac{4M}{b} \right) \left( -\frac{4aM}{b^2} + \frac{2M}{b} - x + x^3 \right) \left( 1 + \frac{4M}{bx} - \frac{4aM}{b^2 x} \right) \\
 &\simeq b^2 \left( -\frac{4aM}{b^2} + \frac{2M}{b} - x + x^3 \right) \left( x + \frac{4M}{b} - \frac{4aM}{b^2} - \frac{4M}{b} \right) \\
 & \quad [\because \text{Neglecting terms } \geq M^2] \\
 &\simeq b^2 \left( -\frac{4aMx}{b^2} + \frac{2Mx}{b} - x^2 + \frac{4aMx}{b^2} + x^4 - \frac{4aMx^3}{b^2} \right) \\
 &= b^2 \left( x^4 - x^2 + \frac{2Mx}{b} - \frac{4aMx^3}{b^2} \right) \\
 &= b^2 \frac{r^4}{b^4} - b^2 \frac{r^2}{b^2} + b^2 \frac{2M}{b} \frac{r}{b} - b^2 \frac{4aM}{b^2} \frac{r^3}{b^3} \\
 & \quad [\because \text{Substituting back } x = \frac{r}{b}] \\
 &= \frac{r^4}{b^2} - r^2 + 2Mr - \frac{4aM}{b^3} r^3 + \mathcal{O}(a^2) \tag{5.49}
 \end{aligned}$$

Now using eq. (5.49) and substituting in eq. (5.40) we can write,

$$\left( \frac{du}{d\phi} \right)^2 = F(u) = u^4 \left( \frac{dr}{d\phi} \right)^2$$

$$\begin{aligned}
 &= u^4 \left[ \frac{r^4}{b^2} - r^2 + 2Mr - \frac{4aM}{b^3} r^3 \right] + \mathcal{O}(a^2 u^4) \\
 \Rightarrow F(u) &= \frac{1}{b^2} - u^2 + 2Mu^3 - \frac{4aM}{b^3} u + \mathcal{O}(a^2 u^4)
 \end{aligned} \tag{5.50}$$

Now if we consider the distance of closest approach  $r = r_0 = \frac{1}{u_0}$  and  $\phi = \frac{\pi}{2}$  then  $\left( \frac{du}{d\phi} \right)_{u=u_0, \phi=\frac{\pi}{2}} = 0$ . We can employ this condition to find the expression of  $b$  from [eq. \(5.50\)](#) as shown below

$$\begin{aligned}
 0 &= \frac{1}{b^2} - u_0^2 + 2Mu_0^3 - \frac{4aM}{b^3} u_0 \\
 \Rightarrow \frac{1}{b^2} &= u_0^2 - 2Mu_0^3 + \frac{4aM}{b^3} u_0 \\
 \Rightarrow \frac{1}{b^2} &= u_0^2 \left[ 1 - \left( 2Mu_0 - \frac{4aM}{b^3 u_0} \right) \right] \\
 \Rightarrow b &= \frac{1}{u_0} \left[ 1 - \left( 2Mu_0 - \frac{4aM}{b^3 u_0} \right) \right]^{-\frac{1}{2}} \\
 &\simeq \frac{1}{u_0} \left[ 1 - \left( -\frac{1}{2} \right) \left( 2Mu_0 - \frac{4aM}{b^3 u_0} \right) \right]
 \end{aligned} \tag{5.51}$$

[ $\cdot$ : Expanding binomially and neglecting the higher order terms]

$$\begin{aligned}
 &= \frac{1}{u_0} \left[ 1 + Mu_0 - \frac{2aM}{b^3 u_0} \right] \\
 &= \frac{1}{u_0} + M - \frac{2aM}{b^3} \frac{1}{u_0^2} \\
 &\simeq \frac{1}{u_0} + M - 2aMu_0
 \end{aligned}$$

[ $\cdot$ : Neglecting the  $\geq M^2$  terms]

$$\Rightarrow b = \frac{1}{u_0} + M - 2aMu_0 + \mathcal{O}(M^2, a^2) \tag{5.52}$$

The expressions for  $\frac{1}{b^2}$  &  $b$  given by [eq. \(5.51\)](#) & [eq. \(5.52\)](#) are of critical importance in determining  $\phi_{VS}$ .

Let us now express  $\frac{1}{\sqrt{F(u)}}$  in a form more conducive to performing the integration. Starting with [eq. \(5.50\)](#) and substituting the value of  $\frac{1}{b^2}$  from [eq. \(5.51\)](#) we get,

$$\begin{aligned}
 F(u) &= \frac{1}{b^2} - u^2 + 2Mu^3 - \frac{4aM}{b^3} u + \mathcal{O}(a^2 u^4) \\
 &= u_0^2 - 2Mu_0^3 + \frac{4aM}{b^3} u_0 - u^2 + 2Mu^3 - \frac{4aM}{b^3} u \\
 &= (u_0^2 - u^2) - 2M(u_0^3 - u^3) + \frac{4aM}{b^3} (u_0 - u) \\
 &= (u_0^2 - u^2) \left[ 1 - \left\{ 2M \left( \frac{u_0^3 - u^3}{u_0^2 - u^2} \right) - \frac{4aM}{b^3} \left( \frac{u_0 - u}{u_0^2 - u^2} \right) \right\} \right] \\
 \Rightarrow \frac{1}{\sqrt{F(u)}} &= \frac{1}{\sqrt{u_0^2 - u^2}} \left[ 1 - \left\{ 2M \left( \frac{u_0^3 - u^3}{u_0^2 - u^2} \right) - \frac{4aM}{b^3} \left( \frac{u_0 - u}{u_0^2 - u^2} \right) \right\} \right]^{-\frac{1}{2}}
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{\sqrt{u_0^2 - u^2}} \left[ 1 + \frac{1}{2} \left\{ 2M \left( \frac{u_0^3 - u^3}{u_0^2 - u^2} \right) - \frac{4aM}{b^3} \left( \frac{u_0 - u}{u_0^2 - u^2} \right) \right\} \right] \\
 &= \frac{1}{\sqrt{u_0^2 - u^2}} + M \frac{(u_0^3 - u^3)}{(u_0^2 - u^2)^{\frac{3}{2}}} - 2aM \frac{(u_0 - u)}{(u_0^2 - u^2)^{\frac{3}{2}}} \times \frac{1}{\frac{1}{u_0^3} (1 + Mu_0 - 2aMu_0^2)^3} \\
 &\quad [\because \text{Substituting } b = \frac{1}{u_0} + M - 2aMu_0] \\
 &\simeq \frac{1}{\sqrt{u_0^2 - u^2}} + M \frac{(u_0^3 - u^3)}{(u_0^2 - u^2)^{\frac{3}{2}}} - 2aM \frac{u_0^3 (u_0 - u)}{(u_0^2 - u^2)^{\frac{3}{2}}} \tag{5.53}
 \end{aligned}$$

All terms  $\geq M^2$  have been neglected while deriving eq. (5.53).

We can now substitute this in eq. (5.40) to derive the final expression for  $\phi_{VS}$  in the case of Kerr metric

$$\begin{aligned}
 \phi_{VS} &= \int_{u_S}^{u_0} \frac{1}{\sqrt{F(u)}} du + \int_{u_V}^{u_0} \frac{1}{\sqrt{F(u)}} du \\
 &= \int_{u_S}^{u_0} \frac{1}{\sqrt{u_0^2 - u^2}} du + \int_{u_S}^{u_0} M \frac{(u_0^3 - u^3)}{(u_0^2 - u^2)^{\frac{3}{2}}} du - \int_{u_S}^{u_0} 2aM \frac{u_0^3 (u_0 - u)}{(u_0^2 - u^2)^{\frac{3}{2}}} du \\
 &\quad + \int_{u_V}^{u_0} \frac{1}{\sqrt{u_0^2 - u^2}} du + \int_{u_V}^{u_0} M \frac{(u_0^3 - u^3)}{(u_0^2 - u^2)^{\frac{3}{2}}} du - \int_{u_V}^{u_0} 2aM \frac{u_0^3 (u_0 - u)}{(u_0^2 - u^2)^{\frac{3}{2}}} du \tag{5.54}
 \end{aligned}$$

Let us denote the three different integrals (irrespective of their limits) as,

$$I_1 = \int_{u_*}^{u_0} \frac{1}{\sqrt{u_0^2 - u^2}} du \tag{5.55a}$$

$$I_2 = \int_{u_*}^{u_0} M \frac{(u_0^3 - u^3)}{(u_0^2 - u^2)^{\frac{3}{2}}} du \tag{5.55b}$$

$$I_3 = \int_{u_*}^{u_0} 2aM \frac{u_0^3 (u_0 - u)}{(u_0^2 - u^2)^{\frac{3}{2}}} du \tag{5.55c}$$

where  $u_*$  represents a stand-in for the limits  $u_S$  &  $u_V$ . We can solve the improper integrals separately by a series of appropriate substitutions as shown below.

For the integral  $I_1$  we proceed as follows, with the substitution  $u = u_0 \sin \theta$

$$I_1 = \int_{u_*}^{u_0} \frac{1}{\sqrt{u_0^2 - u^2}} du = \int_{\sin^{-1} \frac{u_*}{u_0}}^{\frac{\pi}{2}} \frac{1}{\sqrt{u_0^2 - u_0^2 \sin^2 \theta}} u_0 \cos \theta d\theta = \int_{\sin^{-1} \frac{u_*}{u_0}}^{\frac{\pi}{2}} d\theta = \frac{\pi}{2} - \sin^{-1} \left( \frac{u_*}{u_0} \right) \tag{5.56}$$

The integrals  $I_2$  &  $I_3$  are somewhat a little more involved. For the integral  $I_2$  we proceed with the substitution  $u = u_0 \sin \theta$

$$I_2 = \int_{u_*}^{u_0} M \frac{(u_0^3 - u^3)}{(u_0^2 - u^2)^{\frac{3}{2}}} du = M \int \frac{(u_0^3 - u_0^3 \sin^3 \theta)}{(u_0^2 - u_0^2 \sin^2 \theta)^{\frac{3}{2}}} u_0 \cos \theta d\theta = M \int \frac{u_0^3 (1 - \sin^3 \theta) (u_0 \cos \theta)}{u_0^3 \cos^3 \theta} d\theta$$

$$= M u_0 \int \left( \frac{1}{\cos^2 \theta} - \frac{\sin^3 \theta}{\cos^2 \theta} \right) d\theta = M u_0 \left[ \int \sec^2 \theta d\theta - \int \frac{\sin \theta (1 - \cos^2 \theta)}{\cos^2 \theta} d\theta \right] \quad (5.57)$$

Notice, that we haven't substituted the limits for the integral after the substitution yet. The reasoning will become evident in the following steps. Solving the integral requires yet another substitution and it is easier to substitute the limits after fully solving the entire integral. The first part of the integral in eq. (5.57) is readily solved but for the second integral we make the substitution  $\cos \theta = x$ .

$$\begin{aligned} I_2 &= M u_0 \left[ \int \sec^2 \theta d\theta - \int \frac{\sin \theta (1 - \cos^2 \theta)}{\cos^2 \theta} d\theta \right] = M u_0 \left[ \tan \theta - \int \frac{1 - x^2}{x^2} (-dx) \right] \\ &= M u_0 \left[ \tan \theta + \int \left( \frac{1}{x^2} - 1 \right) dx \right] = M u_0 \left[ \tan \theta + \left\{ \frac{x^{-2+1}}{-2+1} - x \right\} \right] = M u_0 \left[ \tan \theta + \left\{ -\frac{1}{x} - x \right\} \right] \\ &= M u_0 \left[ \tan \theta - \left\{ \frac{1}{\cos \theta} + \cos \theta \right\} \right] = M u_0 \left[ \frac{\sin \theta}{\sqrt{1 - \sin^2 \theta}} - \left\{ \frac{1}{\cos \theta} + \cos \theta \right\} \right] \\ &= M u_0 \left[ \frac{\frac{u}{u_0}}{\sqrt{1 - \frac{u^2}{u_0^2}}} - \left\{ \frac{u_0}{\sqrt{u_0^2 - u^2}} + \frac{\sqrt{u_0^2 - u^2}}{u_0} \right\} \right] = M u_0 \left[ \frac{u}{\sqrt{u_0^2 - u^2}} - \frac{u_0}{\sqrt{u_0^2 - u^2}} - \frac{\sqrt{u_0^2 - u^2}}{u_0} \right] \\ &= M u_0 \left[ \frac{u - u_0}{\sqrt{u_0^2 - u^2}} - \frac{\sqrt{u_0^2 - u^2}}{u_0} \right] \end{aligned} \quad (5.58)$$

Applying L' Hospital's rule for the limit  $u \rightarrow u_0$  eq. (5.58) yields 0. Now plugging in the lower limit  $u_*$  we get

$$I_2 = -\frac{M u_0 (u_* - u_0)}{\sqrt{u_0^2 - u_*^2}} + M \sqrt{u_0^2 - u_*^2} = M \left[ \frac{u_0^2 - u_* u_0 + u_0^2 - u_*^2}{\sqrt{u_0^2 - u_*^2}} \right] = M \frac{(2u_0 + u_*)(u_0 - u_*)}{\sqrt{u_0^2 - u_*^2}} \quad (5.59a)$$

$$= M \frac{(2u_0 + u_*)(u_0 - u_*)}{(u_0 - u_*)(u_0 + u_*)} \times \sqrt{u_0^2 - u_*^2} = M \frac{(2u_0 + u_*) \sqrt{u_0^2 - u_*^2}}{(u_0 + u_*)} \quad (5.59b)$$

Similarly, for the integral  $I_3$  we start with the substitution  $u = u_0 \sin \theta$ ,

$$\begin{aligned} I_3 &= \int_{u_*}^{u_0} 2aM \frac{u_0^3 (u_0 - u)}{(u_0^2 - u^2)^{\frac{3}{2}}} du = 2aM \int \frac{u_0^3 \cdot u_0 (1 - \sin \theta)}{(u_0^2 - u^2)^{\frac{3}{2}}} u_0 \cos \theta d\theta = 2aM \int \frac{u_0^2 (1 - \sin \theta) \cos \theta}{\cos^3 \theta} d\theta \\ &= 2aM u_0^2 \int \frac{1 - \sin \theta}{\cos^2 \theta} d\theta = 2aM u_0^2 \left[ \int \sec^2 \theta d\theta - \int \frac{\sin \theta}{\cos^2 \theta} d\theta \right] \end{aligned} \quad (5.60)$$

As in the case of  $I_2$  we substitute  $\cos \theta = x$  to solve the second integral in eq. (5.60).

$$\begin{aligned} I_3 &= 2aM u_0^2 \left[ \int \sec^2 \theta d\theta - \int \frac{\sin \theta}{\cos^2 \theta} d\theta \right] = 2aM u_0^2 \left[ \tan \theta + \int \frac{1}{x^2} dx \right] = 2aM u_0^2 \left[ \tan \theta + \frac{x^{-2+1}}{-2+1} \right] \\ &= 2aM u_0^2 \left[ \tan \theta - \frac{1}{x} \right] = 2aM u_0^2 \left[ \frac{u}{\sqrt{u_0^2 - u^2}} - \frac{u_0}{\sqrt{u_0^2 - u^2}} \right] = 2aM u_0^2 \frac{u - u_0}{\sqrt{u_0^2 - u^2}} \end{aligned} \quad (5.61)$$

Applying L' Hospital's rule for the limit  $u \rightarrow u_0$  we find that the integral vanishes. Hence, plugging in the lower limit  $u_*$  we get,

$$I_3 = -2aMu_0^2 \frac{u_* - u_0}{\sqrt{u_0^2 - u_*^2}} = 2aMu_0^2 \frac{u_0 - u_*}{\sqrt{u_0^2 - u_*^2}} \quad (5.62a)$$

$$= 2aMu_0^2 \frac{(u_0 - u_*)}{(u_0 - u_*)(u_0 + u_*)} \sqrt{u_0^2 - u_*^2} = 2aMu_0^2 \frac{\sqrt{u_0^2 - u_*^2}}{u_0 + u_*} \quad (5.62b)$$

We can finally substitute these integrals back in [eq. \(5.54\)](#) to evaluate  $\phi_{VS}$

$$\begin{aligned} \phi_{VS} &= \int_{u_S}^{u_0} \frac{1}{\sqrt{u_0^2 - u^2}} du + \int_{u_S}^{u_0} M \frac{(u_0^3 - u^3)}{(u_0^2 - u^2)^{\frac{3}{2}}} du - \int_{u_S}^{u_0} 2aM \frac{u_0^3 (u_0 - u)}{(u_0^2 - u^2)^{\frac{3}{2}}} du \\ &\quad + \int_{u_V}^{u_0} \frac{1}{\sqrt{u_0^2 - u^2}} du + \int_{u_V}^{u_0} M \frac{(u_0^3 - u^3)}{(u_0^2 - u^2)^{\frac{3}{2}}} du - \int_{u_V}^{u_0} 2aM \frac{u_0^3 (u_0 - u)}{(u_0^2 - u^2)^{\frac{3}{2}}} du \\ &= \left[ \frac{\pi}{2} - \sin^{-1} \left( \frac{u_S}{u_0} \right) + M \frac{(2u_0 + u_S)(u_0 - u_S)}{\sqrt{u_0^2 - u_S^2}} - 2aMu_0^2 \frac{u_0 - u_S}{\sqrt{u_0^2 - u_S^2}} \right] \\ &\quad + \left[ \frac{\pi}{2} - \sin^{-1} \left( \frac{u_V}{u_0} \right) + M \frac{(2u_0 + u_V)(u_0 - u_V)}{\sqrt{u_0^2 - u_V^2}} - 2aMu_0^2 \frac{u_0 - u_V}{\sqrt{u_0^2 - u_V^2}} \right] \\ &= \left[ \frac{\pi}{2} - \sin^{-1} \left( \frac{u_S}{u_0} \right) + \frac{Mu_0^2 \left( 2 - \frac{u_S}{u_0} - \frac{u_S^2}{u_0^2} \right)}{u_0 \sqrt{1 - \frac{u_S^2}{u_0^2}}} - \frac{2aMu_0^2 \times u_0 \left( 1 - \frac{u_S}{u_0} \right)}{u_0 \sqrt{1 - \frac{u_S^2}{u_0^2}}} \right] \\ &\quad + \left[ \frac{\pi}{2} - \sin^{-1} \left( \frac{u_V}{u_0} \right) + \frac{Mu_0^2 \left( 2 - \frac{u_V}{u_0} - \frac{u_V^2}{u_0^2} \right)}{u_0 \sqrt{1 - \frac{u_V^2}{u_0^2}}} - \frac{2aMu_0^2 \times u_0 \left( 1 - \frac{u_V}{u_0} \right)}{u_0 \sqrt{1 - \frac{u_V^2}{u_0^2}}} \right] \quad (5.63) \end{aligned}$$

In the terms involving  $\sin^{-1}$  we make the substitution  $\frac{1}{u_0} = b - M + 2aMu_0$  from [eq. \(5.52\)](#) and expand it in a Taylor series. As for the remaining terms we make the same substitution and neglect terms  $\geq M^2$ . The resulting equations are derived below.

$$\begin{aligned} \phi_{VS} &= \left[ \frac{\pi}{2} - \sin^{-1} \{u_S (b - M + 2aMu_0)\} + \frac{Mu_0^2 \left( 2 - \frac{u_S}{u_0} - \frac{u_S^2}{u_0^2} \right)}{u_0 \sqrt{1 - \frac{u_S^2}{u_0^2}}} - \frac{2aMu_0^2 \times u_0 \left( 1 - \frac{u_S}{u_0} \right)}{u_0 \sqrt{1 - \frac{u_S^2}{u_0^2}}} \right] \\ &\quad + \left[ \frac{\pi}{2} - \sin^{-1} \{u_V (b - M + 2aMu_0)\} + \frac{Mu_0^2 \left( 2 - \frac{u_V}{u_0} - \frac{u_V^2}{u_0^2} \right)}{u_0 \sqrt{1 - \frac{u_V^2}{u_0^2}}} - \frac{2aMu_0^2 \times u_0 \left( 1 - \frac{u_V}{u_0} \right)}{u_0 \sqrt{1 - \frac{u_V^2}{u_0^2}}} \right] \end{aligned}$$

$$\begin{aligned}
 &= \frac{\pi}{2} - \sin^{-1}(bu_S) + \frac{Mu_S}{\sqrt{1-b^2u_S^2}} - \frac{2aM}{b} \frac{u_S}{\sqrt{1-b^2u_S^2}} - \frac{Mu_S}{\sqrt{1-b^2u_S^2}} + \frac{M(2-b^2u_S^2)}{b\sqrt{1-b^2u_S^2}} \\
 &\quad - \frac{2aM}{b^2} \frac{1}{\sqrt{1-b^2u_S^2}} + \frac{2aM}{b} \frac{u_S}{\sqrt{1-b^2u_S^2}} \\
 &+ \frac{\pi}{2} - \sin^{-1}(bu_V) + \frac{Mu_V}{\sqrt{1-b^2u_V^2}} - \frac{2aM}{b} \frac{u_V}{\sqrt{1-b^2u_V^2}} - \frac{Mu_V}{\sqrt{1-b^2u_V^2}} + \frac{M(2-b^2u_V^2)}{b\sqrt{1-b^2u_V^2}} \\
 &\quad - \frac{2aM}{b^2} \frac{1}{\sqrt{1-b^2u_V^2}} + \frac{2aM}{b} \frac{u_V}{\sqrt{1-b^2u_V^2}} \\
 &= \pi - \sin^{-1}(bu_S) - \sin^{-1}(bu_V) + \frac{M}{b} \left[ \frac{2-b^2u_S^2}{\sqrt{1-b^2u_S^2}} + \frac{2-b^2u_V^2}{\sqrt{1-b^2u_V^2}} \right] \\
 &\quad - \frac{2aM}{b^2} \left[ \frac{1}{\sqrt{1-b^2u_S^2}} + \frac{1}{\sqrt{1-b^2u_V^2}} \right] \quad (5.64)
 \end{aligned}$$

### 5.3.3 Bending angle of a photon trajectory in Kerr metric from OIA formalism

Finally we can now substitute the values of  $\psi_V$  (from eq. (5.47)),  $\psi_S$  (from eq. (5.48)) &  $\phi_{VS}$  (from eq. (5.64)) into eq. (4.6b) to find the expression for deflection angle of a photon trajectory in Kerr spacetime.

$$\begin{aligned}
 \alpha_{D_{\text{Kerr}}}^{\text{pro}} &= \psi_V - \psi_S + \phi_{VS} \\
 &= \sin^{-1}(bu_V) - \frac{bMu_V^2}{\sqrt{1-b^2u_V^2}} + \frac{2aMu_V^2}{\sqrt{1-b^2u_V^2}} - \pi + \sin^{-1}(bu_S) - \frac{bMu_S^2}{\sqrt{1-b^2u_S^2}} + \frac{2aMu_S^2}{\sqrt{1-b^2u_S^2}} \\
 &\quad + \pi - \sin^{-1}(bu_S) - \sin^{-1}(bu_V) + \frac{2M}{b\sqrt{1-b^2u_S^2}} - \frac{bMu_S^2}{\sqrt{1-b^2u_S^2}} + \frac{2M}{b\sqrt{1-b^2u_V^2}} - \frac{bMu_V^2}{\sqrt{1-b^2u_V^2}} \\
 &\quad - \frac{2aM}{b^2} \frac{1}{\sqrt{1-b^2u_S^2}} - \frac{2aM}{b^2} \frac{1}{\sqrt{1-b^2u_V^2}} \\
 &= -\frac{2bMu_S^2}{\sqrt{1-b^2u_S^2}} - \frac{2bMu_V^2}{\sqrt{1-b^2u_V^2}} + \frac{2M}{b} \frac{1}{\sqrt{1-b^2u_S^2}} + \frac{2M}{b} \frac{1}{\sqrt{1-b^2u_V^2}} \\
 &\quad + \frac{2aMu_S^2}{\sqrt{1-b^2u_S^2}} + \frac{2aMu_V^2}{\sqrt{1-b^2u_V^2}} - \frac{2aM}{b^2} \frac{1}{\sqrt{1-b^2u_S^2}} - \frac{2aM}{b^2} \frac{1}{\sqrt{1-b^2u_V^2}} \\
 &= \frac{2M}{b} \frac{1}{\sqrt{1-b^2u_S^2}} (1-b^2u_S^2) + \frac{2M}{b} \frac{1}{\sqrt{1-b^2u_V^2}} (1-b^2u_V^2) \\
 &\quad - \frac{2aM}{b^2\sqrt{1-b^2u_S^2}} (1-b^2u_S^2) - \frac{2aM}{b^2\sqrt{1-b^2u_V^2}} (1-b^2u_V^2) \\
 &= \frac{2M}{b} \left( \sqrt{1-b^2u_S^2} + \sqrt{1-b^2u_V^2} \right) - \frac{2aM}{b^2} \left( \sqrt{1-b^2u_S^2} + \sqrt{1-b^2u_V^2} \right) \\
 \Rightarrow \alpha_{D_{\text{Kerr}}}^{\text{pro}} &= \left( \sqrt{1-b^2u_S^2} + \sqrt{1-b^2u_V^2} \right) \left( \frac{2M}{b} - \frac{2aM}{b^2} \right) + \mathcal{O} \left( \frac{M^2}{b^2} \right) \quad (5.65)
 \end{aligned}$$

The expression presented in [eq. \(5.65\)](#) is actually for the prograde case of the photon orbit in Kerr spacetime. For the retrograde case we simply invert the sign of the term linear in  $a$  giving us the orbit expression as

$$\alpha_{\text{D Kerr}}^{\text{retro}} = \left( \sqrt{1 - b^2 u_S^2} + \sqrt{1 - b^2 u_V^2} \right) \left( \frac{2M}{b} + \frac{2aM}{b^2} \right) + \mathcal{O} \left( \frac{M^2}{b^2} \right) \quad (5.66)$$

Note that for the asymptotic case  $r_S \& r_V \rightarrow \infty$  or  $u_S \& u_V \rightarrow 0$  the deflection angle for a photon trajectory reduces to the form already present in the existing literature [[Chandrasekhar \(1998\)](#), [Epstein and Shapiro \(1980\)](#) & [Ibanez \(1983\)](#)].

$$\alpha_{\text{D Kerr}}^{\text{pro}} \rightarrow \frac{4M}{b} - \frac{4aM}{b^2} + \mathcal{O} \left( \frac{M^2}{b^2} \right) \quad (5.67a)$$

$$\alpha_{\text{D Kerr}}^{\text{retro}} \rightarrow \frac{4M}{b} + \frac{4aM}{b^2} + \mathcal{O} \left( \frac{M^2}{b^2} \right) \quad (5.67b)$$

These same results are obtained by working out the expression presented in [eq. \(4.6a\)](#). This is basically an extension of the formalism initially developed in [Gibbons and Werner \(2008\)](#) and improved on by—[Ono, Ishihara, and Asada \(2017\)](#) & [Ono and Asada \(2019\)](#). The content in [chapter 6](#) presents the detailed exposition of this technique while illustrating the equivalence of the two separate formalism.

*"Well, in our country," said Alice, still panting a little, "you'd generally get to somewhere else—if you run very fast for a long time, as we've been doing."*

*"A slow sort of country!" said the Queen. "Now, here, you see, it takes all the running you can do, to keep in the same place. If you want to get somewhere else, you must run at least twice as fast as that!"*

*—Lewis Carroll*

# 6

## BENDING ANGLE OF LIGHT IN AXISYMMETRIC SPACETIME BY GW-OIA FORMALISM

As mentioned earlier in [section 4.3](#) the bending angle of a photon trajectory can be evaluated in terms of the surface integral of the Gaussian curvature  $K$  and line integral of the geodesic curvature  $\kappa_g$ . Although we have discussed some critical geometric aspects related to photon trajectory & Fermat metric in [section 4.2](#) & [section 5.1](#) any significant discussion on Gaussian and geodesic curvature has not been presented so far. However, in order to fully appreciate the Gibbons-Werner—Ono-Ishihara-Asada formalism it is essential to have a concrete idea of these quantities.

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### 6.1 GAUSSIAN CURVATURE & GEODESIC CURVATURE

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The Gaussian curvature and the geodesic curvature are perhaps the most critical parameters associated with any optical geometry responsible for describing the photon trajectory in any spacetime [[Qiao and Li \(2022\)](#)]. A very concise but succinct definition of these two quantities may be given as :-

- **Gaussian curvature**– It is the intrinsic curvature of a two dimensional surface. This gives us a measure of the departure of the surface from being intrinsically flat.
- **Geodesic curvature**– It gives us a measure of the departure of any one-dimensional curve on a two dimensional curved surface from being a geodesic.

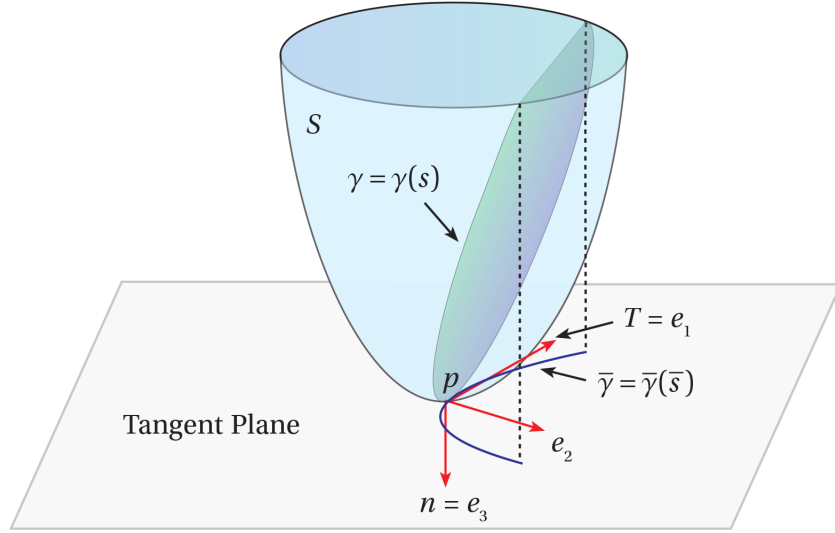


Figure 6.1: Embedding of a three dimensional space with associated curves and surfaces relevant to the geometry.

Therefore, it automatically follows that the Gaussian curvature of an intrinsically flat surface must be zero and the geodesic curvature of any geodesic curve must be zero. It is worth noting that both the Gaussian curvature as well as the geodesic curvature strictly relies on the intrinsic metric of the two dimensional surface, *i.e.*, there is no bearing on the embedding of this surface in higher dimensional spacetime.

For the sake of completeness we must also remember that there is another parameter called the **normal curvature**  $\kappa_n$  which is actually an extrinsic property of the two dimensional surface and depends on the embedding of the two dimensional surface in higher dimensions. Although for our purposes the normal curvature is not a necessary quantity. The **total curvature**  $\kappa$  of a surface is given as  $\sqrt{\kappa_g^2 + \kappa_n^2}$ .

The embedding of a surface  $S$  in three dimensional Euclidean space is depicted in [fig. \[6.1\]](#). The basic features and characteristics of these quantities are tabulated in [table. 6.1](#). The four different flavours of curvature associated with a curve on any particular spacetime geometry are discussed with their respective definitions, dependencies and the null curvature criteria. In particular as discussed on several instances earlier the consequences of the null curvature criteria has significant implications for photon geodesics as well as massive particle trajectories.

**Table 6.1:** Characteristics of different types of curvatures associated with a curve on a 2D surface

|                           | Symbol     | Definition                                                                                  | Dependency                                                                                     | Null curvature criteria                                                                 |
|---------------------------|------------|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <b>Gaussian curvature</b> | $K$        | It is the intrinsic curvature of a two dimensional surface.                                 | Independent of the embedding of surface $S$ into higher dimensional space. Intrinsic property. | $K = 0$ implies that the two dimensional surface $S$ is intrinsically flat.             |
| <b>Geodesic curvature</b> | $\kappa_g$ | The curvature of a curve $\gamma$ in a two dimensional surface $S$ .                        | Independent of the embedding of surface $S$ into higher dimensional space. Intrinsic property. | $\kappa_g = 0$ implies that the curve $\gamma$ on the surface $S$ is a geodesic.        |
| <b>Normal curvature</b>   | $\kappa_n$ | It is the measure of the curvature associated with the surface rather than the curve on it. | Dependent on the embedding of surface $S$ into higher dimensional space. Extrinsic property.   | $\kappa_n = 0$ implies that the curvature of the surface is zero.                       |
| <b>Total curvature</b>    | $\kappa$   | Curvature of a curve $\gamma$ when embedded in a higher dimensional space.                  | Dependent on the embedding of surface $S$ into higher dimensional space. Extrinsic property.   | $\kappa = 0$ implies that the curve $\gamma$ is a geodesic in higher dimensional space. |

## 6.2 SOLVING THE ORBIT EQUATION TO EVALUATE $r_{VS}$

In order to solve the surface integral of the Gaussian curvature we first need to solve the orbit equation derived in [section 5.2.2](#) as eq. (5.28). We simply have to substitute the appropriate values of  $A$ ,  $B$ ,  $D$  &  $H$  to find the solution of the orbit equation.

$$\begin{aligned}
\left(\frac{dr}{d\phi}\right)^2 &= \frac{AD + H^2}{B} \times \frac{D - 2Hb - Ab^2}{(H + Ab)^2} \\
&= \frac{\left(1 - \frac{2Mr}{r^2}\right) \left(r^2 + a^2 + \frac{2a^2Mr}{r^2}\right) + \left(\frac{2aMr}{r^2}\right)^2}{r^2} \\
&\quad \times \frac{r^2 - 2Mr + a^2}{\left(r^2 + a^2 + \frac{2a^2Mr}{r^2}\right) - \frac{4aMrb}{r^2} - \left(1 - \frac{2Mr}{r^2}\right)b^2} \\
&\quad \times \frac{\left[\frac{2aMr}{r^2} + \left(1 - \frac{2Mr}{r^2}\right)b\right]^2}{\left(1 - \frac{2M}{r}\right) \left(r^2 + a^2 + \frac{2a^2M}{r}\right) + \frac{4a^2M^2}{r^2}} \times (r^2 - 2Mr + a^2)
\end{aligned}$$

$$\begin{aligned}
 & \times \frac{\left(r^2 + a^2 + \frac{2a^2 M}{r}\right) - \frac{4aMb}{r} - \left(1 - \frac{2M}{r}\right)b^2}{\left(\frac{2aM}{r} + b - \frac{2Mb}{r}\right)^2} \\
 &= \frac{r^2 + a^2 + \frac{2a^2 M}{r} - 2Mr - \frac{2a^2 M}{r} - \frac{4a^2 M^2}{r^2} + \frac{4a^2 M^2}{r^2}}{r^2} \times (r^2 - 2Mr + a^2) \\
 & \times \frac{r^2 + a^2 + \frac{2a^2 M}{r} - \frac{4abM}{r} - b^2 + \frac{2Mb^2}{r}}{\left[\frac{b^2}{r} \left(\frac{2aM}{b^2} + \frac{r}{b} - \frac{2M}{b}\right)\right]^2} \\
 &= \frac{(r^2 + a^2 - 2Mr)^2}{r^2} \times \frac{r^2 + a^2 + \frac{2a^2 M}{r} - \frac{4abM}{r} - b^2 + \frac{2Mb^2}{r}}{\frac{b^4}{r^2} \left(\frac{2aM}{b^2} + \frac{r}{b} - \frac{2M}{b}\right)^2} \\
 &= \frac{(r^2 + a^2 - 2Mr)^2 \times \left(r^2 + a^2 + \frac{2a^2 M}{r} - \frac{4abM}{r} - b^2 + \frac{2Mb^2}{r}\right)}{b^4 \left(\frac{2aM}{b^2} + \frac{r}{b} - \frac{2M}{b}\right)^2} \\
 &= \frac{b^4 \left(\frac{r^2}{b^2} + \frac{a^2}{b^2} - \frac{2Mr}{b^2}\right)^2 \times \frac{b^3}{r} \left(\frac{r^3}{b^3} + \frac{a^2 r}{b^3} + \frac{2a^2 M}{b^3} - \frac{4aM}{b^2} - \frac{r}{b} + \frac{2M}{b}\right)}{b^4 \left(\frac{2aM}{b^2} + \frac{r}{b} - \frac{2M}{b}\right)^2} \\
 &= \frac{b^2 \left[\frac{r}{b} \left(\frac{r}{b} - \frac{2M}{b}\right) + \frac{a^2}{b^2}\right]^2 \left[\frac{a^2}{b^2} \left(\frac{2M}{b} + \frac{r}{b}\right) - \frac{4aM}{b^2} + \frac{2M}{b} - \frac{r}{b} + \frac{r^3}{b^3}\right]}{\frac{r}{b} \left(\frac{2aM}{b^2} + \frac{r}{b} - \frac{2M}{b}\right)^2} \\
 &= \frac{b^2 \left[\frac{a^2}{b^2} + \frac{r}{b} \left(\frac{r}{b} - \frac{2M}{b}\right)\right]^2 \left[\frac{a^2}{b^2} \left(\frac{2M}{b} + \frac{r}{b}\right) - \frac{4aM}{b^2} + \frac{2M}{b} - \frac{r}{b} + \frac{r^3}{b^3}\right]}{\frac{r}{b} \left(\frac{2aM}{b^2} + \frac{r}{b} - \frac{2M}{b}\right)^2} \\
 &= \frac{b^2 \left[\frac{r}{b} \left(\frac{r}{b} - \frac{2M}{b}\right)\right]^2 \left[-\frac{4aM}{b^2} + \frac{2M}{b} - \frac{r}{b} + \frac{r^3}{b^3}\right]}{\frac{r}{b} \left(\frac{2aM}{b^2} + \frac{r}{b} - \frac{2M}{b}\right)^2} \tag{6.1}
 \end{aligned}$$

Substituting  $\frac{r}{b} = x$  in eq. (6.1) we get,

$$\left(\frac{dr}{d\phi}\right)^2 = \frac{b^2 \left[x \left(x - \frac{2M}{b}\right)\right]^2 \left[-\frac{4aM}{b^2} + \frac{2M}{b} - x + x^3\right]}{x \left[x + \frac{2aM}{b^2} - \frac{2M}{b}\right]^2}$$

$$\begin{aligned}
 &= \frac{b^2 x^2 \left( x^2 + \frac{4M^2}{b^2} - \frac{4Mx}{b} \right) \left( -\frac{4aM}{b^2} + \frac{2M}{b} - x + x^3 \right)}{x^3 \left[ 1 - \left( \frac{2M}{bx} - \frac{2aM}{b^2 x} \right) \right]^2} \\
 &= \frac{\frac{b^2 x^2}{x^3} \left( x^2 - \frac{4Mx}{b} \right) \left( -\frac{4aM}{b^2} + \frac{2M}{b} - x + x^3 \right)}{\left[ 1 - \frac{2M}{bx} \left( 1 - \frac{a}{b} \right) \right]^2} \\
 &= \frac{b^2 \left( x - \frac{4M}{b} \right) \left( -\frac{4aM}{b^2} + \frac{2M}{b} - x + x^3 \right)}{\left[ 1 + \frac{4M^2}{b^2 x^2} \left( 1 - \frac{a}{b} \right)^2 - \frac{4M}{bx} \left( 1 - \frac{a}{b} \right) \right]} \\
 &= \frac{b^2 \left( x - \frac{4M}{b} \right) \left( -\frac{4aM}{b^2} + \frac{2M}{b} - x + x^3 \right)}{1 - \frac{4M}{bx} \left( 1 - \frac{a}{b} \right)} \\
 &= b^2 \left( x - \frac{4M}{b} \right) \left( -\frac{4aM}{b^2} + \frac{2M}{b} - x + x^3 \right) \left( 1 + \frac{4M}{bx} - \frac{4aM}{b^2 x} \right) \\
 &= b^2 \left( -\frac{4aM}{b^2} + \frac{2M}{b} - x + x^3 \right) \left( x + \frac{4M}{b} - \frac{4aM}{b^2} - \frac{4M}{b} \right) \\
 &\simeq b^2 \left( -\frac{4aMx}{b^2} + \frac{2Mx}{b} - x^2 + \frac{4aMx}{b^2} + x^4 - \frac{4aMx^3}{b^2} \right) \\
 &= b^2 \left( x^4 - x^2 + \frac{2Mx}{b} - \frac{4aMx^3}{b^2} \right) \\
 &= b^2 \frac{r^4}{b^4} - b^2 \frac{r^2}{b^2} + b^2 \frac{2M}{b} \frac{r}{b} - b^2 \frac{4aM}{b^2} \frac{r^3}{b^3} \quad [\text{Substituting } x = \frac{r}{b}] \\
 &= \frac{r^4}{b^2} - r^2 + 2Mr - \frac{4r^3 aM}{b^3} + \mathcal{O}(a^2) \tag{6.2}
 \end{aligned}$$

We have neglected any terms involving  $M^2$ . In terms of the reciprocal coordinates we can write [eq. \(6.2\)](#) as

$$\begin{aligned}
 \left( \frac{du}{d\phi} \right)^2 &= F(u) = u^4 \left( \frac{dr}{d\phi} \right)^2 \\
 &= u^4 \left[ \frac{r^4}{b^2} - r^2 + 2Mr - \frac{4r^3 aM}{b^3} \right] + \mathcal{O}(a^2 u^4) \\
 &= \frac{1}{b^2} - u^2 + 2Mu^3 - \frac{4aMu}{b^3} + \mathcal{O}(a^2 u^4) \tag{6.3}
 \end{aligned}$$

The solution of the [eq. \(6.3\)](#) can be solved iteratively by taking [eq. \(6.3\)](#) and truncating it. We proceed by finding the zeroth order solution first followed by linear order solution so on and so forth.

The zeroth order solution is found by solving the equation,

$$\left( \frac{du}{d\phi} \right)^2 = \frac{1}{b^2} - u^2 + \mathcal{O}(Mu^3, aMu^4, a^2 u^4)$$

$$\begin{aligned}
 \Rightarrow du &= \sqrt{\frac{1}{b^2} - u^2} d\phi \\
 \Rightarrow \frac{b}{\sqrt{1 - b^2 u^2}} du &= d\phi \\
 \Rightarrow \sin^{-1}(bu) &= \phi + C \quad [\text{Integrating both sides}]
 \end{aligned} \tag{6.4}$$

Since the distance of closest approach in the present case is given by  $r = r_0 = \frac{1}{u_0}$ ,  $\phi = \frac{\pi}{2}$  we can employ the boundary condition  $\left(\frac{du}{d\phi}\right)_{\phi=\frac{\pi}{2}} = 0$ . This in turn yields the value of the constant of integration as 0. Hence, the zeroth order solution is

$$u_0 = \frac{\sin \phi}{b} \tag{6.5}$$

Similarly, we can assume that the linear order solution with  $M$  is in the form

$$\begin{aligned}
 u &= u_0(\phi) + u_1(\phi)M \\
 \Rightarrow u &= \frac{\sin \phi}{b} + u_1(\phi)M
 \end{aligned} \tag{6.6}$$

In order to find  $u_1(\phi)$  we simply need to substitute [eq. \(6.6\)](#) in [eq. \(6.3\)](#) with the terms bearing a linear dependency on  $M$ . This leads upto the desired solution as follows,

$$\begin{aligned}
 \left(\frac{du}{d\phi}\right)^2 &= \frac{1}{b^2} - u^2 + 2Mu^3 + \mathcal{O}(aMu^4, a^2u^4) \\
 \Rightarrow \left[\frac{d}{d\phi} \left(\frac{\sin \phi}{b} + u_1(\phi)M\right)\right]^2 &= \frac{1}{b^2} - \frac{\sin^2 \phi}{b^2} + u_1^2(\phi)M^2 + 2\frac{\sin \phi}{b}u_1(\phi)M \\
 &\quad + 2M \left[\frac{\sin^3 \phi}{b^3} + u_1^3(\phi)M^3 + \frac{3M}{b} \sin \phi u_1(\phi) \left(\frac{\sin \phi}{b} + u_1(\phi)M\right)\right] \\
 \Rightarrow \left[\frac{\cos \phi}{b} + u_1'(\phi)M\right]^2 &\simeq \frac{1}{b^2} - \frac{\sin^2 \phi}{b^2} - \frac{2M}{b} \sin \phi u_1(\phi) + \frac{2M}{b^3} \sin^3 \phi \\
 \Rightarrow \frac{\cos^2 \phi}{b^2} + \frac{2M}{b} \cos \phi u_1'(\phi) &= \left(\frac{1}{b^2} - \frac{\sin^2 \phi}{b^2}\right) - \frac{2M}{b} \sin \phi u_1(\phi) + \frac{2M}{b^3} \sin^3 \phi \\
 \Rightarrow \frac{2M}{b} \cos \phi u_1'(\phi) &= -\frac{2M}{b} \sin \phi \left[u_1(\phi) - \frac{\sin^2 \phi}{b^2}\right] \\
 \Rightarrow \cos \phi \frac{du_1(\phi)}{d\phi} &= -\sin \phi \left[u_1(\phi) - \frac{\sin^2 \phi}{b^2}\right] \\
 \Rightarrow \frac{du_1(\phi)}{d\phi} + \tan \phi u_1(\phi) - \frac{1}{b^2} \tan \phi \sin^2 \phi &= 0
 \end{aligned} \tag{6.7}$$

We can easily solve the differential equation in [eq. \(6.7\)](#) by employing the boundary condition  $\frac{du_1}{d\phi} = 0$  at

$\phi = \frac{\pi}{2}$ . The equation yields the linear order solution of  $u$  as

$$u_1(\phi) = \frac{1}{b^2} (1 + \cos^2 \phi) \quad (6.8)$$

Therefore, the solution with  $a$  can be said to be

$$u = \frac{\sin \phi}{b} + \frac{M}{b^2} (1 + \cos^2 \phi) + u_2(\phi)a \quad (6.9)$$

However, it is evident from [eq. \(6.3\)](#) that there are no terms which bear a linear dependency on  $a$ . As such we can conclude that  $u_2(\phi) = 0$ . Finally, considering the term involving  $aM$  we assume the solution to be

$$u = \frac{\sin \phi}{b} + \frac{M}{b^2} (1 + \cos^2 \phi) + u_3(\phi)aM \quad (6.10)$$

Let us now substitute [eq. \(6.10\)](#) in [eq. \(6.3\)](#) to find  $u_3(\phi)$ .

$$\begin{aligned} \left( \frac{du}{d\phi} \right)^2 &= \frac{1}{b^2} - u^2 + 2Mu^3 - \frac{4uM}{b^3} + \mathcal{O}(a^2u^4) \\ \Rightarrow \left[ \frac{d}{d\phi} \left\{ \frac{\sin \phi}{b} + \frac{M}{b^2} (1 + \cos^2 \phi) + u_3(\phi)aM \right\} \right]^2 &= \frac{1}{b^2} - \frac{\sin^2 \phi}{b^2} - M^2 \left[ \frac{1}{b^2} (1 + \cos^2 \phi) + u_3(\phi)a \right]^2 - \frac{2 \sin \phi M}{b} \left[ \frac{1}{b^2} (1 + \cos^2 \phi) + u_3(\phi)a \right] \\ &\quad + 2M \frac{\sin^3 \phi}{b^3} - \frac{4aM \sin \phi}{b^3} \\ \Rightarrow \left[ \frac{\cos \phi}{b} - \frac{2M}{b^2} \sin \phi \cos \phi + aM u'_3(\phi) \right]^2 &\simeq \frac{1}{b^2} - \frac{\sin^2 \phi}{b^2} - \frac{2M}{b^3} (1 + \cos^2 \phi) \sin \phi - \frac{2aM}{b} \sin \phi u_3(\phi) + \frac{2M}{b^3} \sin^3 \phi - \frac{4aM}{b^4} \sin \phi \\ \Rightarrow \frac{\cos^2 \phi}{b^2} + \frac{2aM}{b} \cos \phi u'_3(\phi) - \frac{4M}{b^3} \sin \phi \cos^2 \phi &= \frac{1}{b^2} - \frac{\sin^2 \phi}{b^2} - \frac{2M}{b^3} \sin \phi - \frac{2M}{b^3} \sin \phi \cos^2 \phi - \frac{2aM}{b} \sin \phi u_3(\phi) + \frac{2M}{b^3} \sin^3 \phi - \frac{4aM}{b^4} \sin \phi \\ \Rightarrow \frac{2aM}{b} \cos \phi u'_3(\phi) &= -\frac{2aM}{b} \sin \phi u_3(\phi) - \frac{4aM}{b^4} \sin \phi \\ \Rightarrow \frac{aM}{b} \cos \phi u'_3(\phi) + \frac{aM}{b} \sin \phi u_3(\phi) + \frac{2aM}{b^4} \sin \phi &= 0 \\ \Rightarrow u'_3(\phi) + \tan \phi u_3(\phi) &= -\frac{2}{b^3} \tan \phi \\ \Rightarrow u'_3 \sec \phi + \sec \phi \tan \phi u_3(\phi) &= -\frac{2}{b^3} \sec \phi \tan \phi \\ \Rightarrow \frac{d}{d\phi} [u_3(\phi) \sec \phi] &= -\frac{2}{b^3} \sec \phi \tan \phi \\ \Rightarrow u_3(\phi) \sec \phi &= -\frac{2}{b^3} \sec \phi + C_3 \end{aligned} \quad (6.11)$$

Applying the boundary condition we find the value of the integration constant to be zero. Thus, the solution of  $u_3$  is

$$u_3 = -\frac{2}{b^3} \quad (6.12)$$

The complete solution of  $u$  is therefore given as,

$$\begin{aligned} u &= u_0 + Mu_1 + au_2 + aMu_3 \\ &= \frac{\sin \phi}{b} + \frac{M}{b^2} (1 + \cos^2 \phi) - \frac{2aM}{b^3} + \mathcal{O}\left(\frac{M^2}{b^3}, \frac{a^2}{b^3}\right) \end{aligned} \quad (6.13)$$

We can further manipulate [eq. \(6.13\)](#) with the help of Taylor expansion to get a solution of  $\phi$  as demonstrated below,

$$\begin{aligned} ub &= \sin \phi + \frac{M}{b} (1 + 1 - \sin^2 \phi) - \frac{2aM}{b^2} \\ &= \sin \phi + \frac{2M}{b} - \frac{M}{b} \sin^2 \phi - \frac{2aM}{b^2} \\ \Rightarrow \phi &= \arcsin(bu) - \frac{2M}{b} \frac{d}{du} [\arcsin(bu)] + \frac{M}{b} b^2 u^2 \frac{d}{du} [\arcsin(bu)] + \frac{2aM}{b^2} \frac{d}{du} [\arcsin(bu)] \\ &= \arcsin(bu) - \frac{2M}{b} \frac{1}{\sqrt{1-b^2u^2}} + \frac{Mb^2u^2}{b\sqrt{1-b^2u^2}} + \frac{2aM}{b^2\sqrt{1-b^2u^2}} \end{aligned} \quad (6.14)$$

The domain of  $\phi$  can be chosen to lie in the range  $-\pi \leq \phi < \pi$  without compromising generality. With consideration to the source location the angular coordinate  $\phi_s$  will be contained in the domain  $-\frac{\pi}{2} \leq \phi_s < \frac{\pi}{2}$  whereas for the viewer location the angular coordinate  $\phi_v$  will abide by the constraint  $|\phi_v| > \frac{\pi}{2}$ . Therefore, within the specific domains the solution of  $\phi$  will be,

**For the domain**  $\left[|\phi| < \frac{\pi}{2}\right]$

$$\phi = \arcsin(bu) + \frac{-2 + b^2u^2}{b\sqrt{1-b^2u^2}}M + \frac{2aM}{b^2\sqrt{1-b^2u^2}} + \mathcal{O}\left(\frac{M^2}{b^3}, \frac{a^2}{b^3}\right) \quad (6.15a)$$

**For the domain**  $\left[\frac{\pi}{2} < |\phi|\right]$

$$\phi = \pi - \arcsin(bu) - \frac{-2 + b^2u^2}{b\sqrt{1-b^2u^2}}M - \frac{2aM}{b^2\sqrt{1-b^2u^2}} + \mathcal{O}\left(\frac{M^2}{b^3}, \frac{a^2}{b^3}\right) \quad (6.15b)$$

It is worth noting in this context that  $|bu| < 1$  since the square roots involved in [eq. \(6.15\)](#) cannot be imaginary or equate to zero, and of course both  $b$  and  $u$  should be positive. As such  $bu$  should abide by the condition  $0 < bu < 1$ . We can now proceed with evaluating the surface integral of the Gaussian curvature which requires the values  $\phi_v$  and  $\phi_s$  as the limits of the integral.

### 6.3 SURFACE INTEGRAL OF THE GAUSSIAN CURVATURE FOR KERR METRIC (IN EQUATORIAL PLANE)

For the purpose of the present discussion the mathematical formulation of the Gaussian curvature can be defined in terms of the 2-dimensional Riemann tensor and the Fermat metric as follows,

$$K = \frac{R_{r\phi r\phi}}{\gamma_{ij}^{(F)}} = \frac{1}{\sqrt{\det \gamma_{ij}^{(F)}}} \left[ \frac{\partial}{\partial \phi} \left( \frac{\sqrt{\det \gamma_{ij}^{(F)}}}{\gamma_{rr}^{(F)}} \Gamma_{rr}^\phi \right) - \frac{\partial}{\partial r} \left( \frac{\sqrt{\det \gamma_{ij}^{(F)}}}{\gamma_{rr}^{(F)}} \Gamma_{r\phi}^\phi \right) \right] \quad (6.16)$$

The terms  $R_{r\phi r\phi}$  and  $\Gamma_{jk}^i$  can be expressed in terms of the Fermat metric  $\gamma_{ij}^{(F)}$  as defined on the equatorial plane. The surface element is defined in terms of the Fermat metric as  $dS = \sqrt{\det \gamma_{ij}^{(F)}} dr d\phi$ . This helps us to express the surface integral of the Gaussian curvature as required by eq. (4.6a) in the form,

$$\iint_{\phi_S \square_S^\infty} K dS = \int_{\phi_S}^{\phi_V} \int_{r_{VS}}^{\infty} K \sqrt{\det \gamma_{ij}^{(F)}} dr d\phi \quad (6.17)$$

where  $r_{VS}$  represents the solution of the orbit equation.

Evaluating the Christoffel symbols we find that  $\Gamma_{rr}^\phi = 0$  and  $\Gamma_{r\phi}^\phi = \frac{1}{2} \times \frac{A^2}{AD + H^2} \times \frac{\partial}{\partial r} \left( \frac{AD + H^2}{A^2} \right)$ . Upon substituting these values in eq. (6.16) we find that

$$K = \sqrt{\frac{A^3}{B(AD + H^2)}} \times \frac{\partial}{\partial r} \left[ \frac{1}{2} \sqrt{\frac{A^3}{B(AD + H^2)}} \times \frac{\partial}{\partial r} \left( \frac{AD + H^2}{A^2} \right) \right] \quad (6.18)$$

A simple but lengthy bit of algebra is required to give the expression for the Gaussian curvature as

$$K = \frac{M [-6r(a^2 + M^2) + 6a^2M + 7Mr^2 - 2r^3]}{r^5(r - 2M)} \quad (6.19)$$

Under the weak-field and slow rotation approximations eq. (6.19) can be further simplified to the form

$$K = -\frac{2M}{r^3} + \mathcal{O} \left( \frac{M^2}{r^4}, \frac{a^2 M}{r^5} \right) \quad (6.20)$$

The surface element of the Fermat metric  $dS = \sqrt{\det \gamma_{ij}^{(F)}} dr d\phi$  can be similarly simplified as follows

$$dS = \sqrt{\det \gamma_{ij}^{(F)}} dr d\phi$$

$$\begin{aligned}
 &= \sqrt{\frac{B(AD + H^2)}{A^3}} dr d\phi \\
 &= \sqrt{\frac{\frac{r^2}{r^2 - 2Mr + a^2} \times \left(1 - \frac{2M}{r}\right) \left(r^2 + a^2 + \frac{2a^2 M}{r}\right) + \left(\frac{2aM}{r}\right)^2}{\left(1 - \frac{2M}{r}\right)^3}} dr d\phi \\
 &= \sqrt{\frac{r^2 \left(1 - \frac{2M}{r}\right) \left(1 + \frac{2M}{r}\right)}{\left(1 - \frac{2M}{r}\right)^3}} dr d\phi \\
 &= r \times \left(1 + \frac{3}{2} \times \frac{2M}{r}\right) dr d\phi \\
 &\Rightarrow dS = \left[r + 3M + \mathcal{O}\left(\frac{M^2}{r}\right)\right] dr d\phi
 \end{aligned} \tag{6.21}$$

where we have neglected the terms involving  $a^2$  &  $M^2$ .

Now, we can finally proceed with evaluating the surface integral of the Gaussian curvature on the equatorial plane, by substituting the values of  $r_{VS}$ ,  $\phi_V$  &  $\phi_S$  at the appropriate vacancies.

$$\begin{aligned}
 \iint_{\infty \square_S^\infty} K dS &= \int_{\phi_S}^{\phi_V} \int_{r_{VS}}^{\infty} K \sqrt{\det \gamma_{ij}^{(F)}} dr d\phi \\
 &= - \int_{\phi_S}^{\phi_V} \int_{\infty}^{r_{VS}} K \sqrt{\det \gamma_{ij}^{(F)}} dr d\phi \\
 &= - \int_{\phi_S}^{\phi_V} \int_{r_{VS}}^{\infty} -\frac{2M}{r^3} (r + 3M) dr d\phi + \mathcal{O}\left(\frac{M^2}{b^2}, \frac{aM^2}{b^3}, \frac{a^2 M}{b^3}\right) \\
 &= -2M \int_{\phi_S}^{\phi_V} \int_0^{\frac{\sin \phi}{b} + \frac{M}{b^2} (1 + \cos^2 \phi) - \frac{2aM}{b^3}} du d\phi + \mathcal{O}\left(\frac{M^2}{b^2}, \frac{aM^2}{b^3}, \frac{a^2 M}{b^3}\right) \\
 &= -2M \int_{\phi_S}^{\phi_V} \frac{\sin \phi}{b} d\phi + \mathcal{O}\left(\frac{M^2}{b^2}, \frac{aM^2}{b^3}, \frac{a^2 M}{b^3}\right) \quad [\because \text{Neglecting terms involving } M^2] \\
 &= -\frac{2M}{b} [\cos \phi_S - \cos \phi_V] + \mathcal{O}\left(\frac{M^2}{b^2}, \frac{aM^2}{b^3}, \frac{a^2 M}{b^3}\right) \\
 &= -\frac{2M}{b} \left[\sqrt{1 - b^2 u_S^2} + \sqrt{1 - b^2 u_V^2}\right] + \mathcal{O}\left(\frac{M^2}{b^2}, \frac{aM^2}{b^3}, \frac{a^2 M}{b^3}\right)
 \end{aligned} \tag{6.22}$$

## 6.4 EVALUATING THE PATH INTEGRAL OF GEODESIC CURVATURE (IN EQUATORIAL PLANE)

In order to establish the final form of the deflection angle we now need to evaluate the path integral of the geodesic curvature. In this regard, the geodesic curvature [do Carmo (1976), Oprea (2019)] can be vectorially expressed as

$$\kappa_g \equiv \vec{T}' \cdot (\vec{T} \times \vec{N}) \quad (6.23)$$

where,

$\vec{T}$  – unit tangent vector for the curve after reparameterizing the curve using its arc length.

$\vec{T}'$  – derivative of  $T$  w.r.t the parameter

$\vec{N}$  – unit normal to the surface under consideration

Let us now revisit eq. (5.13) & eq. (5.14) and define the travel time of a photon from the source to the viewer in terms of the integral of eq. (5.5),

$$T = \int_{t_s}^{t_v} dt = \int_S^V \left( \sqrt{\gamma_{ij} de^i de^j} + \beta_i de^i \right) d\ell_{ph} \quad (6.24)$$

In terms of the travel time of a photon the Fermat's principle dictates that  $\delta T = 0$ . Therefore, the Lagrangian for a photon can be expressed as

$$\mathcal{L} = \sqrt{\gamma_{ij} e^i e^j} + \beta_i e^i \quad (6.25)$$

Let us now determine the Euler Lagrange equation. At first we determine the value of  $\frac{\partial \mathcal{L}}{\partial x^k}$  as

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial x^k} &= \beta_{i,k} \dot{x}^i + \frac{1}{2} \frac{1}{\sqrt{\gamma_{ik} \dot{x}^i \dot{x}^k}} \gamma_{ij,k} \dot{x}^i \dot{x}^j \\ &= \beta_{i,k} e^i + \frac{1}{2} \gamma_{ij,k} e^i e^j \end{aligned} \quad (6.26)$$

as  $\gamma_{ik} e^i e^k = 1$ .

Similarly,  $\frac{d}{d\ell_{ph}} \left( \frac{\partial \mathcal{L}}{\partial e^k} \right)$  can be evaluated as shown below,

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial e^k} &= \frac{\partial}{\partial e^k} \left( \sqrt{\gamma_{ij} e^i e^j} + \beta_i e^i \right) \\ &= \beta_k + \frac{1}{2} \frac{1}{\sqrt{\gamma_{ij} e^i e^j}} \frac{\partial}{\partial e^k} (\gamma_{ij} e^i e^j) \end{aligned}$$

$$\begin{aligned}
 &= \beta_k + \frac{1}{2} \frac{1}{\sqrt{\gamma_{ij} e^i e^j}} \left[ \gamma_{ij} \delta_k^i e^j + \gamma_{ij} \delta_k^j e^i \right] \\
 &= \beta_k + \frac{1}{2} \frac{1}{\sqrt{\gamma_{ij} e^i e^j}} \left[ \gamma_{kj} e^j + \gamma_{ik} e^i \right] \\
 &= \beta_k + \frac{1}{2} \frac{1}{\sqrt{\gamma_{ij} e^i e^j}} \left[ \gamma_{ki} e^i + \gamma_{ik} e^i \right] \\
 &= \beta_k + \frac{1}{2} \frac{1}{\sqrt{\gamma_{ij} e^i e^j}} \times 2 \gamma_{ik} e^i \\
 &= \beta_k + \gamma_{ik} e^i
 \end{aligned} \tag{6.27}$$

Now, using  $\frac{d}{d\ell_{ph}} \equiv \frac{d}{dx^i} \frac{dx^i}{d\ell_{ph}} \equiv \frac{d}{dx^i} e^i$  we can proceed as follows-

$$\begin{aligned}
 \frac{d}{d\ell_{ph}} \left( \frac{\partial \mathcal{L}}{\partial e^k} \right) &= \frac{\partial \beta_k}{\partial x^i} e^i + \frac{\partial \gamma_{ik}}{\partial x^a} e^a e^i + \frac{\partial e^i}{\partial x^a} e^l \gamma_{ik} \\
 &= \beta_{k,i} e^i + \gamma_{ik} e_{,a}^i e^a + \gamma_{ik,a} e^i e^a
 \end{aligned} \tag{6.28}$$

Finally, we can plug in [eq. \(6.26\)](#) & [eq. \(6.28\)](#) in the Euler-Lagrange equation to get the equation for the photon orbit.

$$\begin{aligned}
 &\frac{d}{d\ell_{ph}} \left( \frac{\partial \mathcal{L}}{\partial e^k} \right) - \frac{\partial \mathcal{L}}{\partial x^k} = 0 \\
 &\Rightarrow \gamma_{ik} e_{,a}^i e^a + \gamma_{ik,a} e^i e^a + \beta_{k,i} e^i - \frac{1}{2} \gamma_{ij,k} e^i e^j - \beta_{i,k} e^i = 0 \\
 &\Rightarrow e_{,a}^i e^a \gamma_{ik} + \gamma_{ik,a} e^i e^a - \frac{1}{2} \gamma_{ij,k} e^i e^j + \beta_{k,i} e^i - \beta_{i,k} e^i = 0 \\
 &\Rightarrow e_{,a}^i e^a + \gamma^{kj} \left[ \gamma_{ik,a} e^i e^a - \frac{1}{2} \gamma_{ia,k} e^i e^a \right] = \gamma^{kj} (\beta_{a,k} e^a - \beta_{k,a} e^a) = \gamma^{kj} (\beta_{a,k} - \beta_{k,a}) e^a \\
 &\Rightarrow e_{,a}^i e^a + \gamma^{ia} \left[ \gamma_{ja,k} e^j e^k - \frac{1}{2} \gamma_{jk,a} e^j e^k \right] = \gamma^{ij} (\beta_{k,j} - \beta_{j,k}) e^k \\
 &\Rightarrow \frac{de^i}{d\ell_{ph}} + \gamma^{ia} \left( \gamma_{aj,k} - \frac{1}{2} \gamma_{jk,a} \right) e^j e^k = \gamma^{ij} (\beta_{k,j} - \beta_{j,k}) e^k
 \end{aligned} \tag{6.29}$$

We can now utilise [eq. \(6.29\)](#) to express the geodesic equation as follows-

$$\begin{aligned}
 e_{;j}^i e^j &= \frac{de^i}{d\ell_{ph}} + \tau_{jk}^i e^j e^k \\
 &= \frac{de^i}{d\ell_{ph}} + \gamma^{ia} \left( \gamma_{aj,k} - \frac{1}{2} \gamma_{jk,a} \right) e^j e^k \\
 &= \gamma^{ij} (\beta_{k,j} - \beta_{j,k}) e^k \equiv a^i
 \end{aligned} \tag{6.30}$$

where ; indicates covariant derivative with respect to the Fermat metric  $\gamma_{ij}^{(F)}$  and  $\tau_{jk}^i$  represents the Christoffel symbol in terms of  $\gamma_{ij}^{(F)}$ .

The acceleration vector  $a^i$  is also expressed in terms of [eq. \(6.30\)](#).

The eq. (6.23) can be tensorially expressed as

$$\kappa_g = \varepsilon_{ijk} N^i a^j e^k \quad (6.31)$$

where  $\vec{T}$  and  $\vec{T}'$  are in correspondence with  $e^k$  and  $a^j$ .

The geodesic curvature can therefore be expressed as shown below. We simply need to substitute  $\gamma_{ij} e^i e^j = 1$  and  $\gamma_{ij} e^i N^j = 0$  at the appropriate junctures.

$$\begin{aligned} \kappa_g &= \varepsilon_{ijk} N^i \gamma^{jl} (\beta_{n;l} - \beta_{l;n}) e^n e^k \\ &= \gamma^j a N^i e^k e^b \varepsilon_{ijk} \varepsilon_{sab} \varepsilon^{sml} \beta_{l;m} \\ &= N_i e_k e^b (\delta_s^i \delta_b^k - \delta_b^i \delta_s^k) \varepsilon^{sml} \beta_{l;m} \\ &= -\varepsilon^{ijk} N_i \beta_{j;k} \end{aligned} \quad (6.32)$$

The unit vector normal to the equatorial plane is  $N_p = \frac{1}{\sqrt{\gamma^{\theta\theta}}} \delta_p^\theta$ . We have chosen the upward direction over here without compromising generality. Since, we are concerned with the equatorial plane the term  $\varepsilon^{\theta xy} \beta_{y;x}$  can be expressed as  $= -\frac{1}{\sqrt{\gamma}} \beta_{\phi,r}$  since  $\varepsilon^{\theta r \phi} = -\frac{1}{\sqrt{\gamma}}$  and  $\beta_{r,\phi}$  vanishes on account of axisymmetry. Now, substituting all of this back into eq. (6.32) we get the expression for geodesic curvature on the equatorial plane as

$$\kappa_g = -\frac{1}{\sqrt{\gamma^{\theta\theta}}} \beta_{\phi,r} \quad (6.33)$$

As for the line element  $d\ell_{\text{ph}}$  we can easily define it for the equatorial plane ( $\theta = \frac{\pi}{2}$ ) with the help of  $d\ell_{\text{ph}}^2 \equiv \gamma_{ij} dx^i dx^j$  as

$$d\ell_{\text{ph}} = \sqrt{\gamma_{rr} \left( \frac{dr}{d\phi} \right)^2 + \gamma_{\phi\phi} d\phi} \quad (6.34)$$

Recalling that for the Kerr metric  $\gamma_{ij}^{\text{F}}$  can be expressed as

$$\gamma_{ij}^{(\text{F})} = \frac{\Sigma^2}{\Delta (\Sigma - 2Mr)} dr^2 + \frac{\Sigma^2}{(\Sigma - 2Mr)} d\theta^2 + \left( r^2 + a^2 + \frac{2a^2 Mr \sin^2 \theta}{\Sigma - 2Mr} \right) \frac{\Sigma \sin^2 \theta}{\Sigma - 2Mr} \quad (6.35)$$

Using this we can get the required expression for  $\gamma^{\theta\theta}$  as

$$\gamma^{\theta\theta} = \frac{\Sigma - 2Mr}{\Sigma^2} = \frac{r^2 + a^2 \cos^2 \theta - 2Mr}{(r^2 + a^2 \cos^2 \theta)^2} = \frac{r^2 - 2Mr}{r^4} = \frac{1}{r^2} - \frac{2M}{r^3} \quad (6.36)$$

and that of  $\gamma$  as

$$\gamma = \frac{\Sigma^2}{\Delta (\Sigma - 2Mr)} \frac{\Sigma^2}{\Sigma - 2Mr} \left( r^2 + a^2 + \frac{2a^2 Mr \sin^2 \theta}{\Sigma - 2Mr} \right) \frac{\Sigma \sin^2 \theta}{\Sigma - 2Mr}$$

$$\begin{aligned}
 &= \frac{(r^2 + a^2 \cos^2 \theta)^2}{\Delta (r^2 + a^2 \cos^2 \theta - 2Mr)} \frac{(r^2 + a^2 \cos^2 \theta)^2}{r^2 + a^2 \cos^2 \theta - 2Mr} \left( r^2 + a^2 + \frac{2a^2 Mr \sin^2 \theta}{r^2 + a^2 \cos^2 \theta - 2Mr} \right) \frac{(r^2 + a^2 \cos^2 \theta) \sin^2 \theta}{(r^2 + a^2 \cos^2 \theta - 2Mr)} \\
 &\simeq \frac{r^4}{(r^2 - 2Mr + a^2)(r^2 - 2Mr)} \frac{r^4}{r^2 - 2Mr} \left( r^2 + a^2 + \frac{2a^2 Mr}{r^2 - 2Mr} \right) \frac{r^2}{r^2 - 2Mr} \\
 &= \frac{r^{10}}{(r^2 - 2Mr)^3 (r^2 - 2Mr + a^2)} \left( r^2 + a^2 + \frac{2a^2 Mr}{r^2 - 2Mr} \right)
 \end{aligned} \tag{6.37}$$

Therefore,  $\gamma\gamma^{\theta\theta}$  assumes the form,

$$\begin{aligned}
 \gamma\gamma^{\theta\theta} &= \frac{r^2 - 2Mr}{r^4} \frac{r^{10}}{(r^2 - 2Mr)^3 (r^2 - 2Mr + a^2)} \left( r^2 + a^2 + \frac{2a^2 Mr}{r^2 - 2Mr} \right) \\
 &= \frac{r^6}{(r^2 - 2Mr)^2 (r^2 - 2Mr + a^2)} \left( r^2 + a^2 + \frac{2a^2 Mr}{r^2 - 2Mr} \right) \\
 &= \frac{r^6}{r^2 (r - 2M)^2 (r^2 - 2Mr + a^2)} \frac{r^3 - 2Mr^2 + a^2 r - 2Ma^2 + 2Ma^2}{r - 2M} \\
 &= \frac{r^4}{(r - 2M)^3 (r^2 - 2Mr + a^2)} r (r^2 - 2Mr + a^2) \\
 &= \frac{r^5}{(r - 2M)^3}
 \end{aligned} \tag{6.38}$$

On the other hand,  $\beta_\phi$  in the equatorial plane will reduce down to,

$$\beta_\phi = -\frac{2aMr \sin^2 \theta}{r^2 + a^2 \cos^2 \theta - 2Mr} = -\frac{2aM}{r - 2M} \tag{6.39}$$

Hence,  $\beta_{\phi, r}$  becomes

$$\beta_{\phi, r} = -\frac{\partial}{\partial r} \frac{2aM}{r - 2M} = \frac{2aM}{(r - 2M)^2} \tag{6.40}$$

Finally, we can plug in all these expressions to find the final expression of the geodesic curvature,

$$\begin{aligned}
 \kappa_g &= -\frac{1}{\sqrt{\gamma\gamma^{\theta\theta}}} \beta_{\phi, r} \\
 &= -\frac{1}{\sqrt{\frac{r^5}{(r - 2M)^3}}} \frac{2aM}{(r - 2M)^2} \\
 &= -\frac{2aM}{r^2 \sqrt{r(r - 2M)}} = -\frac{2aM}{r^3} \frac{1}{\sqrt{1 - \frac{2M}{r}}} \\
 &= -\frac{2aM}{r^3} \left( 1 - \frac{2M}{r} \right)^{-\frac{1}{2}} \\
 &= -\frac{2aM}{r^3} \left( 1 + \frac{M}{r} + \dots \right) \\
 &= -\frac{2aM}{r^3} + \mathcal{O} \left( \frac{aM^2}{r^4} \right)
 \end{aligned} \tag{6.41}$$

We have neglected any terms involving  $a^n M$  over here for  $n \geq 2$ . For the line element  $d\ell_{\text{ph}}$  in eq. (6.34) we can similarly substitute the expressions of  $\gamma_{rr}$  to reduce it to the form

$$d\ell_{\text{ph}} = \left[ \frac{b}{\sin^2 \phi} + \mathcal{O}(M) \right] d\phi \quad (6.42)$$

Now, we can finally amalgamate the two expressions presented in eq. (6.41) & eq. (6.42) to get the expression of the path integral of the geodesic curvature as follows

$$\begin{aligned} \int_V \kappa_g d\ell_{\text{ph}} &= \int_S \frac{2aM}{r^3} d\ell_{\text{ph}} + \mathcal{O}\left(\frac{aM^2}{r^4}\right) \\ &= \int_{\phi_S}^{\phi_V} 2aM \times \frac{\sin^3 \phi}{b^3} \times \frac{b}{\sin^2 \phi} d\phi + \mathcal{O}\left(\frac{aM^2}{r^4}\right) \\ &= \frac{2aM}{b^2} \int_{\phi_S}^{\phi_V} \sin \phi d\phi + \mathcal{O}\left(\frac{aM^2}{r^4}\right) \\ &= \frac{2aM}{b^2} \left[ \sqrt{1 - b^2 u_V^2} + \sqrt{1 - b^2 u_S^2} \right] + \mathcal{O}\left(\frac{aM^2}{b^3}\right) \end{aligned} \quad (6.43)$$

Tracing back to eq. (4.6a) we can see that the angle of deviation as evaluated from the expressions involving the summation of the surface integral of Gaussian curvature and the line integral of geodesic curvature is

$$\begin{aligned} \alpha_{D_{\text{Kerr}}}^{\text{pro}} &= - \iint_{\square_S^\infty} K dS + \int_S \kappa_g d\ell_{\text{ph}} \\ &= \frac{2M}{b} \left[ \sqrt{1 - b^2 u_V^2} + \sqrt{1 - b^2 u_S^2} \right] - \frac{2aM}{b^2} \left[ \sqrt{1 - b^2 u_V^2} + \sqrt{1 - b^2 u_S^2} \right] + \mathcal{O}\left(\frac{M^2}{b^2}\right) \\ &= \left[ \sqrt{1 - b^2 u_V^2} + \sqrt{1 - b^2 u_S^2} \right] \left[ \frac{2M}{b} - \frac{2aM}{b^2} \right] \end{aligned} \quad (6.44)$$

which agrees exactly with eq. (5.65). And of course tweaking the sign of the term containing  $a$  gives the expression for the retrograde trajectory for the Kerr case as before which is the same as eq. (5.66). Therefore, the equivalency of the two formalism is firmly established for the generic case of the axisymmetric scenario by taking the Kerr metric.

*Tyger Tyger, burning bright,  
In the forests of the night;  
What immortal hand or eye,  
Could form thy fearful symmetry?*

—William Blake, *The Tyger*

# 7

## DEFLECTION OF LIGHT IN STATIC SPHERICALLY SYMMETRIC SPACETIME BY OIA FORMALISM

We have already established that the Gibbons-Werner and the Ono-Ishihara-Asada formalism are equivalent to each other for the axisymmetric spacetimes. Furthermore, the fact that the results reduce to the more familiar expressions of deflection angle under the consideration of  $r_S, r_{SV} \rightarrow \infty$  gives validation to the GW-OIA formalism. For, the sake of thoroughness we can now step down to the case of spherically symmetric spacetime and derive the more generic expression of deflection angle in that context. For the current study we restrict ourselves to the Schwarzschild metric only.

The generic representation of a spherically symmetric spacetime is expressed as

$$\begin{aligned} ds^2 &= g_{ij} dx^i dx^j \\ &= -A(r) dt^2 + B(r) dr^2 + r^2 (d\theta^2 + \sin^2 \theta d\phi^2) \end{aligned} \quad (7.1)$$

As usual we restrict ourselves to studying the photon orbit in the equatorial plane and evaluate the impact parameter as done earlier in [section 5.2.2](#). Under the specified conditions the energy  $E$  and the azimuthal component of angular momentum  $J$  are conserved along the geodesics and hence can be expressed as

$$E = -g_{tt}\dot{t} - g_{t\phi}\dot{\phi} = A \quad (7.2)$$

and

$$J = g_{\phi t}\dot{t} + g_{\phi\phi}\dot{\phi} = r^2\dot{\phi} \quad (7.3)$$

Therefore, the impact parameter of the photon trajectory under the influence of a Schwarzschild metric is

$$b = \frac{J}{E} = \frac{r^2}{A(r)} \frac{d\phi}{dt} = \frac{r^3}{r-2M} \frac{d\phi}{dt} \quad (7.4)$$

From, the mathematical framework already developed in [section 5.1](#) as [eq. \(5.7\)](#) & [eq. \(5.8\)](#) we can derive the Fermat metric for the Schwarzschild case as follows

$$\begin{aligned} \gamma_{ij}^{(\text{F})} dx^i dx^j &= \frac{\frac{r}{r-2M}}{\frac{r}{r-2M}} dr^2 + \frac{\frac{r^2}{r-2M}}{\frac{r}{r-2M}} d\theta^2 + \frac{\frac{r^2 \sin^2 \theta}{r-2M}}{\frac{r}{r-2M}} d\phi^2 \\ &= \frac{r^2}{(r-2M)^2} dr^2 + \frac{r^3}{r-2M} d\theta^2 + \frac{r^3 \sin^2 \theta}{r-2M} d\phi^2 \end{aligned} \quad (7.5)$$

As there are no cross terms in the Schwarzschild metric we have  $\beta_i dx^i = 0 \Rightarrow \beta_i = 0$ .

Now, starting with the Schwarzschild metric in the form expressed below we can derive the equation for the photon trajectory by setting  $ds^2 = 0$ .

$$ds^2 = - \left(1 - \frac{2M}{r}\right) dt^2 + \frac{dr^2}{\left(1 - \frac{2M}{r}\right)} + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2 \quad (7.6)$$

Therefore,

$$\begin{aligned} 0 &= -A\dot{t}^2 + B\dot{r}^2 + D\dot{\phi}^2 \\ \Rightarrow \dot{r}^2 &= \frac{A}{B}\dot{t}^2 - \frac{D}{B}\dot{\phi}^2 \\ \Rightarrow \left(\frac{dr}{d\ell_{\text{ph}}}\right)^2 &= \frac{A}{B}\left(\frac{dt}{d\ell_{\text{ph}}}\right)^2 - \frac{D}{B}\left(\frac{d\phi}{d\ell_{\text{ph}}}\right)^2 \\ \Rightarrow \left(\frac{dr}{d\phi}\right)^2 &= \frac{A}{B}\left(\frac{dt}{d\ell_{\text{ph}}}\right)^2 \left(\frac{d\ell_{\text{ph}}}{d\phi}\right)^2 - \frac{D}{B} \\ \Rightarrow \left(\frac{dr}{d\phi}\right)^2 + \frac{r^2 \sin^2 \theta}{B} &= \frac{A}{B} \frac{1}{\left(\frac{d\phi}{dt}\right)^2} = \frac{A}{B} \frac{1}{\left(\frac{Ab}{r^2}\right)^2} = \frac{r^4}{ABb^2} \\ \Rightarrow \left(\frac{dr}{d\phi}\right)^2 + \frac{r^2}{B} &= \frac{r^4}{b^2 AB} \end{aligned}$$

$$\Rightarrow \left( \frac{dr}{d\phi} \right)^2 = \frac{r^4}{b^2 AB} - \frac{r^2}{B} \quad (7.7)$$

This expression will come in useful while evaluating the value of  $\phi_{VS}$ . As has been illustrated in the case of Kerr metric earlier determining the deflection angle by evaluating the surface integral of the Gaussian curvature and the line integral of the geodesic curvature is comparatively tedious. As such we proceed with the OIA formalism.

The unit tangential vector along the photon trajectory is given by  $\tau^i = \frac{1}{\zeta} \left( \frac{dr}{d\phi}, 0, 1 \right)$ . Now, developing from the condition  $\gamma_{ij}^{(F)} \tau^i \tau^j = 1$  we get the following equation,

$$\begin{aligned} \gamma_{ij}^{(F)} \tau^i \tau^j &= 1 \\ \Rightarrow \gamma_{rr} \tau^r \tau^r + \gamma_{\phi\phi} \tau^\phi \tau^\phi &= 1 \\ \Rightarrow \frac{r^2}{(r-2M)^2} \left( \frac{1}{\zeta} \frac{dr}{d\phi} \right)^2 + \frac{r^3 \sin^2 \theta}{r-2M} \frac{1}{\zeta^2} &= 1 \\ \Rightarrow \frac{B}{A} \left( \frac{1}{\zeta} \frac{dr}{d\phi} \right)^2 + \frac{r^2}{A} \frac{1}{\zeta^2} &= 1 \\ \Rightarrow \frac{1}{\zeta^2} \left[ \frac{B}{A} \left( \frac{dr}{d\phi} \right)^2 + \frac{r^2}{A} \right] &= 1 \end{aligned} \quad (7.8)$$

Substituting the expression for  $\left( \frac{dr}{d\phi} \right)^2$  from eq. (7.7) in eq. (7.8) we get,

$$\begin{aligned} \frac{1}{\zeta^2} \left[ \frac{B}{A} \left( \frac{r^4}{b^2 AB} - \frac{r^2}{B} \right) + \frac{r^2}{A} \right] &= 1 \\ \Rightarrow \frac{1}{\zeta^2} \left[ \frac{r^4}{b^2 A^2} \right] &= 1 \\ \Rightarrow \frac{1}{\zeta} &= \frac{bA}{r^2} \end{aligned} \quad (7.9)$$

Similarly, in reference to the unit radial vector  $\rho^i = (\rho^r, 0, 0)$  along the photon trajectory we can proceed from the condition  $\gamma_{ij}^{(F)} \rho^i \rho^j = 1$  to derive the expression for  $\rho^r$

$$\begin{aligned} \gamma_{ij}^{(F)} \rho^i \rho^j &= 1 \\ \Rightarrow \gamma_{rr} \rho^r \rho^r + \gamma_{\theta\theta} \rho^\theta \rho^\theta + \gamma_{\phi\phi} \rho^\phi \rho^\phi &= 1 \\ \Rightarrow \gamma_{rr} (\rho^r)^2 &= 1 \\ \Rightarrow \rho^r &= \frac{1}{\sqrt{\gamma_{rr}}} \end{aligned} \quad (7.10)$$

Now, in order to evaluate the jump angles  $\psi_V$  &  $\psi_S$  we need to determine the expression for  $\sin \psi$ . This is easily done by working through the mathematical formulation for  $\cos \psi$

$$\cos \psi = \gamma_{ij}^{(F)} \rho^i \tau^j$$

$$\begin{aligned}
 &= \gamma_{rr} \rho^r \tau^r \\
 &= \gamma_{rr} \frac{1}{\sqrt{\gamma_{rr}}} \frac{1}{\zeta} \frac{dr}{d\phi} \\
 &= \sqrt{\gamma_{rr}} \frac{bA}{r^2} \frac{dr}{d\phi} \\
 \Rightarrow \cos^2 \psi &= \gamma_{rr} \frac{b^2 A^2}{r^4} \left( \frac{r^4}{b^2 AB} - \frac{r^2}{B} \right) \\
 &= \gamma_{rr} \frac{b^2 A^2}{r^2} \frac{1}{B} \left( \frac{r^2}{b^2 A} - 1 \right) \\
 &= \frac{B}{A} \frac{b^2 A^2}{r^2 B} \left( \frac{r^2}{b^2 A} - 1 \right) \\
 &= 1 - \frac{b^2 A}{r^2} \\
 \Rightarrow 1 - \cos^2 \psi &= \frac{b^2 A}{r^2} \\
 \Rightarrow \sin \psi &= \frac{b\sqrt{A}}{r} = \frac{b}{r} \left( 1 - \frac{2M}{r} \right)^{-\frac{1}{2}} = bu - bMu^2
 \end{aligned} \tag{7.11}$$

Expanding in Taylor's series we can now express the required expressions for  $\psi_V$  &  $\psi_S$  as

$$\psi_V = \sin^{-1}(bu_V) - \frac{bMu_V^2}{\sqrt{1 - b^2u_V^2}} \tag{7.12}$$

and

$$\pi - \psi_S = \sin^{-1}(bu_S) - \frac{bMu_S^2}{\sqrt{1 - b^2u_S^2}} \tag{7.13}$$

The remaining contribution to the deflection angle comes from the term  $\phi_{VS}$ . We can easily achieve this by restructuring [eq. \(7.7\)](#) in terms of reciprocal coordinates.

$$\left( \frac{du}{d\phi} \right)^2 = \frac{1}{b^2 AB} - \frac{u^2}{B} = \frac{1}{b^2} - u^2 + 2Mu^3 = F(u) \tag{7.14}$$

For the distance of closest approach  $u = u_0$  and  $\frac{du}{d\phi} = 0$  which for the Schwarzschild case leads to

$$\begin{aligned}
 0 &= \frac{1}{b^2} - u_0^2 + 2Mu_0^3 \\
 \Rightarrow \frac{1}{b^2} &= u_0^2 [1 - 2Mu_0] \\
 \Rightarrow b &= \frac{1}{u_0} (1 - 2Mu_0)^{-\frac{1}{2}} = \frac{1}{u_0} + M
 \end{aligned} \tag{7.15}$$

Substituting the expression for the impact parameter in terms of the distance of closest approach in [eq. \(7.14\)](#)

$$\begin{aligned}
 F(u) &= \frac{1}{b^2} - u^2 + 2Mu^3 \\
 &= u_0^2 - 2Mu_0^3 - u^2 + 2Mu^3
 \end{aligned}$$

$$\begin{aligned}
 &= (u_0^2 - u^2) - 2M (u_0^3 - u^3) \\
 &= (u_0^2 - u^2) \left[ 1 - 2M \left( \frac{u_0^3 - u^3}{u_0^2 - u^2} \right) \right] \\
 \Rightarrow \frac{1}{\sqrt{F(u)}} &= \frac{1}{\sqrt{u_0^2 - u^2}} \left[ 1 - 2M \left( \frac{u_0^3 - u^3}{u_0^2 - u^2} \right) \right]^{-\frac{1}{2}} \\
 &= \frac{1}{\sqrt{u_0^2 - u^2}} \left[ 1 + \frac{2M}{2} \left( \frac{u_0^3 - u^3}{u_0^2 - u^2} \right) \right] \\
 &= \frac{1}{\sqrt{u_0^2 - u^2}} + M \frac{u_0^3 - u^3}{(u_0^2 - u^2)^{\frac{3}{2}}}
 \end{aligned} \tag{7.16}$$

The expression for  $\frac{1}{\sqrt{F(u)}}$  when substituted in [eq. \(5.40\)](#) for the Schwarzschild case yields the same integrals as [eq. \(5.55a\)](#) & [eq. \(5.55b\)](#). As such we can write down the expression for  $\phi_{VS}$  pertaining to Schwarzschild case as

$$\phi_{VS} = \pi - \sin^{-1}(bu_V) - \sin^{-1}(bu_S) + \frac{M}{b} \frac{(2 - b^2 u_V^2)}{\sqrt{1 - b^2 u_V^2}} + \frac{M}{b} \frac{(2 - b^2 u_S^2)}{\sqrt{1 - b^2 u_S^2}} \tag{7.17}$$

Finally we can combine [eq. \(7.12\)](#), [eq. \(7.13\)](#) & [eq. \(7.17\)](#) to get the expression for deflection angle of a photon in Schwarzschild spacetime as

$$\begin{aligned}
 \alpha_D^{\text{Sch}} &= \psi_V - \psi_S + \phi_{VS} \\
 &= \frac{2M}{b} \left[ \sqrt{1 - b^2 u_V^2} + \sqrt{1 - b^2 u_S^2} \right]
 \end{aligned} \tag{7.18}$$

As usual under the limits  $u_V \rightarrow 0$  &  $u_S \rightarrow 0$  [eq. \(7.18\)](#) reduces to the familiar form of  $\alpha_D^{\text{Sch}} \rightarrow \frac{4M}{b}$ .

*If you have built castles in the air, your work need not be lost; that is where they should be. Now put the foundations under them.*

—Henry David Thoreau

*With this discovery, we humans are embarking on a marvelous new quest: the quest to explore the warped side of the universe—objects and phenomena that are made from warped spacetime. Colliding black holes and gravitational waves are our first beautiful examples.*

—Kip Thorne

# 8

## CONCLUSION & FUTURE PROSPECTS

General relativity has a long-standing history of coming up with theoretical predictions that would be experimentally verified only after several decades. And this is a subject which has a plethora of such theories. Indeed the celebrated detection of gravitational waves came to fruition after the incessant efforts of over five decades. Gravitational lensing in this context has been comparatively lucky as several of its theoretical predictions could be verified relatively quickly. However, the fine tuning of bending angle of light accounting for finite distance effects is something that requires precision in our measurements hitherto not achieved. However, with the innovations developed in HST, SKA, Euclid and several other upcoming observational facilities the measurements related to gravitational lensing are becoming evermore precise and accurate.

The finite distance correction to the deflection angle of photon trajectories may possibly be verified by future instruments. It is worth noticing in this context that we have dealt with Schwarzschild and Kerr metric. The Kerr metric is widely accepted as the most realistic model of a black hole since it has the angular momentum component as well. Since most galaxies are conjectured to harbour a supermassive black hole at their centre the prospect of verifying the treatments presented here are a distinct possibility.

The strong deflection regime of gravitational lensing, for which the finite distance corrections already exist in the literature [[Ishihara et al. \(2016\)](#)]. As a future endeavour a followup on laying out the detailed

framework of these techniques is planned. Furthermore, the applications of these methodologies and drawing comparisons with already available observational data. On the other hand purely theoretical studies related to spacetimes describing entities like wormholes, di-holes, global monopoles etc can be carried out.



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